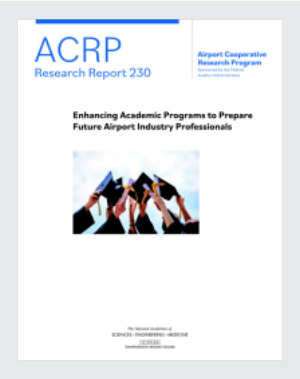
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ISBN 978-0-309-46570-0 | DOI 10.17226/26140

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SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2021. *Enhancing Academic Programs to Prepare Future Airport Industry Professionals*. Washington, DC: The National Academies Press. https://doi.org/10.17226/26140.

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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 230

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Subscriber Categories
Education and Training • Aviation

Research sponsored by the Federal Aviation Administration

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SCIENCES • ENGINEERING • MEDICINE
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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP RESEARCH REPORT 230

Project 06-07
ISSN 2572-3731 (Print)
ISSN 2572-374X (Online)
ISBN 978-0-309-67385-3
Library of Congress Control Number 2021934135

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Published research reports of the

AIRPORT COOPERATIVE RESEARCH PROGRAM

are available from

Transportation Research Board
Business Office
500 Fifth Street, NW
Washington, DC 20001

and can be ordered through the Internet by going to

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COOPERATIVE RESEARCH PROGRAMS

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AUTHOR ACKNOWLEDGMENTS

The ACRP Project 06-07 research team would like to acknowledge the more than 100 professionals at airports, airport consulting firms, and academia for participating in surveys, interviews, and focus groups associated with this work. Those listed herein granted permission to share their names. Thank you.

Pamela Belalcazar	TY Lin and Associates, Inc.
Fin Bonset	VHB, Inc.
Nick Brown	Crawford, Murphy, and Tilly, Inc.
Charles Butcher	Northwest Arkansas Regional Airport
Gene Conrad	Lakeland Linder International Airport, Florida
Todd Cox, CM	Avon, Inc.
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AUTHOR ACKNOWLEDGMENTS (*Continued*)

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Gel Le Bris	WSP USA
Ryan Leick	Utah Valley University
David Mathias	Michael Baker International, Inc.
Mary Niemczyk	Arizona State University
Aaron Organ	Utah Valley University
Paul Priegel	Stillwater Airport, Oklahoma
Carlos Ruiz-Coll	Greenville Spartanburg International Airport
Joseph Von Bargen	Cincinnati Northern Kentucky International Airport, Kentucky
Autumn Ward	Environmental Science Associates
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FOREWORD

By Theresia H. Schatz
Staff Officer
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ACRP Research Report 230: Enhancing Academic Programs to Prepare Future Airport Industry Professionals provides guidance to assist academia in preparing graduates for careers as airport industry professionals. The guidance identifies and evaluates the current and future skills needed for airport industry professionals and how educational programs align with those skills. As a means to prepare students for career tracks in the airport industry, this research provides a better understanding of the skills needed and develops model academic curriculum templates to provide a foundation for career success, bridging the gaps that may exist.

Airports and their support industries have changed significantly over the past several decades. It is imperative that academic programs continue to evolve with these changes to better prepare the next generation of airport industry professionals. Emphasis is placed on an enhanced, comprehensive approach to careers in the airport industry to include topic areas such as finance and administration, planning and engineering, operations and facilities, technology, law, humanities, and social sciences.

ACRP Project 06-07 was led by The Ohio State University in association with Quadrex Aviation, ICF, Inc., Pavlik & Associates, Texas A&M Transportation Institute, and DPrather Aviation Solutions. They reviewed the current and future workforce needs of the airport industry, provided many current academic curriculum outlines, and identified the gaps and challenges that exist between what academic programs offer and what the airport industry needs to keep up with the demands of an exceptional workforce. Their work was conducted through surveys and focus groups from over 100 airport industry professionals, including academic faculty, students, and alumni currently employed in the airport sector.

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SECTION I

Introduction and Motivation

The guidebook is based on a year's worth of research into the current and future workforce needs of the airport industry; present-day offerings of a spectrum of academic institutions; gaps and challenges that exist between what academic programs offer and what the industry needs to keep up with workforce demands; and best practices existing in academia today. The study involved extensive review of current aviation and academic literature; examination of academic program curricula at universities and colleges nationwide; and the conduct of surveys and focus groups in which more than 100 airport industry professionals, academic faculty, students, and alumni now in the airport sector participated.

Challenges that the airport industry faces in hiring entry-level and mid-career professionals are addressed. The guidance provided in this guidebook includes detailed recommendations for ensuring that industry needs are met now and in the future.

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CHAPTER 1

Introduction

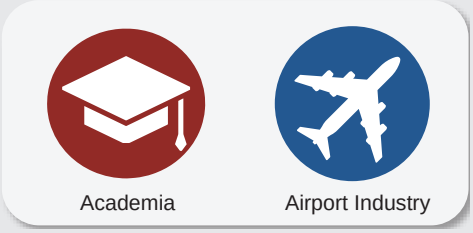
The challenge of airports in the United States to recruit professionals with the skills, education, and experience necessary is significant. The current airport environment has evolved rapidly since the beginning of the 21st century. Airport business models and practices continue to move from traditional public agency operations to highly complex business administrations much like the corporate world, often creating partnerships between the public and private sectors. Airports have become savvy in evolving financial practices because they must be profit-driven to operate without a significant tax base. The federal regulatory environment continues to generate rules and practices governing the safety and security of airport operations. Information technology has opened up the ability for airports to enhance the entirety of their operations, from planning and engineering to social media and marketing.

An existing general perception is that the nation's institutions of higher education are lagging in the quantity and quality of aviation-related and airport-specific academic programming. Of the more than 6,000 colleges and universities in the United States, approximately 160 offer degree programs in aviation, and a small subset of those have courses that focus on airports. Although several of these have been formally recognized for their quality through aviation academic program accreditations, such programming is not accredited for curricula specifically addressing airport needs. New courses relevant to today's industry are needed. Experiential learning—through project work, internships, and cooperatives (co-ops)—requires expansion, and learning methods need to be adapted.

A disconnect between the needs of the industry and academic programs is possibly related to a general lack of industry collaboration. Although the industry may often look to aviation programs from which to recruit educated and trained professionals, many more non-aviation programs are educating future professionals in key topics of relevance to airports. However, too often their courses do not apply specifically to the airport environment. Thus, the industry is required to further train these graduates once they enter the workforce.

A Guidebook for Academia and Industry

Two primary audiences will benefit from the guidebook: (1) academic institutions wishing to enhance their airport-oriented academic programming, and (2) airport industry professionals desiring to work with academic institutions in pursuit of their own airport-oriented academic programming goals. Both academia and industry must play significant roles in this effort.



The guidance provided herein applies to academic institutions of any type, from community colleges to research-focused universities, from

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programs that currently offer degree programs specifically in airports to others with no aviation course content whatsoever. Any academic institution has the potential to contribute to the airport industry through even minor enhancements to its curricula. The guidebook is also a resource for any airport industry professional, from the airport manager/CEO to the entry-level professional.

Guidebook Contents and Format

The guidebook is divided into four sections, within which are six chapters and a compilation of reference materials. Icons for academia (a graduation cap) and for airport industry (an airplane) appear throughout the guidebook to identify guidance relevant to these audiences. Two chapters are written explicitly for academia in which case the academia icon is emphasized, and two for industry in which case the airport industry icon is emphasized. Other chapters are written for both audiences in which case both icons are emphasized. With that in mind, familiarity with all sections on the part of both audiences is recommended. Of great benefit, for example, would be if academic institutions better understood the needs of the airport workforce, and how these institutions may ensure these needs are addressed. Conversely, it would be helpful for industry to better understand how academic programs are developed and administered, so they are better prepared to contribute to the processes and are aware of the unique policies and practices within the academic environment. The following outlines the guidebook's content.



Section I Introduction and Motivation

Section I applies to all audiences. It introduces the guidebook, provides background on the guidebook's development, and motivates the reader to pursue opportunities and strategies to enhance academic programming.

Chapter 1: Introduction. This chapter provides introductory, welcoming, and motivating commentary about how academia can meet the workforce needs of the airport industry.

Chapter 2: Motivation: Addressing the Needs of the Airport Workforce. This chapter describes the needs of the airport industry workforce. These needs include not only the need for future professionals to have a strong knowledge base of traditional airport-industry subject matter but also to have the ability to apply general academic topics to the industry. This chapter also describes the need for students to gain practical experience within their academic programs, and how professional and personal effectiveness skills are needed by future airport professionals.



Section II Guidance to Academia

Section II provides a comprehensive understanding of the current state of airport academic offerings and broad guidance on how to enhance academic programming, ranging from institutions that offer no airport content to those that have degree programs in airport management.

Chapter 3: The State of Airport Education in Academia. This chapter provides a comprehensive understanding of the wide variety of airport-oriented educational programming that exists within academia, ranging from traditional aviation programs to non-aviation programs, such as business and engineering programs that offer some airport content. This chapter describes how airport content fits into various degree programs, how extracurricular and experiential learning

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contributes to educational strategies, how these programs are meeting the needs of the industry, and how these programs may be enhanced.

Chapter 4: Enhancing Academic Programming. This chapter provides guidance to academia by introducing a set of strategic building blocks, curriculum structures, course offerings, and experiential learning opportunities to position curricula to better meet the needs of the industry. This includes the following:

- Expanding the depth of programming by adding courses in specialized topics and increasing the depth of existing courses.
- Expanding the breadth of programming by adding airport content to non-airport and non-aviation courses within a course of study.
- Developing and expanding experiential learning programs, such as internships and extra-curricular activities.



Section III Guidance to Industry

Given that industry involvement by collaborating and partnering with academia is crucial to enhancing academic programs to meet industry needs, this section describes opportunities for accomplishing this.

Chapter 5: How Academia Works for Non-Academics. This chapter concisely describes the process for designing, creating, maintaining, delivering, and enhancing academic programming so that the industry better understands and appreciates the process.

Chapter 6: Opportunities to Participate in the Academic Process. Industry representatives can directly participate in academic programming by contributing content, participating as instructors, hosting site visits, supervising course projects, and participating in program advisory boards.



Section IV Reference Materials

This section contains a comprehensive set of materials that academia is encouraged to apply to enhance their aviation-related programs.

- An **annotated bibliography** of existing literature, including research reports, academic publications, and trade journals, covering the ongoing topics of importance to the airport workforce.
- A portfolio of **template course syllabi covering key airport topics**. These templates include topic descriptions, outlines of subtopics, descriptions of possible course assignments, and available external materials, such as applicable research reports.
- **Guides and templates for developing internships** with various industry partners, including a variety of departments within airports as well as with airport service industry professional firms.
- A **collection of airport-specific topic summaries and activities** that may be added to courses that are not necessarily airport or aviation specific.

These materials are provided in a generic format, allowing modifications to meet the needs of specific programs.

Topic Identifying Icons

The guidebook is designed for use within all types of academic programs to enhance airport-related content, from those that already have airport major degree programs to programs that do not cover any airport-related material. Subject matter icons within the guidebook identify relevant

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ACADEMIC ENVIRONMENT ICONS	
	Business/Finance
	Engineering/Planning
	Operations
	Interacting with Industry

guidance for each academic topic of interest, including business and finance, engineering and planning, and operations.

The guidebook identifies strategies applicable to participating in the enhancement of academic programming for those in industry, ranging from upper management at an airport looking to enhance its workforce from a strategic perspective, to entry-level and junior management professionals who wish to share their new industry experience with their potential future colleagues, to those in airport professional service firms, such as engineering, planning, or management consulting.

The application of all, some, or simply one of the elements included herein is important for both academic institutions and airport industry professionals to contribute to the educational development of the airport industry workforce.

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CHAPTER 2

Motivation: Addressing the Needs of the Airport Workforce

Key Takeaways

- A gap exists between what is taught in academic programs and the technical skills and knowledge required for new airport professionals entering the workforce. Academic programs provide students with a basic background in a variety of essential topic areas but have limited coverage of specific topic areas.
- There is a crucial need for students to be exposed to fields of study that fall outside traditional aviation management programs, including finance, business acumen, project management, property management, and engineering. Students should also be provided opportunities to gain skills relevant to mission-critical occupations (MCO) identified in *ACRP Web-Only Document 28: Identifying and Evaluating Airport Workforce Requirements* (Cronin et al. 2016).
- Academic programs and airports should work with each other to strengthen existing relationships and build new ones. Increased connectivity between airports and academia helps increase the availability of internships and encourages the involvement of airport managers in the classroom.
- Recent graduates often overestimate their ability to apply their education to the world outside of the classroom. Furthermore, recent graduates may lack personal effectiveness skills, such as communication and interpersonal skills, business acumen, and public speaking.
- Airports have difficulty finding and hiring skilled applicants for their available positions, in part due to a large demand for a limited number of aviation degree program graduates. As a result, airports may hire graduates who lack an aviation degree but have an interest in aviation.
- Many of the skills and topic areas that airports indicate are challenging to hire for are frequently not covered in students' programs, or at least not in a standalone airport-related course (see Table 2.1).

The airport environment is constantly evolving in response to the global demands associated with business and personal air travel. Given the ever-changing landscape, airports must react to advancements in technology, evolving passenger-experience preferences, and changing business models. At the same time, airport industry leaders are concerned about the lack of talent available in the labor market, at a time when more than half of airport executive staff are estimated to be retirement eligible, and the labor pool of workers available to fill these roles is shrinking.



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Other trends that have implications for the airport workforce include the rapid emergence of new technologies, increased financial and commercial pressures, political sensitivities, new regulatory requirements, challenges recruiting in local job markets, and a growing shortage of skilled trade workers (ICF 2018).

Now and into the future, airports must be leaner and more creative in terms of establishing new revenue streams and managing their resources, while continuing timely and safe operations. For example, as of the writing of this guidebook, the COVID-19 global pandemic was sending shock waves throughout the air transportation industry, swinging the pendulum wildly from an airport environment focused on infrastructure expansion due to record high demand to a world with vastly reduced global travel, air carriers facing tens of thousands of job reductions, and airports struggling to maintain revenues. To survive and be ready for the next economic recovery, airports must remain in a proactive, adaptive mode that is best served by maintaining talent pools that possess diversified skill sets.

Specifically, as airports consolidate resources, airport workers will need to perform a variety of different and cross-functional roles within the airport's operations structure. Similarly, students coming from aviation programs will need a broader spectrum of airport-related expertise to "hit the ground running" and function quickly within the varying job roles that employees will likely occupy. This, in turn, suggests that academia should grow well-rounded students with a broader understanding of the airport ecosystem.

In *ACRP Web-Only Document 28,* Cronin et al. identified eight MCOs for the next 5 to 10 years:

- Airport Development
- Airport Operations
- Airport Security
- Electrician
- Engineering (Civil, Electrical, Mechanical)
- Financial Analysis and Planning
- Information Technology (IT), and
- Project Planning

For each MCO occupation, more than 60% of survey respondents (746 airport leaders) reported that additional education and development opportunities are needed to adequately equip the workforce for these specific responsibilities.

Given the limited size and number of academic programs offering airport-oriented curricula, current training and education options for any given airport-relevant subject tend to be limited. Content for students may also be more focused on the past needs of the industry, rather than what is needed to prepare the future workforce to succeed in the rapidly changing airport environment. Furthermore, existing training and educational content lack sufficient coverage of many of the core competencies required for MCO jobs integral to the continued functioning and growth of airports.

The first step to ensure airports are adequately prepared to face current and future challenges starts with understanding what is needed of individuals entering the airport workforce, both now and into the future (Byers 2016). Academic programs can then be tailored, and new programs developed, to meet industry needs by providing a pipeline of graduates with the appropriate knowledge and skill sets to excel within the airport industry.

In preparation for the writing of this guidebook, the research team thoroughly examined the existing literature on the workforce needs of the industry; parsed through a spectrum of airport position announcements and descriptions, and conducted focus groups and surveys in which more than 200 airport industry professionals participated. From this research, the team identified four key areas that encompass many of the needs and challenges of entry-level airport workforce professionals. The four areas describe the need to





1. Better prepare graduates for careers in the airport industry;
2. Strengthen collaboration between the airport industry and academia;
3. Ensure recent graduates have appropriate and readily applicable skills for the workplace; and
4. Address existing challenges in hiring recent graduates.

A comprehensive overview of these topic areas is included in this guidebook.

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Creating More and Broader Opportunities for Recent Graduates

Limited Entry-Level Opportunities for Recent Graduates

The airport industry provides promising career opportunities; however, there are currently a limited number of avenues by which recent graduates can enter these fields. A review of job postings from general Internet sources (such as indeed.com as well as the American Association of Airport Executives Career Center website) **found that the most frequently posted jobs targeted at recent graduates are for entry-level airport operations positions (specialists and coordinators).** The majority of other available jobs required several years of experience in the field.

Depth of Program Subject Areas

> *Coursework that is currently developed at universities is not detailed enough in any area for a new employee to be able to hit the ground running practically.*
>
> – Manager, Airport Operations and Facilities

The tendency for most entry-level airport jobs to be in airport operations may be attributed to the typical skills and abilities of recent graduates from aviation and airport management programs. Over the years, these requirements have changed, becoming more diverse, challenging, and technical (Quilty 2005). **This has resulted in a gap between what is taught in academic programs and the unique technical skills and knowledge required for new airport professionals entering the workforce.** Most academic programs provide students with a basic background in a variety of different topic areas that are essential for a career in aviation, such as airport management and airport operations, but they are not specific to any particular topic areas nor do they focus on specific trades. Academic programs have many general education requirements that preclude students from taking a significant number of courses tailored to specific content areas. Additionally, many aviation management programs are small in size and, therefore, do not have enough students to offer these types of courses. As a result, recent graduates often lack an understanding of many of the finer details and technical skills required for a successful airport career. Research indicates that new graduates are not learning or acquiring all of the unique skills required to thrive in the aviation sector. For example, the entry-level airport workforce often lacks airport-specific knowledge or experience, including knowledge of standard operating procedures and aviation regulations (Young 2010). Although students graduate with a general understanding of the industry, significant on-the-job learning is often required post-graduation to gain proficiency in their work responsibilities.

Incorporation of Outside Fields of Study

In addition to the need for programs to dive deeper into airport-specific topic areas, such as landside operations, safety, and trades (e.g., electrical skills); there is a critical need for students to be exposed to fields of study that fall outside traditional aviation management programs. These other fields, essential to planning and operating an airport, include the following:

- Finance (e.g., budgeting and accounting),
- Business acumen,
- Project management,
- Property management (e.g., real estate development, contracts), and
- Engineering [e.g., AutoCAD, Geographic Information System (GIS)].

Students' depths of knowledge should also be expanded to equip them with the appropriate skills to fill the eight MCO jobs/professions identified in *ACRP Web-Only Document 28* (Cronin et al. 2016). These occupations include the following:

- **Airport Development:** Involves the design and development of airport property and facilities as well as real estate transactions. Example job titles include Air Terminal Development Manager and Aviation Real Estate Manager.



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- **Airport Operations:** Ensures everything is in place on airfield and airside environments for safe and efficient flight operation. Example job titles include Airfield Operations Specialist and Ground Services Manager.
- **Airport Security:** Ensures airport facilities, employees, and passengers are protected from potential threats. Example job titles include Airport Security Inspector and Transportation Inspector.
- **Electrician:** Is responsible for installation, maintenance, inspection, and repair of airport electrical systems, including power distribution, security, and communication. Example job titles include Electrical Technician and Airport Electrician.
- **Engineering (Civil, Electrical, and Mechanical):** Involves all aspects of airport planning, including design, construction, and inspection. Example job titles include Senior Engineer and Civil Engineer.
- **Financial Analysis and Planning:** Is responsible for financial and economic business planning, development, and management. Example job titles include Assistant Finance Manager and Airport Economic Planner.
- **Information Technology (IT):** Includes the design, installation, and management of security mechanisms to protect airport security networks and systems from cybersecurity threats and attacks. Example job titles include Network Administrator and IT Services Specialist.
- **Project Planning:** Involves all aspects of airport planning, including creating airport plans, developing master plans, and understanding future needs of the airport itself and its customers to ensure these needs can be met. Example job titles include Project Planner and Airport Planner.

Optimally, academic courses can be airport-focused or, at least, airport-specific use cases can be incorporated to help students gain the skills that are essential for conducting business in the unique airport ecosystem.

Engagement between airports and academia is limited.

15.3% of airport respondents and 14.3% of professional firm respondents indicated that their airport/firm engages in guest speaker activities at academic institutions.

18.4% of airport respondents and 7.7% of professional firm respondents host site visits for student organizations from academic programs.

Strengthening Collaboration Between the Airport Industry and Academia

Many airport professionals who participated in focus groups for this study expressed the need for increased outreach and collaboration between the airport industry and academia. While some airports have formal partnerships or informal connections with local colleges and universities, others are still working to increase engagement and foster these connections. One key objective of this guidebook is to encourage the development of new relationships between academic programs and airports and to strengthen existing relationships. These partnerships not only benefit airports and academia but also students in aviation and airport management programs through the opportunity to increase the availability of internships and to grow the involvement of airport management in the classroom.

Importance of Internships



To succeed in the airport industry, new professionals not only need theoretical knowledge, which they obtain through expanded academic curricula but also practical knowledge (i.e., how to apply what they have learned within the demands and constraints of the real world). Internships and co-ops are essential ways for students to experience a bit of what they can expect before they graduate. They allow students to apply what they are learning in class and also help students learn new skills and identify potential career paths available to them.

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Airport professionals perceive a lack of adequate internship opportunities. A limited number of airports have established programs, while others are in the process of developing them. For the most part, students graduate without having participated in any type of internship or co-op, placing them at a disadvantage when entering their first job. Although the need is clear, the future availability of internships and co-ops is dependent on local airports and academic institutions coming together to develop opportunities that meet the needs of students, while also taking into consideration the constraints faced by airports.

21.6% of airport survey respondents and **23.6% of professional firm respondents** offer internships, co-ops, or seasonal positions that they fill with college/university students.

Over **71% of airport respondents** and **81% of professional firm respondents** did **not** participate in an internship or co-op at an airport/airport-related professional service firm while enrolled in their own program of study.

Airport On-Campus Involvement

Outreach and collaboration between airports and academia can also go beyond internships to include airport involvement on campus, both inside and outside the classroom. This includes opportunities such as airport leaders serving as guest speakers in classes. Focus group participants highlighted instances in which airport leaders bring some of the real-world problems they are facing at their airport into the classroom for students to review. These challenges are then integrated into the class curriculum as case studies, with student teams assigned to spend part of the semester devising a solution to the problem at hand and then presenting this solution to the airport. These case studies provide another avenue for students to gain insight into the real world of airports, outside of participating in an internship. In addition to participating in the classroom, airports should also aim to connect with students by attending university-sponsored industry nights or other events held by student organizations, such as local American Association of Airport Executives (AAAE) student chapters.

Ensuring Recent Graduates Have Appropriate Skills for the Workplace

The skills of recent graduates reflect what they have learned in the classroom along with their involvement in extracurricular activities, internships, or co-ops. The study identified three important areas that are crucial for recent graduates:

- Knowledge of the airport industry
- Technological skills
- Personal effectiveness and professional skills

*The biggest detriment [is graduates] coming out of college programs saying **"we know this." But do they know how to apply this knowledge in the real world?** Or what you need to do to actually get things done?*

– Airport Operations Agent

Knowledge of the Airport Industry

Airport professionals agree that recent graduates entering the workforce generally have a sufficient knowledge base to begin their career within an airport organization, along with a passion for aviation and desire to continue learning. However, new employees within both MCO and business-related occupations do not have exposure to the airport-specific aspects of their roles (Cronin et al. 2016). Therefore, in some instances, graduates may overestimate their ability to apply their classroom knowledge within a real airport situation. For example, recent graduates learn about important airport and aviation regulatory issues, but they lack knowledge of which regulations apply to various situations, or how to apply the regulations to job tasks within an actual airport environment. Ultimately, graduates typically work in the industry for several years before they develop sufficient skills to be able to take on a specialized role.



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Technological Skills

The airport industry is experiencing fundamental shifts due to advances in technology, resulting in the need for new skills that previously did not exist (Wensveen 2017). Rapid changes and advancements require employees to be familiar with both current technology and to continually learn the new technical skills required as new systems are implemented. Required technical capabilities include general computer skills, knowledge of computer software programs, IT skills, and experience with data analytics. A key strength of recent graduates entering the workforce is their ability to efficiently and effectively use technology, including different systems and software for finding and gathering information. However, some graduates lack the problem-solving and analytical skills necessary to analyze the data they gather and use it as a basis for informed decision-making. This gap should be addressed, given the growing emphasis on using data to track passenger experiences and leveraging big data to make key operational decisions.

Personal Effectiveness and Professional Skills

Technical skills alone do not lead to successful job performance. New professionals must also have the personal characteristics and behaviors that support a productive and cooperative workplace (Phillips et al. 2006). This includes basic interpersonal, written, and verbal communication skills as well as the ability to work in teams. Other non-technical skills include critical thinking, problem-solving, decisionmaking, rapport building, negotiation, and a willingness to adapt. Although many of these skills are ultimately required to lead and plan for an airport's future, they are covered in less than one-third of airport training and education programs (ICF 2018). One of the weaknesses seen in recent graduates is their lack of these skills. For example, many of the airport professionals interviewed perceive that recent graduates do not feel comfortable communicating with others or to groups and have difficulty conveying technical information.

> *People feel uneasy interacting and communicating – **we need people who can communicate with each other and groups** – those people end up being very successful.*
>
> – Airport Deputy Executive Director, Development

Addressing Existing Challenges in Hiring Recent Graduates

Beyond ensuring that graduates are adequately prepared for gainful employment, airports also face the challenge of finding and hiring skilled applicants. Airports prefer to hire entry-level employees out of aviation degree programs that have a background in aviation management or airport administration. However, airport representatives who participated in this study have experienced difficulty over the past several years in finding and hiring recent graduates out of these programs. This is due, in part, to the large demand for these types of students but also to the limited number of aviation degree programs from which to pull new talent. As a result, airports are now hiring recent graduates who do not have a degree from an aviation program but still have a passion or interest in aviation. Generally, airports have success with this approach and can teach the necessary aviation-specific skills in house.





Despite these efforts, a survey of 137 airport professionals and airport consultants points to several skill areas from which it is "extremely challenging" or "very challenging" to hire, including the following:

- Technical Communication (report writing, etc.)
- Environmental Issues

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- Public Policy
- Airport Engineering
- Geographic Systems (GIS)
- Airport Finance

Survey respondents reported that many of these skill areas are not frequently covered in their programs, or not covered in a standalone, airport-related course. Table 2.1 highlights the skills and topics that were identified as either "extremely" or "very challenging" to find in job applicants. Data is also provided regarding the type(s) of classes in which respondents indicated each of these skills or topics was covered.

Table 2.1. Skills and topics highlighted in hiring challenges and academic coverage.

Skill/Topic Area	At least 25% of respondents indicated as "Extremely Challenging" or "Very Challenging" to hire ✓ = airports *= professional firms	Coverage in Coursework			
		% of Airport respondents indicating this topic was **not** covered in their program	% of Airport respondents indicating this topic was covered in a standalone airport-related course	% of Airport respondents indicating this topic was covered in an aviation course not specific to airports	% of Airport respondents indicating this topic was covered in a course not specific to aviation
Geographic Information System (GIS)	✓*	80.8%	1.9%	3.9%	13.5%
Airport Engineering	✓*	69.2%	19.2%	7.7%	3.9%
Public Policy	✓	40.4%	7.7%	21.2%	30.8%
Emergency Mgmt		40.4%	17.3%	26.9%	15.4%
Airport Design	✓*	38.5%	50.0%	5.8%	5.8%
Airport-Airspace Compatibility	✓*	38.5%	42.3%	17.3%	1.9%
Safety Management		38.5%	21.2%	25.0%	15.4%
Airport Planning	✓*	34.6%	51.9%	11.5%	1.9%
Environmental Issues	✓*	34.6%	23.1%	19.2%	23.1%
Airport Finance	✓	34.0%	37.7%	13.2%	15.1%
Airport Operations: Airside	✓*	32.7%	46.2%	21.2%	0.0%
Airport Admin		32.7%	42.3%	17.3%	7.7%
Airport Operations: Terminal/Landside	*	32.7%	44.2%	17.3%	0.0%
Airport Security		30.8%	44.2%	17.3%	7.7%
Airport Regulations	✓*	28.9%	50.0%	19.2%	1.9%
Technical Communications (report writing, etc.)	*	26.9%	7.7%	9.6%	55.8%
Professionalism	✓*	15.1%	11.3%	18.9%	54.7%



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Summary of Data and Findings

There are many instances in which coordination and communication between academic institutions and the airport industry is limited or lacking. As a result, **a disconnect exists between the industry workforce needs and the education that academic programs currently provide.** Rather than academic institutions dictating the needs of the industry, academia and the airport industry may enhance academic programming by working together to revise curricula content that is more focused on learning outcomes that meet the needs of the industry. Likewise, airports will benefit from partnerships with academic programs that prepare graduates with the knowledge and skills required for all MCOs and relevant skilled-trade jobs.



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SECTION II

Guidance to Academia

This section provides information to the academic community on the current state of airport-oriented education in academia and opportunities to enhance academic programming. As such, the primary audience for this section is the academic community itself, particularly faculty, administrators, and even students who wish to enhance their airport academic programming.

The information herein was derived from extensive study of domestic and international academic degree programs. The research team conducted surveys with faculty, students, and alumni of programs offering specific airport-oriented content within their curricula, and industry professionals with the desire to see graduates of these programs enter the workforce with greater depth and breadth of education as well as an enhanced level of real-world experience.

Chapter 3 discusses the current state of airport education in academia. The vast majority of airport-oriented educational programming lies within what is commonly referred to as "collegiate aviation" programs, while to a lesser extent, airport-oriented education is found within programs of other core disciplines, mostly business, public policy, and engineering. Also discussed is the current structure of airport-oriented academic programming, focusing on how such curricula are designed. Several constraints and opportunities are pointed out, paving the way for an informative discussion on how these programs can be enhanced.

Chapter 4 describes methods to enhance airport-oriented academic programming based on the findings from Chapters 2 and 3. A curriculum structure is presented that increases both breadth and depth of airport education, allows for real-world experience, and contributes to professional development through curricular and extracurricular activities. The methods described apply to programs of all types, from fully established airport management degree programs to programs without any aviation or airport content. All institutions have the potential to enhance their programming for the benefit of the airport industry.

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CHAPTER 3

The State of Airport Education in Academia

Key Takeaways

- Airport academic programming is found within a small subset of the nation's colleges and universities, mostly within traditional aviation, undergraduate programs. They are typically 4-year undergraduate, baccalaureate programs with roots in collegiate flight education. They tend to have a focus in airport operations or airport management.
- A handful of programs offering airport content exists in engineering programs. These programs tend to focus on airport planning and design.

  Few programs offer more than two airport-specific courses. The most common courses offered are introductory courses in airport operations, management, and planning and design. Less frequently, additional courses are offered that go deeper into these topics or focus on other topics relevant to the airport industry.
- Aviation programs generally include some airport content within other aviation courses that focus on flight, aviation security, and aviation regulations and law.
- Airport offerings are limited due to the curricular constraints that require a certain number of general education and core courses to be part of the curriculum.
- Most programs offer opportunities for additional learning through internships, although few programs require them. There is a wide range of how internships are structured among programs.
- Less than 30% of programs offering airport content report having formal relationships with airport industry professional organizations, such as AAAE or participate in regionally or nationally sponsored airport academic endeavors such as the ACRP Airport Design Competition for Universities.
- Less than 30% of programs offering airport content are accredited by the Aviation Accreditation Board International (AABI), which focuses its accreditation on flight and aviation management—not specifically on airport curricula.
- Less than 20% of the programs offering airport content have full-time faculty who have any formal education in airports, are industry accredited, or have significant practical airport industry experience.
- Ample opportunities exist to enhance airport-oriented, academic programming by creating deeper and broader course curricula; increasing extracurricular and experiential learning opportunities; adding airport content to programs at colleges and universities that do not currently offer such content; enhancing faculty supply; and creating deeper and broader interactions between academia and airport industry professionals.

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As described in Chapter 2, the technical needs of the current and future airport workforce are both deep and wide-ranging, covering a full spectrum of topics, from business to technology. Many of the skills required are rooted in fundamental academic topics, such as economics; finance and accounting; law; physical sciences and engineering; and many other disciplines offered at institutions of higher education. The majority of the academic programs, however, do not include any airport- or aviation-related content, either in the form of aviation- or airport-specific courses or simply using aviation case studies or applications within their courses.

Relative to the totality of existing higher education programs, only a small group offers programming in aviation, and only a small subset of those programs offer content specific to airports. These programs vary widely in the depth of airport content, ranging from complete airport major degree programs to offering only one introductory airport course. They do, however, offer a base for evaluating the state of academia on how well it, on the whole, is helping to prepare the next generation of airport professionals.

This chapter describes an investigation of these programs, and based on the findings, offers insight into the current state of airport academic programming.

Education that is relevant to the airport industry is wide-ranging because airports require a vast array of skills and knowledge to operate. Engineering expertise is required to ensure that airports are designed and maintained with the appropriate level of safety and infrastructure. Business and administrative skills are required to successfully operate a complex organization serving a wide range of customers and community constituents. Communication skills are required to effectively share information with airport users and the wider community. Of course, front and center are the particulars of how airports operate, from design and certification requirements to unique strategies for managing complex operating budgets to the financing associated with capital improvements.

Perhaps the greatest distinction between technical training and academic education, particularly at degree-granting institutions of higher education, is that university and college academic programs have the mission of providing well-rounded education in a full spectrum of subject areas (Prather and California Baptist University 2019). Although non-academic training programs focus on specific technical material only, collegiate degree programs include a wider variety of course requirements. Their mission statements often indicate that they intend to produce well-rounded graduates who will be successful in their professions and contributors to the greater community.

Less than 3% of the nation's 6,000+ institutions of higher education offer courses in aviation, and only a subset of these institutions offers courses specific to airports.

The findings presented herein are based on a multifaceted investigation of academic programs throughout the United States as well as select international programs with airport-oriented content.

A Survey of Academic Programs with Airport-Related Content

Most academic programs offer content relevant to the airport industry, at least indirectly. For example, architecture programs educate students in skills relevant to designing and constructing airport facilities. Public policy programs educate students in how public agencies operate, which is relevant to all of the nation's airports operating under some type of public governance. Even programs in the arts are relevant to airports because they continue to evolve their public image through public good programs, such as public displays of art, history, and musical performances. **For the most part, however, most traditional degree programs do not offer any content specific to how their core disciplines apply to the airport environment.** Cases where this does exist are limited to a few engineering,

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technology, business, and policy programs, or to faculty with some knowledge or expertise from previous involvement or relationship with airports.

The study conducted for this guidebook identified an exhaustive list of academic institutions offering at minimum some sort of aviation (but not necessarily airport-specific) content, including those with programs that focus nearly entirely on flight education. Of the nation's 6,000+ institutions of higher education, 157 institutions (less than 3%) were found to offer programming that ranges from a single aviation course offering up to a 2-year (associate) or 4-year (baccalaureate) degree in aviation. The team examined the depth and breadth of these aviation offerings, with a specific look at how they may or may not offer any airport-specific content. Also highlighted were whether these institutions have industry accreditation for their aviation degree programs or some sort of formal relationship with the airport industry.

Of the 157 institutions investigated, 65 programs were found to have at least one course specific to airports. Of these 65 institutions, 42 (65%) were public institutions and 23 (35%) were private institutions. Public institutions tend to draw students from their respective states, whereas private institutions tend to have more of a national draw. Cost of tuition is one determining factor for students selecting a school. For public institutions, tuition rates significantly favor in-state residents. The average in-state resident semester tuition (minimum 12 credit hours) for the public institutions in the study was approximately $4,275. Out-of-state tuition averaged $9,580 per academic term. Private institutions make no distinction regarding residency. The average tuition for private schools is approximately $15,400 per term.

Of the 65 programs, 41 were found to have a readily available curriculum roadmap and catalog of courses for which to further study the depth of airport-related programming. For these **41 "representative" programs**, an in-depth investigation of their curriculum structures was performed, and surveys, focus groups, and interviews of a sampling of the programs' faculty, students, and alumni were conducted. This thorough investigation provides a comprehensive look at the current state of airport education in academia. A list of these programs is found below. A summary of the curricula of the representative programs is provided in spreadsheet form in Appendix A.

Universities with Representative Aviation Programs*

- Arizona State University, AZ
- Auburn University, AL
- Bowling Green State University, OH
- California Baptist University, CA
- California State University, Los Angeles, CA
- Central Washington University, WA
- Delta State University, MS
- Eastern Kentucky University, KY
- Embry-Riddle Aeronautical University, Daytona Beach, FL
- Embry-Riddle Aeronautical University, Prescott, AZ
- Farmingdale State College, NY
- Florida Institute of Technology, FL
- Florida Memorial University, FL
- Hampton University, VA
- Indiana State University, IN
- Jacksonville University, FL
- Kansas State University – Polytechnic, KS
- Kent State University, OH
- Lewis University, IL
- Louisiana Tech University, LA
- Metropolitan State University of Denver, CO
- Middle Georgia State University, GA
- Middle Tennessee State University, TN
- Minnesota State University, Mankato, MN
- The Ohio State University, OH
- Oklahoma State University, OK
- Purdue University, IN
- Rocky Mountain College, MT
- St. Louis University, MO

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Universities with Representative Aviation Programs* (*Continued*)

- Southern Illinois University, IL
- Southern New Hampshire University, NH
- Texas Southern University, TX
- University of Alaska, Anchorage, AK
- University of Central Missouri, MO
- University of Dubuque, IA
- University of Nebraska, Omaha, NE
- University of North Dakota, ND
- University of Oklahoma, OK
- Utah Valley University, UT
- Vaughn College of Aeronautics and Technology, NY
- Western Michigan University, MI

** Aviation management-oriented curriculum that is AABI accredited and/or affiliated with the AAAE via a recognized student chapter, offers at least one airport course, and has an accessible full curriculum map for study.*

Aviation Programs

Most specific airport-oriented content is found within what are known as traditional aviation academic programs within institutions organized as a college, school, or department with aviation in their name, denoting the recognition of its mission to focus on aviation as a degree program. Approximately 66% of the programs in the United States offering airport-specific content are colleges, departments, or academic centers of aviation or aeronautics. Approximately 33% of programs that provide some aviation education do so within wider academic units ranging from Applied Arts & Sciences to Public Health.

Academic units (colleges, departments, schools) offering airport content include the following:

- Aviation
- Aeronautics
- Applied Arts and Sciences
- Business
- Engineering
- City and Regional Planning
- Education
- Public Administration
- Public Health and Science
- Technology

The aviation-related degrees offered at these institutions include Air Traffic Management, Aviation/Aerospace Management, Aviation Business Administration, Aircraft Maintenance/Management, and Aerospace Technology along with Professional Pilot, Flight Technology, and other degree names for collegiate flight education programs. Among the newest aviation-related degree programs are those that involve unmanned aerial systems (UAS).

Aviation academic programs, regardless of the specific degree names, are generally categorized as either **flight education or aviation management.** Within the representative programs studied, most airport-oriented content is found within the aviation management degree programs. Flight education programs tend to have ancillary airport content relevant to pilots, and airport courses offered in the aviation management programs are available for students in flight education as well.

Flight Education Degree Programs

Discussions with current airport professionals who have aviation education backgrounds reveal that a significant percentage began their aviation collegiate education by enrolling in a flight education degree program, validating previous research that suggests young adults and older teenagers, initially attracted to aviation as a career, focus on becoming a licensed pilot.





Along the way, however, many students find that flight training is not for them, and they seek another path to a career in aviation. They often transfer from a flight education program to an aviation management program, either within their current institution or elsewhere.

Reasons for this realization are numerous and understanding them may provide opportunities and motivation for academic programs to further enhance their non-flight curricula, including increasing offerings in airport-oriented content to retain these students.

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Flight Student and Pilot Attrition

Student Aptitude and Attitude

A collegiate flight training program is demanding, with physical, mental, logistical, and financial pressures. Some may quickly determine that flight training is not what they expected, or they may find they do not have the aptitude or motivation to operate in a cockpit environment. They may consider changing majors to a non-flight aviation program or elect to drop out of the program altogether.

Federal Policy

One factor adversely affecting student enrollment and retention is the result of federal legislation passed in 2013 that requires all commercial airline pilots to hold an airline transport pilot (ATP) certificate that includes a minimum of 1,500 hours of flight experience. Before this, licensed commercial pilots with as little as 250 hours of flight time could serve as first officers with an airline. The legislation does allow provisions for collegiate flight students receiving professional instruction. Baccalaureate graduates from an accredited university are eligible for restricted ATP (R-ATP) license with a minimum of 1,000 hours of flight time, while associate degree graduates can qualify with 1,250 hours.

Many students in a 4-year professional flight program recognize that they will have only 300 to 500 flight hours by graduation and thus, will need to find ways to build flight time, often by serving as flight instructors or through other flying assignments (e.g., banner and glider towing, skydiving, etc.). This gap in flight experience coupled with student loan repayments often makes the difficulty of achieving an R-ATP license rating a driving force to reconsider their career goals.

Another example of how federal policy changes can seriously affect aviation programs was demonstrated by the FAA's shift in the selection process for air traffic control specialists (ATCSs). Beginning in 1991, the FAA partnered with five aviation colleges to form the Controller Training Initiative (CTI) program designed to generate student interest in air traffic control (ATC) as a career choice and to provide portions of the initial controller training before entering the FAA's Training Academy. Over the next 20 years, the CTI program grew to include 36 participating colleges and universities. Graduates of an academic-degreed CTI program were considered prime candidates for selection because they were motivated and possessed an ATC specialized education. As a result, CTI graduates were allowed to skip the initial 5-week Stage I, En Route Air Traffic Basics course.

However, in 2014, FAA changed the candidate screening and selection process, relying primarily on a biographical assessment that includes personality attributes, attitudes, experiences, interests, skills, and abilities. Under this new practice, CTI graduates who had previously applied to the FAA were required to take the new assessment and as a result, some were not accepted. Even though FAA retained the CTI program, enrollments declined, and some institutions dropped out. Legislation attempted to reinstate CTI graduate preferences. However, in 2020, the program had yet to regain the status it had before the hiring policy changes.

Socioeconomic Conditions

The aviation industry is vulnerable to national and international economic and other conditions that directly affect the career opportunities and interests for aviation students. Over the past 20 years, macro-economic cycles have also impacted the number of students enrolling in flight education programs compared with other aviation degree programs. The terrorist attacks of September 11, 2001, the global economic recession of 2008, and the 2020 COVID-19



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global pandemic have had negative effects on the industry, which have resulted in aviation students needing to adjust their career goals and transferring to non-flight programs that are still connected to the aviation world.

Career Compromise

Regardless of the specific influence, the concept of career compromise in career development theory suggests that when faced with the denial of one's dream job, the next step will be based on the individual's motivation for initially pursuing the career. For flight students, the driver may be ego-driven by the perceived prestige of becoming an airline pilot. On the other hand, career compromise also identifies "interest" as a strong driver for some. In this case, if flying professionally is not looking like a realistic option, the person may take an alternate position that allows for staying in the field of aviation.

Implications and Considerations for Collegiate Aviation Programs

Depending on how the curriculum is organized, some students who elect to discontinue flight training may be able to change their major to non-flight aviation or other available degree option. For some programs, the change may be easily accommodated by curriculum design, or the student may be required to take several additional courses to meet the degree requirements. In extreme cases, the inability of a professional flight student to change majors without a significant increase in coursework, time, and associated costs may result in the student dropping out.

Aviation academic programs that provide flight training may adjust or redesign their curriculum to (1) provide flight students a better understanding of the policies and practices in business and operational aspects of aviation; (2) provide flight students the ability to easily change majors to a non-flight degree depending on circumstances; and (3) ensure that graduating flight students are not so heavily invested in flight education that they are disadvantaged or unable to compete for non-flight aviation positions if necessary as a result of the economy, layoffs, medical issues, or other factors that prevent them from pursuing flight as a career. **A course curriculum in airports may be such an option.**

Aviation Management Degree Programs

Among the 157 institutions investigated, 81 (52%) offer a 4-year degree program in aviation management, administration, or other similar business-oriented offerings. There are another 31 institutions that offer a standalone, non-flight associate degree program.

Although no two programs are alike, the curriculum of most aviation management programs is found to share common attributes with other degrees, aviation or otherwise. The distinctions generally emphasize core courses that may or may not have an aviation theme. Most aviation management programs are created as alternatives to flight education programs. As such, they draw most aviation content from the professional flight perspective. Also, these programs are found to have been developed by leveraging existing business administration or other general curriculum courses, such as accounting, finance, computer applications, statistics, and business law found within the institution. Others are developed with a more technical orientation emphasizing mathematics (calculus), physics, electrical/electronic systems, intermodal transportation/logistics, and other fields.



Most aviation management programs are broadly aviation industry-focused, with course material covering all aspects of management, such as general aviation operators, ATC, airlines, and in most but not all cases, airports. On rare occasions, there is a specific emphasis, specialization, concentration, minor, or major degree program focusing specifically on airports. This

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presents an opportunity for programs to enhance their curricula by providing more airport-focused material that leads to an airport-specific degree or specialization (ACRP 2017).

Airport-Specific Degree Programs and Specializations

Among the institutions investigated, two were specifically noted to have airport management degree programs. The University of North Dakota, through an intercollegiate partnership between its Nistler College of Business & Public Administration and the John D. Odegard School of Aerospace Sciences, offers a bachelor of business administration degree in airport management. The program requires more than 30 credit hours of business courses ranging from accounting to information systems, along with 30 credit hours of aviation required courses, including courses in airport operations, planning, and safety management systems, and culminating experiences that include preparation for professional industry certification. Vaughn College of Aeronautics and Technology offers a bachelor of science degree in airport management. More than half of the curriculum course requirements are aviation or management-focused with airport-specific courses in airport administration, development, and management, in addition to college requirements in English, foreign languages, and technical writing as well as math, statistics, and physics requirements.

A few representative programs offer a concentration or specialization in airport management as part of the general aviation management degree. For example, the Aviation Administration Bachelor of Science degree program at Farmingdale State College offers a specialization in airport management. Through this specialization, students enroll in advanced courses in airport capacity; delay and airspace modeling; airport management and finance; airport planning; and airport operations, in addition to a core curriculum with required courses in business and aviation. California Baptist University offers a concentration in Airport Management and Operations within its Aviation Management Bachelor of Science degree program. The concentration emphasizes airport management, planning and design, and finance and consulting, with a capstone course that prepares students for professional industry certification.

Approximately 40% of the representative aviation management programs offer multiple airport-specific courses as part of an aviation management degree but not as a concentration or specialization. The remaining 60% offer one course specific to airports.

Institutions with Airport-Specific Majors, Specializations, or Concentrations

- California Baptist University, CA
- Farmingdale State College, NY
- Vaughn College of Aeronautics and Technology, NY
- University of North Dakota, ND

Engineering Programs with Airport Course Offerings

- University of California, Berkeley, CA
- Georgia Institute of Technology, GA
- Purdue University, IN
- North Carolina State University, NC
- Virginia Polytechnic Institute and State University, VA
- Massachusetts Institute of Technology, MA
- The Ohio State University, OH
- University of Maryland, MD
- University of South Florida, FL

Non-Aviation Programs with Airport Courses

In addition to aviation programs, a handful of non-aviation programs offer airport courses. Some non-aviation programs include aviation/airport content in their curriculum, many of which are found at the graduate level, mostly in engineering programs offering one or more courses in airport planning and design. Although these are not part of the representative programs, the specialized nature of these courses holds a unique place in the overall spectrum of airport-oriented academic programming. Specifically, courses like these attract students to the airport industry as engineers, planners, and other professionals.



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Curriculum Structure of Representative Programs

The representative degree programs that were studied generally follow a curriculum structure standard to baccalaureate programs throughout academia, with an average of 120 credit hours of instruction organized along broad categories that include general education, core, and courses on topics specific to the degree program. The typical breakdown of credit-hour requirements by these categories is illustrated in Table 3.1.

Although aviation management programs usually follow this outline, core courses may be subdivided into aviation and non-aviation topics. Airport-specific courses are most often within the aviation core and technical elective offerings.

A description of these course categories provides insights into where airport-related content is presented within these curriculum structures.

General Education. General education (Gen Ed) courses are found in most undergraduate programs. They are intended to provide a broadened perspective of the relationships among the various topics universal to higher learning and for building critical thinking skills. Although these courses rarely contain much technical content directly associated with airports, some may be included in these courses. For example, a course in global issues may briefly discuss matters associated with airport-relevant content, such as global security.

Non-Aviation Core. A variety of core courses that are not aviation-related are often required by all students pursuing degrees. For example, for the aviation management programs found within a business administration department, core courses in business are included in the program. Analysis of the representative programs shows the following most commonly offered non-aviation core courses in their curricula:

- Principles of Management
- Organizational Administration
- Macroeconomics and Microeconomics
- Accounting and Finance
- Business Law
- Information Systems and Technology
- Business Statistics and Quantitative Analysis
- Human Relations
- Strategic Planning and Marketing

This suggests that most curricula contain core principles of importance to the airport industry. However, since these sources are "non-aviation" in nature, direct examples and applications of these principles to the airport environment may not necessarily be included in these courses. **This provides an opportunity to enhance programming through the addition of airport-specific content into these core courses.**

Table 3.1. Average breakdown of representative programs.

Course Category	Avg. Credit Hours	# Courses
General Education	24	6
College (Non-Aviation) Core	32	7
Major (Aviation) Core	32	8
Technical Elective	21	6
Open Elective	9	4
Extracurricular Activity	2	1
Total	120	32

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Aviation Core. The representative aviation management programs studied tend to have a significant number of courses that have a direct relationship with the aviation industry. The average number of aviation core credit hours is 45 among the representative programs with a high of 72 hours and a low of 15 hours. These courses include introductory overviews of the aviation industry in topics ranging from aviation history, regulations, and aviation sectors including ATC, airlines, general aviation, and airports. The most common aviation core courses in the representative programs include the following:

- Historical Perspectives
- Federal Regulations
- Meteorology
- National Air Transportation System
- Aviation Safety

This cadre of courses indicates that the focus of these programs has traditionally been flight-oriented. As such, the content is more focused on the impact of these topics on flight operations, and tangentially, airport operations. It is from here that the focus on airport operations education is derived. **This speaks to the opportunity to broaden the aviation core to include more aviation core topics outside of an operations focus.**

Technical Electives. Other more advanced aviation courses may be part of an aviation core requirement or may be offered as technical electives within the curriculum. The most common advanced core and technical elective courses found within the representative programs include the following:

- Airspace and Air Traffic Control
- Airline Operations & Management
- Fixed Base Operator (FBO) and Corporate Aviation Operations and Management
- Aviation Safety and Security
- Aviation Law and Regulations
- Air Cargo and Logistics
- Aviation Economics and Management
- Aviation Marketing and Communications
- Global Aviation Systems

In addition to this list, all the representative programs studied have at least one airport-specific course in the curriculum, most often with the title "airport operations," "airport administration," or "airport management," intended to cover an introductory overview of airports as a component of the National Air Transportation System. Engineering programs offering one course tend to offer a course in "airport planning" or "airport design" rather than an operations- or business-focused course.

In nearly all representative programs, the number of credit hours allocated to airport-specific content was small. There was an average of 6.6 airport-specific credit hours among the total number of aviation management programs. Nearly 40% of the programs have only one 3-credit-hour airport course, while one program has 27 credit hours. Table 3.2 illustrates the range of total airport-specific credit hours offered by aviation management programs.

Airport-specific courses constitute on average 5% of the total credit hours offered within typical aviation management degree programs.

A variety of other airport-specific courses offered by many representative programs provides a more detailed exposure to the airport industry. Examples of the most common courses include the following:

- Airport Planning
- Airport Design

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Table 3.2. Aviation management programs with airport-specific credit hours.

Airport Credit Hours	No. of Programs	Percentage
3	26	40%
4 to 6	20	31%
7 to 9	8	12%
10 to 14	6	9%
15 +	5	8%
Total	65	100%

- Environmental Aspects of Airports
- Airport Economics and Finance
- Airport and Airspace Capacity
- Airport Management (including certification)

Programs that offer a concentration in airport management generally include a minimum of 3 to 4 airport-specific courses (9 or more credit hours). From this analysis, **it is clear that opportunities exist for more programs to deepen their airport-specific content through additional courses offered either as aviation core or technical electives.**

Open Electives. Open electives are offered to allow the student to gain a broad education. These electives may also allow for internships and work-study programs. Open electives may present an opportunity for students to take additional airport-oriented courses, even if not offered by the program if the course or courses offered by another institution can be approved for credit.

Of the representative programs, open elective opportunities are limited, primarily due to the limited number of credit hours allocated as such. Despite the limited number of allocated credit hours, there are opportunities to further enhance airport-related content by taking advantage of these elective hours, through additional coursework or experiential learning opportunities, such as internships.

Internships. Internships and co-op employment serve as experiential learning opportunities. They can also be a component of the broader accreditation requirements. Of the representative programs, 79% do not require an internship. Relatively few programs accept an internship as an elective (3 to 12 credit hours) for fulfilling the degree requirements.

A Deeper Look at Airport-Specific Courses

Most aviation management programs have at least one airport-specific course. Usually, the course is titled airport operations, airport administration, or airport management and is intended to cover only the basics of airports as a component of the National Air Transportation System.



Airport-specific courses, defined as those courses with "airport" in the course title and whose syllabi suggest that the majority of the course content is specific to airports, most often cover an overview of the management and administration of airports, the planning and design of airports, and airport operations. They may be part of an aviation core or offered as a technical elective.

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Fundamental/Introductory Airport Courses

Nearly all curricula among representative programs have at least one airport course that is usually programmed to deliver an overview of airports as an air transportation facility. Topics typically include a historical perspective as well as legislation that directly influence airport operations and development. Characteristics of airports including activity metrics (operations and enplanements); FAA classifications; governance; administration and organizational structures; fiscal activity; and various other topics are also included. Table 3.3 presents a compilation of basic airport courses among the 65 aviation management programs.

Throughout academia, the most prolific airport course topic is one that provides an overview of airport management. Course titles that include topics related to airport operations and administration may include elements of management, including introductions to airport administration, operations, and planning. Most engineering programs that offer an airport course focus on airport planning and design.

Airport Planning and Development Courses

The second most common airport course topic found in the curriculum of aviation management programs involves the planning and development of airports. Usually, these topics are especially relevant to airport consultancies, which provide entry-level employment, especially for programs known for providing qualified and capable graduates with a certain level of knowledge regarding the airport planning process, FAA airport design criteria, and other technical areas. Table 3.4 presents a compilation of courses related to airport planning and development among the 65 aviation management programs.

In addition to the number of programs found in Table 3.4, most non-aviation programs offering at least one airport course have a course in airport planning and design that aligns with their planning and/or engineering degree programs.

The topics of some airport planning and development courses are combined, such as airport planning and design, airport planning and management, and so forth. These courses may be designed around a specific text, such as Young and Wells, *Airport Planning and Management* (2019) or Horonjeff, McKelvey, Sproule, and Young, *Planning and Design of Airports* (2010), or Prather, *Airport Management* (2015).

Other Airport-Specific Course Topics

Other airport-specific courses are aligned with common topics that can be viewed as a specialty. Airport marketing is a business-oriented topic that usually includes airlines in the coursework. Airport Security has been a prominent topic since 9/11. Airports share a responsibility with Transportation Security Administration (TSA) for passenger activity; however, security

Table 3.3. Basic airport topics (65 programs).

Topic	No. of Courses*	No. of Programs Offering Course	Percentage of Total Programs
Airport Management	50	46	71%
Airport Operations	26	34	52%
Airport Administration	14	12	18%

** There are several courses that include multiple topics, such as airport operations and administration or airport operations and management, and they may be counted twice.*

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Table 3.4. Airport planning and development topics (all programs).

Topic	No. of Courses*	No. of Programs Offering Course	Percentage of Total Programs
Airport Planning	37	34	52%
Airport Design	16	15	23%
Airport Finance	9	9	14%
Airport Environmental	5	3	5%

** There are several courses that include multiple topics, such as airport planning and design or airport planning and management, and they may be counted twice.*

of the airfield and other areas of the airport is the primary concern of airport management. Courses also include the study of international airports as a function of the global air transportation network. Table 3.5 presents a summary of the most common of these courses among the 65 aviation management programs.

Generally, airport-specific courses are offered in later stages of an overall curriculum. In the case of programs where one airport course is offered, that course may be most commonly taken during the student's final year of study. As a result, by the time the students are exposed to airports in this context, they are ready to graduate without having the opportunity to learn more. **This presents an opportunity for programs to offer airport content, particularly fundamental/introductory airport courses, earlier in their curricula,** as early as the first year in some cases.

Aviation Courses with Airport-Specific Content

Aviation courses that are not specific to airports often cover material that is airport-specific, within a wider syllabus of topics. For example, most aviation programs offer a course in aviation regulations, which typically includes airport-related regulatory issues. The relationship between airline operations and the airports is often discussed in airline management and airline operations courses. Airspace around the airport is addressed in the context of obstructions to runway approaches in ATC courses. FBO, corporate, and general aviation operations often have material covering airport activity as a topic.

Table 3.6 provides the percentage of the representative programs in which aviation courses that are not airport-specific may contain airport-related topics and provide opportunities to include airport content in the course.

Most aviation programs also offer courses focusing on career preparation and professional development. Professional skills, such as communications, public speaking, and technical writing, are also pertinent to the aviation workforce. The most commonly offered courses include those in professional communications and career development.

Table 3.5. Other airport topics.

Topic	No. of Courses*	No. of Programs Offering Course	Percentage of Total Programs
Airport Security	10	9	14%
Airport Safety	3	3	5%
Airport Marketing	3	3	5%
Airport Law	2	2	3%

** There are several courses that include multiple topics, such as airport safety and security or airport administration and finance, and they may be counted twice.*

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Table 3.6. Aviation courses with airport-specific content.

Topic	Percentage of Total Programs
Aviation Regulations and Law	88%
Aviation Safety and Security	72%
Airspace and Air Traffic Control	36%
Aviation Economics and Finance	32%

Career Development, Capstone, and Culminating Experiences

General accreditation criteria for higher education are moving institutions to ensure that their program curricula include a capstone course or culminating experience. Although a capstone project is not quite at the level of a thesis, it is generally organized to provide an opportunity for students to apply their collective knowledge, experience, and analytical and communication skills gained during their matriculation to a real-world topic. A capstone project may require a technical report about the identification, evaluation, and summation of the topic and delivery of an oral presentation or participation in a poster session. The capstone project may be an airport-specific issue.

Among the 41 representative programs, **24 have a specific course listed as a capstone project or culminating experience.** It should be noted that AABI accreditation requires evidence of a culminating experience, either in the curriculum or in practice. Some programs hold out internships as a culminating experience; however, they generally include a formal report and presentation requirement. Other course topics may include career preparation activities. One example includes preparing for the AAAE's Certified Member (CM) examination. The CM credential is widely recognized by the airport industry as evidence of a comprehensive level of understanding of airports. The exam is based on a broad variety of topics including the following:

- Operations, Security, Maintenance
- Planning, Construction, and Environmental
- Finance and Administration
- Communications and Community Relations

There are four modules aligned with these topics that provide the body of knowledge for preparation for the 180-question exam. At least three aviation programs have a course specifically for preparing for the exam in the curriculum. These are usually offered near the end of the program and are sometimes included as a culminating experience.

Another program that is used by some aviation programs as a culminating experience is the student's participation in the ACRP's University Design Competition. ACRP annually sponsors a national competition among colleges and universities for innovative approaches that address airport issues in categories including airport operations and maintenance; runway safety/runway incursions and excursions; airport environmental interactions; and airport management and planning.

The competition offers cash prizes for first, second, and third place winners. Some schools have a course syllabus designed to allow aviation and non-aviation students to participate in the competition for credit.

Other Activities Supporting Airport-Specific Education

Other sanctioned or ad hoc student activities relevant to airports are part of the collegiate aviation education experience.

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All the 65 aviation management programs studied have at least one aviation-related student organization or club. The most common of these include the following:

- Alpha Eta Rho Aviation - Professional Fraternity (AHP)
- Women in Aviation International (WAI)
- Flight Safety Foundation (FSF)
- National Intercollegiate Flying Association (NIFA)
- National Gay Pilots Association (NGPA)
- Organization of Black Aerospace Professionals (OBAP)
- International Society of Air Safety Investigators (ISASI)
- American Association of Airport Executives (AAAE)

Through its academic relations committee, AAAE promotes the establishment of student chapters for supporting student interest in the airport industry. AAAE's mission for the student chapters is centered on promoting the professional development of students engaged in the study of airport operations, administration, management, and development. The organization also seeks to instill the understanding that professional airport management embodies both technical knowledge and professional skills and responsibilities among student members.

Benefits for these academic chapters include member access to the organization's website and select material and resources. Complementary registration for the annual conference for the faculty advisor and the chapter president, and reduced registration is also offered. Currently, **there are 25 AAAE student chapters among the 41 representative aviation management programs.**

Through these chapters, students have opportunities to learn more about the airport profession through guest speakers, site visits, and participation at industry conferences and meetings. A number of the airport-specific courses use AAAE training materials as class resources, such as the Airport Certified Employee (ACE) and the CM body of knowledge modules.

Other national associations, such as the Airports Council International–North America (ACI–NA), National Business Aviation Association (NBAA), and University Aviation Association (UAA), have informal relationships with aviation management students and faculty through participation in their annual conference events. NBAA hosts a career day at its annual meetings that includes a substantially reduced registration rate and a career seminar with a sponsored lunch.

Faculty

Faculty serve as the focal point of most academic programs, and aviation management is no exception. All educational institutions require an adequate number of qualified and capable faculty to teach the topics of each course, some requiring specific and detailed technical knowledge and experience. For the institutions investigated for this research, supplying full-time faculty with an airport focus to meet the educational needs of the industry was a challenge.

Less than 50% of the 41 representative programs investigated have airport courses taught by full-time faculty that have both terminal degrees and professional experience in airports. Of the programs that do have full-time experienced faculty teaching airport content, the quality of the faculty is generally perceived to be excellent. These faculty are published authors of research papers and textbooks, hold senior roles in industry professional organizations, and have strong relationships with the airport industry. Their stories are unique; however, as colleagues, faculty share a passion for the industry, their enthusiasm for teaching about its various aspects, and for helping graduates achieve entry-level employment positions.



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Most faculty teaching airport courses are adjunct (part-time) positions. These adjunct faculty generally have direct airport experience, some of whom are still employed full-time in the airport industry. Although many are experienced in the field, they do not generally have terminal degrees, nor have they been formally trained in teaching undergraduate courses. Because airport subject matter is highly specialized, aviation academic programs have traditionally relied on adjunct faculty to teach airport-specific courses, usually the more common airport management, operations, and administration courses.

Less than 50% of the 41 representative programs investigated have airport courses taught by full-time faculty that have both terminal degrees and professional experience in airports.

Adjuncts are generally accessible locally through senior leadership at the community's airport. In some cases, they may not have been formally trained in academic instruction, and thus, they may have varying levels of experience with teaching skills involving multiple learning styles.

There is a need to expand the availability of faculty to teach airport-oriented courses that will enable students to gain a well-balanced aviation education to prepare for entering the workforce on an airport-related career path. Recruitment from the airports and other related industries may be a start. Yet, some may need to enhance their teaching skills for an advancing technological environment and the evolving learning styles of new college students. Graduate students in aviation programs may be another source for expanding the number of faculty. Mentoring graduate students, engaging in research projects, and providing opportunities to gain practical experience in the airport field can enhance their capabilities to teach airport-related content.

As opposed to most other academic disciplines that generate faculty through graduate programs, there are a limited number of such programs in aviation. Of the representative programs studied, less than one-third offer master's-level degree programs, and only Embry-Riddle Aeronautical University and Florida Institute of Technology offer doctorate degrees in aviation. There are no graduate programs that offer airport master's or doctorate degrees. Generating faculty relies on graduates from other disciplines, such as engineering or education, who also have an interest and familiarization (through research or professional practice) in airports. This is a rare combination. Thus, this presents opportunities to create pathways for current faculty to become more experienced in airports and for existing airport professionals to be trained as future instructors, and to create a pipeline of graduates with the education and experience to become the next generation of airport-oriented faculty.

Program Effectiveness in Preparing Graduates for Airport Careers

As validated in this research, current academic programs have various levels of coverage pertinent to the eight identified MCO occupations of the airport industry. Table 3.7 provides a summary of the percentage of programs that offer at least an introductory level of education in these identified occupations.

Table 3.7 illustrates several aviation management programs covering two or more course topics that address the wide range of skill sets that industry perceives are required for meeting mission-critical needs (Cronin et al. 2016, p. 94-96). The table also shows that there is little coverage for airport-specific civil engineering and airfield electrical systems among the eight critical occupations. These may be out of the realm of aviation management curricula but should still be encouraged for consideration at engineering schools.

Some programs are successful in providing at least introductory levels of education in what was identified as MCO and other airport professional topics. However, many lack the depth and breadth of topical areas needed by the industry.



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Table 3.7. Academic programs addressing industry needs.

Occupational Category	Programs Offering Some Level of Coverage
Airport Operations	71%
Airport Development	57%
Airport Planning	50%
Airport Finance	43%
Airport Security	29%
Information Technology	15%
Airport Engineering*	< 1%
Airport Electrical Technology**	< 1%

** Topic areas covered are limited to engineering-specific degree programs.*
*** Research did not discover any academic courses specific to airport electrical technology.*

Informal Emphasis on Professional Development and Communications

Investigation of the programs indicates that cursory attention is paid to educating students on being a professional and effective communicator in the industry. Professional development is said to include what are commonly known as soft skills or business acumen, including such topics as adapting to corporate culture, following up on task assignments, being a proactive contributing member of an organization, and the like. Effective communication skills range from the ability to give formal presentations and deliver clear and effective technical written reports, to informal communications skills, such as writing correspondence or posting on social media. The academic programs investigated tend to concentrate on communications skills learning in early non-aviation core and general education courses, such as English Composition. Rarely are these courses designed to specifically address the culture of the airport industry. Advanced courses in the curriculum attempt to address the need to improve professional and communications skills, but often this becomes ancillary to the core content of the course. This realization offers great opportunities to enhance professional development and communications education through the addition of course content in this area that leverages early-year courses and applies directly to the airport industry (Young et al. 2013).

> *I think the best way to enhance curricula is to engage industry so that they are aware of what is being taught and why so that they understand the context for making recommendations for changes.*
>
> – Professor, Aviation Management Program

Limited Experiential Learning and Real-World Applications

Although nearly all the programs investigated offer what may be considered experiential learning and real-world applications, the level and depth to which these are offered vary significantly among the programs. For example, many of the programs offer these opportunities as extracurricular options for internships or co-op opportunities, without earning credits toward graduation. Other programs focus experiential learning opportunities on site visits, field trips, and visits to campus by industry professionals. Yet, others have deep and broad experiential learning opportunities, ranging from for-credit study toward industry certification exams to participation in national research competitions to industry capstone projects. This speaks to the opportunity for academic programs to further engage with industry and create more formal experiential learning opportunities for their students.

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From this detailed examination of academic programs, it can be seen that there are opportunities to enhance airport-oriented topic coverage at all levels of a curriculum. Airport-related content may be included in general education and core courses, and a more robust study of specialized airport topics may be added through higher-level technical and general elective courses. All programs can benefit their students by providing more real-world experience that may be gained through in-class project assignments, site visits, internships, and extracurricular activities.



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CHAPTER 4

Enhancing Academic Programming

Key Takeaways

Airport academic programming may be enhanced by the following:

- Increasing the depth and breadth of existing course curricula through offering a set of broad fundamental and cross-cutting courses, and a set of advanced courses in the areas of airport operations and maintenance; airport business and administration; and airport planning, design, and engineering.
- Expanding airport education to colleges and universities that do not currently offer airport content through the addition of fundamental airport courses and adding airport content to existing course curricula in relevant programs, such as business, engineering, and technology programs.
- Increasing experiential learning opportunities through culminating experiences, such as capstone courses, internships, extracurricular activities, site visits to industry, participation at industry conferences and meetings, research opportunities, and program-sponsored symposia and guest lectures.
- Modernizing course delivery methods using online learning management systems and associated digital platforms.
- Enhancing interaction with industry through guest lecture opportunities, adjunct faculty appointments, capstone and research project mentorship, and participation on academic program advisory boards.

Although most existing aviation-oriented academic programs give some cursory level of airport-oriented education, opportunities exist to enhance these curricula to better meet the needs of industry. Also, programs that do not traditionally offer course content in aviation- or airport-related topics, such as traditional programs in business, public policy, and engineering, open up great opportunities to introduce airport content into these programs. Finally, there are ample opportunities to provide students in academic programs with more practical real-world experience. This chapter provides illustrative examples of how academic programs of all types can contribute to enhancing airport-oriented education.

Two of the most important insights for academia that surfaced from this study's research on the airport workforce are as follows:

- Many MCO jobs are not addressed by airport disciplines, but rather by fields of study that may have little exposure to real-world airport settings; and

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- Academia lacks an expansive or in-depth education in traditional airport topics that are particularly relevant to the execution of occupations, such as airport management and airport operations.

Furthermore, as airports increasingly operate under a business enterprise model rather than the historical government agency model, airport academic programs may consider enhancing their curricula by including more business and policy-oriented topics—not limiting topics to traditional operations, planning, and development courses. Even courses in airport administration need to evolve from the traditional paradigm of teaching about the airport industry in the context of business; that to thrive, airports must be competitive, lean, efficient, and customer-friendly. Airports are a dynamic environment, with constantly evolving implementations of technology, budget models, and customer solutions. Airports are also constantly adapting to widely changing environments, from economic swings to socio-demographic and political landscapes, to global pandemics, to changes in travel behavior. As such, academic programs should be both more flexible to the changing environments and more hands-on practice, so that students matriculating through these programs can have real-world experience before graduation.

Cronin et al. (2016) note that among the eight airport occupations identified as MCOs to the industry by more than 700 industry stakeholders, half of those occupations are served by courses and programs that are not within aviation and might have little to no intersection with the aviation schools. For example, airport jobs related to IT, financial analysis, electrical, and engineering are often filled by talent that has yet to be exposed to the airport environment and the unique work demands of this ecosystem. Further, a fifth occupation among the eight MCOs, project planning, often brings in graduates from business schools and other programs outside of aviation.

Two reasonable conclusions from revisiting this body of research are as follows:

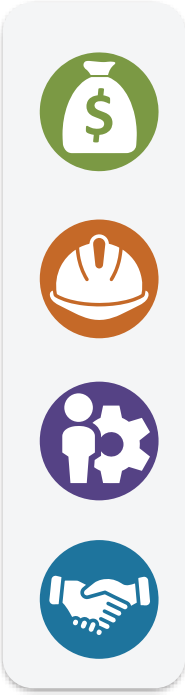
- Airport/aviation programs may create curricula within an existing course structure or as a separate course to provide education in these technical disciplines that is specific to the work requirements performed on the job; and/or
- Airport/aviation programs may work to cultivate strong partnerships with other disciplines to encourage the use of airport-specific case studies within those disciplines that allow students to apply their technical knowledge to airport scenarios.

Cronin et al. (2016) also conducted a detailed review of airport training and education programs to determine the extent to which essential competencies required to perform the eight MCOs are addressed. For example, the report indicates only 50% of programs focus on the interpretation of airport regulations and safety and security procedures that are fundamental to the airport environment. Further, mission-critical topics, including IT security risk/cyber security, are not covered by most airport-oriented academic programs. These findings are validated by the current research effort.

To speak to the second observation, which reveals most aviation management and airport operations courses are too narrow in scope, the current research suggests programs need to feature coursework and supplemental projects as well as experiential learning opportunities that expose students to the types of work activities in which they will be engaged in their future job roles. This might include providing shadowing opportunities for students to learn what landside operations entail or providing case study scenarios as part of coursework or capstone projects that involve safety challenges that students have to work collaboratively to solve. Course projects that require students to respond to airport challenges and apply current regulatory requirements can also serve as a means of developing the requisite knowledge and skills required for the future workforce.



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To this end, enhancing airport curricula requires a multifaceted approach to meet these needs:

Enhancing Depth. Most airport topics, with the exception of airport operations, tend to get cursory, if any, coverage in current curricula. There are opportunities to deliver advanced courses in airport operations, business, policy, and planning.

Enhancing Breadth. Most airport curricula do not include the vast breadth of core education in areas relevant to airports as business enterprises, important public transportation centers, and complex engineered systems. There are opportunities to broaden curricula to business, policy, and engineering topics.

Enhancing Practical Real-World Experiences. One of the most commonly heard themes during this study's surveys, interviews, and focus groups is that graduates of academic programs need more real-world experiences within their programs. There are opportunities to enhance curricula with internships and capstone projects that engage industry as well as other activities that allow students to interact with the airport industry in learning about and addressing current aviation issues.

This chapter provides ideas and concepts about how any academic program, whether it is one with already established airport curricula or one shaping a new program, can become stronger by increasing its depth, breadth, and experiential learning for the benefit of the airport industry. **This is presented through a series of building blocks for an enhanced academic airport program, as illustrated in Figure 4.1.**

This chapter discusses options for curriculum structure, course offerings, delivery methods, and experiential learning options, such as internships and extracurricular opportunities. A series of building blocks illustrates how programs may be enhanced at any level, from those that have zero aviation content to those that already have a degree program or specialization in airports. The building blocks are foundational and build by introducing airport content into an existing curriculum, broadening a curriculum with fundamental foundational courses in topics of importance to airport professionals, and deepening a curriculum in focus areas that align with the mission-critical needs of the industry through advanced core and technical elective courses. The building blocks also include methods for programs at any level of enhancement to support academic programming through innovating delivery methods and activities that encourage experiential learning and ways to champion programs with faculty leadership and industry-partnered activities.

Curricula Structure and Course Offerings

To meet industry needs, a wide spectrum of opportunities exists to improve educational depth and breadth by introducing courses in curricula that address the operational, business, and engineering/planning needs of airports. Furthermore, curricula structures can be envisioned that allow students to focus on a given area of interest, while also providing a well-rounded airport education. To achieve this, a curriculum structure of a series of fundamental and cross-cutting courses; three curriculum tracks that provide depth in their respective topics; and capstone, culminating experiences, internships, and extracurricular offerings is presented.



This enhanced curriculum structure follows the set of building blocks with fundamental and cross-cutting courses taken relatively early in the curriculum. A set of core and elective courses following a particular track to provide depth to the program is offered in later years, and an

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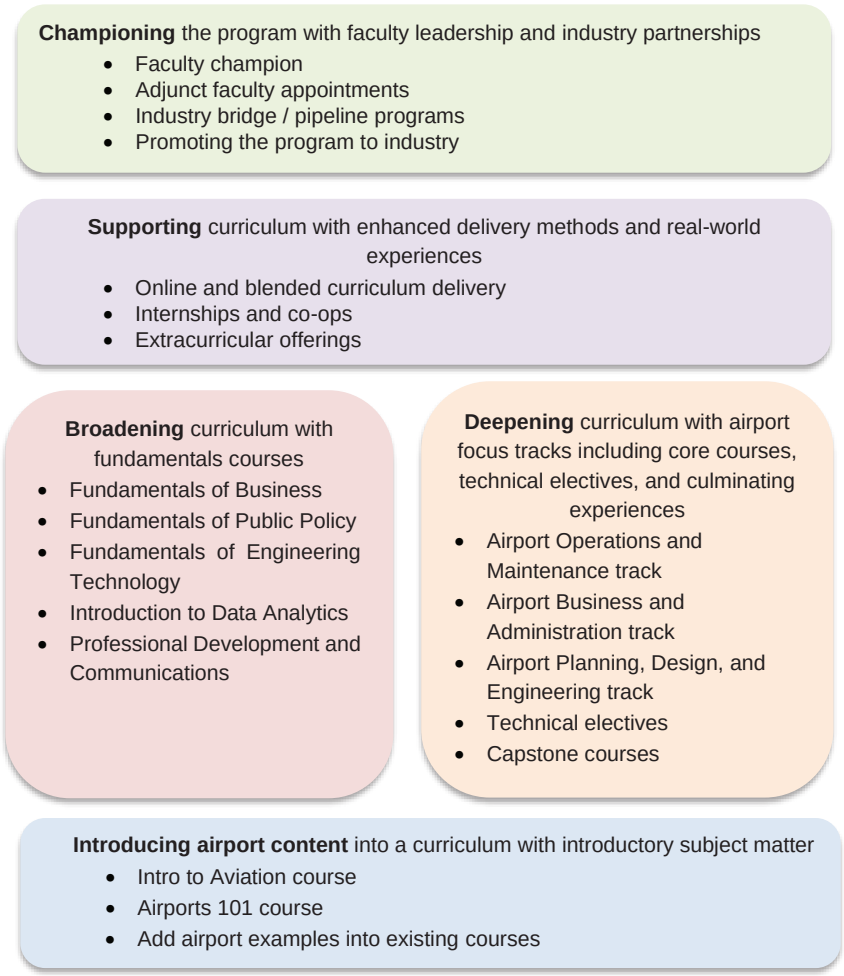


Figure 4.1. Building blocks for an enhanced academic airport program.

assortment of culminating experiences follows to add real-world applications to the program. This is illustrated in Figure 4.2.

The keys to this structure include the following:

- Intro to Aviation and Airports 101 offered early in curriculum
- Fundamentals courses are prerequisites to track's core
- Technical electives build on core (various prerequisite requirements)
- Multiple internship opportunities
- Multiple culminating experiences in 4th year

*Extracurricular activity is encouraged throughout the curriculum. Extracurricular activity in later years may be more formalized (study for industry certification, participation in national research competitions, or work-study).

This curriculum structure is detailed below. **It is important to note that this proposed structure is not intended to be a one-size-fits-all curriculum. Rather, the information provided within this structure is intended to be offered as "items on the shelf" from which any academic program can select to enhance its curriculum.**

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Year 1		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	College (Aviation) Core	
Gen Ed	College (Aviation) Core	
Gen Ed	Gen Ed	
College (Aviation) Core	**Fundamentals Course**	
Intro to Aviation	**Fundamentals Course**	
Year 2		
Autumn Term	**Spring Term**	**Summer Term**
College (Aviation) Core	College (Aviation) Core	**AIRPORT ROTATIONAL INTERNSHIP**
Gen Ed	College (Aviation) Core	
College (Aviation) Core	Gen Ed	
Fundamentals Course	**Fundamentals Course**	
Fundamentals Course	**Airports 101**	
Year 3		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	Gen Ed	**AIRPORT SPECIALIZED INTERNSHIP**
College (Aviation) Core	General Elective	
Airport Track Core Course	**Airport Track Core Course**	
Airport Track Core Course	**Airport Technical Elective**	
Year 4		
Autumn Term	**Spring Term**	**Summer Term**
General Elective	General Elective	
Airport Technical Elective	**Airport Technical Elective**	
Airport Technical Elective	**Airport Capstone**	
Extracurricular Activity		

Figure 4.2. Example enhanced program curriculum structure.

Introductory Aviation and Airport Content

Introducing airport content into a curriculum with introductory subject matter

- Intro to Aviation course
- Airports 101 course
- Add airport examples into existing courses

First and foremost, enhancing curricula with airport-oriented content starts with at least the most introductory courses on the aviation industry and airports. These courses include the following:

Intro to Aviation—An overview of the aviation industry, to include an understanding of all sectors, including government structures (FAA, National Airspace System, ATC, etc.), civil users (commercial airlines; cargo; charter and corporate aviation; general aviation; aeromedical; agricultural aviation; future users of UAS), and airports.

Airports 101—An introduction to the airport industry, to include an understanding of the types of airports that exist, their roles within the national airspace system, their roles within a community, the customers they serve, the organizational structures that they fit in, and operational components. The course would also provide introductory material of the topics covered more in-depth in advanced courses in a deeper curriculum, including airport operations; finance and administration; marketing and air service development; maintenance; and planning and design.

Most aviation and airport programs already offer these courses; however, for those institutions that do not have aviation content and are interested in enhancing their curricula in this area, these two courses are the first to add.

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Adding Airport Examples into Existing Courses

The airport environment has applications to nearly all areas of study in academia. Simply adding airport examples into existing classes can provide both airport education and motivation for further study.

Fundamental and Cross-Cutting Courses

Fundamental and cross-cutting courses applicable to all interests and elements of airports include the following. Although these courses are intended to be fundamental, and not necessarily specific to the airport environment, use of airport examples can be most beneficial in these courses.

Broadening curriculum with fundamentals courses

- Fundamentals of Business
- Fundamentals of Public Policy
- Fundamentals of Engineering Technology
- Introduction to Data Analytics
- Professional Development and Communications

- **Fundamentals of Business**—An introduction to the elements of a business environment, to include intro to economics, finance, accounting, marketing, communications, organizational structures, operations, customer service, human resources management, and the nature of professionalism in industry. *Example airport exercise: Become familiar with an airport's strategic plan and annual report.*
- **Fundamentals of Public Policy**—An introduction to how public policy operates, to include high-level education on federal, state, and local government levels, laws and regulations, and governmental procedures. *Example airport exercise: Attend a local airport town hall meeting or government meeting where the airport is on the agenda.*
- **Fundamentals of Engineering Technology**—An introduction to engineering principles with a focus on properties of mechanics, aerodynamics, and electrical systems; IT; and an overview of engineering specializations to include mechanical, civil, and electrical engineering as well as IT. *Example airport exercises: Study the operational performance of an aircraft that commonly uses a nearby airport; take a site visit to an airport and tour the airfield; study the runway configuration, pavement construction, and electrical lighting systems.*
- **Introduction to Data Analytics**—Introduction and application of data analytics tools, methods, and purposes. Apply data analytics software tools (spreadsheets, visualization tools, programming packages) to airport data to seek business and operational insights. *Example airport exercise: Review and analyze historical airport operations data, create charts, and illustrate possible future demand levels.*
- **Professional Development and Communications**—A course focusing on developing professional skills needed for success in the airport industry, to include the development of oral, written, and presentation communications skills, project planning, time management, and an understanding of professional culture. *Example airport exercise: Create a short professional presentation about an airport, discussing one of its challenges.*

These courses already exist in some form or fashion at most academic institutions, although they may not contain any aviation or airport content. If this is the case, conversations with faculty teaching these courses may yield the possibility of adding such content. Airport content integrated into the courses would be of great benefit. In addition, these courses introduce the topics described. Courses throughout any curriculum would then advance and reinforce these topics. This is particularly important in the area of professional development and communications.

These fundamental cross-cutting courses form the basis of a curriculum that would serve as a foundation for an enhanced airport academic program.



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Focus Tracks, Technical Electives, and Experiences for Deeper Learning

Deepening curriculum with airport focus tracks with core courses and technical electives

- Airport Operations & Maintenance track
  - Airport Airside Operations
  - Airport Terminal and Landside Operations
  - Airport Safety and Security
- Airport Business & Administration track
  - Airport Accounting and Finance
  - Airport Policy and Administration
  - Airport Marketing and Communications
- Airport Planning, Design, and Engineering track
  - Airport Design
  - Airport Master Planning
  - Airport Land Use Compatibility Planning
- Technical electives
- Capstone courses

In addition to gaining education in these fundamental core courses, an enhanced curriculum might include tracks of deeper core and elective courses focusing on the following areas:

- **Airport Operations and Maintenance**
- **Airport Business and Administration**
- **Airport Planning, Design, and Engineering**

Each track consists of a set of core courses and a selection of proposed technical elective courses. Core courses provide a required depth in the track, while the technical electives are designed to build on the core courses to provide the student with the opportunities to go even deeper into specialized topics.

In most curricula, technical electives are available for any student, regardless of the track they choose. However, it would be common for the core courses for each track to be prerequisites for technical electives relevant to that track because **the topics presented in technical electives are intended to be advanced study based on introductory material found in the core courses.**

Within each of the following courses are topics and activities that may not only be applied in a standalone course but also included in other courses on relevant topics. For example, the use of social media by airports can be relevant to any course in social media communications or marketing.

Focus Track: Airport Operations and Maintenance

This track focuses on the daily management of the physical airport environment, to include the airfield and nearby airspace; terminals and other buildings; and landside (roadways, parking, transit systems, etc.) facilities and infrastructure. A working knowledge of how these systems operate safely and should be maintained is stressed. **This track is most similar to existing airport curricula of programs that exist within aviation management programs**. These programs focus their curricula on aviation-related topics and specifically focus their airport-related content on airport operations (see Figure 4.3).

Core courses for this track include the following:

- **Airport Certification.** This course would cover the requirements for an airport to achieve FAA certification to accommodate commercial-service aircraft, covering more than 30 key items in the Airport Certification Manual, per 14 CFR Part 139.
- **Airport Terminal and Landside Operations.** This course would cover the issues concerning the effective management of airport terminals and landside facilities.
- **Airport Safety and Security.** This course would cover the essential elements of airport operational safety and security.

Technical electives pertinent for this track may include but are not limited to the following:

- **Obstruction Evaluation/Airport-Airspace Analysis.** This course covers the procedures for creating published arrival, approach, and departure procedures for airports, known as United States Standard for Terminal Instrument Procedures (TERPS), and the airspace protection requirements to ensure these procedures operate in safe and navigable airspace, per Federal Aviation Regulations (FAR) Part 77 and TERPS airspace clearance requirements.

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Year 1		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	College (Aviation) Core	
Gen Ed	College (Aviation) Core	
Gen Ed	Gen Ed	
College (Aviation) Core	**Fundamentals of Business**	
Intro to Aviation	**Fundamentals of Engineering Technology**	
Year 2		
Autumn Term	**Spring Term**	**Summer Term**
College (Aviation) Core	College (Aviation) Core	**AIRPORT ROTATIONAL INTERNSHIP**
Gen Ed	College (Aviation) Core	
College (Aviation) Core	Gen Ed	
Professional Development and Communications	**Fundamentals of Public Policy**	
Data Analytics	**Airports 101**	
Year 3		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	Gen Ed	**AIRPORT OPERATIONS INTERNSHIP**
College (Aviation) Core	General Elective	
Airport Safety and Security	**Airport Terminal and Landside Operations**	
Airport Certification	**Airport Electrical Systems**	
Year 4		
Autumn Term	**Spring Term**	**Summer Term**
General Elective	General Elective	
Wildlife Hazard Management	**Airport Winter Operations**	
Irregular Operations	**Airport Capstone**	
Extracurricular Activity		

Figure 4.3. Example enhanced curriculum: Airport Operations and Maintenance track.

- **Pavement and Infrastructure Maintenance.** This course covers the practice of maintaining airport pavements and other airfield infrastructure.
- **Operational Safety and Safety Management System (SMS).** This course covers the aspect of maintaining the operational safety of airport operations, including the implementation of airport safety management systems.
- **Airfield lighting, signage, markings, and electrical systems.** This course provides in-depth coverage of the operation of airfield lighting, signage, markings, and electrical systems.
- **Airport Security.** This course covers the spectrum of issues, rules, regulations, and practices associated with maintaining the security of airports.
- **Airport Winter Operations.** This course covers the elements of maintaining the safe operation of airports during winter conditions, to include snow and ice removal, aircraft deicing, and other conditions associated with winter weather.
- **Environmental and Wildlife Hazard Management.** This course covers the requirements to maintain safe and land use compatible airport operations in the face of environmental challenges, including wildlife hazards, air quality, water quality, and noise impacts.

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- **Management of Operations During Airfield Construction.** This course covers the strategies used to manage safe airport operations during periods of construction and other unusual conditions on an airfield.
- **Commercial Terminal Airline Tenant Operations.** This course covers the strategies associated with managing a commercial airline terminal, to include gate operations management, concession operations, passenger processing, and other operational components of the terminal environment.
- **Curbside and Parking Operations.** This course covers the strategies for operating an airport's curbside and parking facilities.
- **Terminal Facilities Maintenance.** This course covers the strategies for maintaining airport terminal facilities, such as gate areas, jet bridges, concessions, restrooms, IT infrastructure, and other building infrastructure.
- **Operation of Surface Roads and Transit Systems.** This course covers the strategies for managing airport surface roads and transit systems.
- **Management of Irregular Operations.** This course covers the strategies to manage an airport under irregular operating conditions, such as severe weather, external events, diverted aircraft, and the like.
- **General Aviation Airport Operations.** This course covers the particulars of managing a general aviation airport, including understanding of small aircraft operations, operations at non-towered airports, FBO operations, and hangar operations.
- **Airport Information Technology.** This course covers the management of IT systems at airports, including flight information systems, security systems, internal administrative systems, public Wi-Fi access, and the like.



Focus Track: Airport Business and Administration

This track focuses on managing the airport as a business enterprise, most often as a public agency, to include applications of policy, finance, marketing, capital planning, revenue management, customer service, communications, and other business interests (see Figure 4.4).

Core courses for this track include the following:

- **Airport Accounting and Finance**—An in-depth investigation of the accounting and finance strategies used by airports for the day-to-day financial management and long-term capital needs of the airport.
- **Airport Policy and Administration**—An in-depth investigation of the administrative practices and policy strategies of airports, to include airport governing structures, interaction with local, state, and federal governmental agencies.
- **Airport Marketing and Communications**—An in-depth investigation of how airports market and communicate with their customers and community partners. Topics include air service development, social media communications, and so forth.

Technical electives pertinent for this track may include but are not limited to the following:

- **Airport Properties Management**—An in-depth look at managing airport properties. Topics include management of FBOs, hangar management, and non-aeronautical properties.
- **Airport Concessions Management**—An in-depth look at how to manage an airport concessions program, to include retail; food and beverage; entertainment; car rental; and other passenger services. Topics include tenant management; contracts; rates and charges; customer service; branding; quality control; and so forth.
- **Airport Consulting**—A study of the role of consultants and contractors in the airport industry. Topics include requests for proposals, requests for qualifications, and requests for bids as well





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Year 1		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	College (Aviation) Core	
Gen Ed	College (Aviation) Core	
Gen Ed	Gen Ed	
College (Aviation) Core	**Fundamentals of Business**	
Intro to Aviation	**Fundamentals of Engineering Technology**	
Year 2		
Autumn Term	**Spring Term**	**Summer Term**
College (Aviation) Core	College (Aviation) Core	**AIRPORT ROTATIONAL INTERNSHIP**
Gen Ed	College (Aviation) Core	
College (Aviation) Core	Gen Ed	
Professional Development and Communications	**Fundamentals of Public Policy**	
Data Analytics	**Airports 101**	
Year 3		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	Gen Ed	**AIRPORT MARKETING INTERNSHIP**
College (Aviation) Core	General Elective	
Airport Accounting and Finance	**Airport Administration and Policy**	
Airport Marketing and Communications	**Airport Properties Management**	
Year 4		
Autumn Term	**Spring Term**	**Summer Term**
General Elective	General Elective	
Social Media Strategies	**Community Engagement**	
Air Service Development	**Airport Capstone**	
Extracurricular Activity		

Figure 4.4. Example enhanced curriculum: Airport Business and Administration track.

as FAA requirements for a competitive selection process and best practices for consultants and contractors.

- **Air Service Development**—An investigation of strategies to acquire air service into an airport that meets the needs of the community. Topics include strategic demand analysis, matching demand to proposed service, identifying providers, marketing, incentive programs, and so forth.
- **Community Engagement**—A study of how airport management may effectively engage with its community of neighbors, tenants, and governing bodies, and other stakeholders.
- **Social Media Strategies**—Investigating how airports may leverage social media platforms to assist in public communications, marketing, advertising, and handling unusual situations. Example applications include Facebook, Twitter, Instagram, YouTube, and interactive website development.

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- **Airport Business Analytics**—An advanced study in applying data analytics toward gaining airport business insights, to include operational and revenue performance, customer service quality measurement, and competitive performance.
- **Airport Communications Under Irregular Operations**—Advanced study in how to effectively communicate the state of the airport and directions for airport users in the face of unusual circumstances, such as extreme weather, infrastructure disrepair, external sociopolitical events, and pandemics.
- **Global Airport Systems**—Focusing on the enterprise models of airports around the world.

Focus Track: Airport Planning, Design, and Engineering

This track focuses on the future of the physical airport. Working knowledge of the engineering and planning of airport infrastructure is the desired goal (see Figure 4.5).

Core courses for this track include the following:

- **Airport Design**—A comprehensive overview of the elements of airport design. Elements include geometric design of the runways, taxiways, and safety areas; placement of airfield

Year 1		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	College (Aviation) Core	
Gen Ed	College (Aviation) Core	
Gen Ed	Gen Ed	
College (Aviation) Core	**Fundamentals of Business**	
Intro to Aviation	**Fundamentals of Engineering Technology**	
Year 2		
Autumn Term	**Spring Term**	**Summer Term**
College (Aviation) Core	College (Aviation) Core	**AIRPORT ROTATIONAL INTERNSHIP**
Gen Ed	College (Aviation) Core	
College (Aviation) Core	Gen Ed	
Professional Development and Communications	**Fundamentals of Public Policy**	
Data Analytics	**Airports 101**	
Year 3		
Autumn Term	**Spring Term**	**Summer Term**
Gen Ed	Gen Ed	**AIRPORT PLANNING INTERNSHIP**
College (Aviation) Core	General Elective	
Airport Design	**Airport Land Use Compatibility Planning**	
Airport Master Planning	**Electrical Systems Design**	
Year 4		
Autumn Term	**Spring Term**	**Summer Term**
General Elective	General Elective	
Airport Noise Modeling	**Terminal Design**	
Capacity and Delay Analysis	**Airport Capstone**	
Extracurricular Activity		

Figure 4.5. Example enhanced curriculum: Airport Planning, Design, and Engineering track.

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signs, lights, and markings; runway orientation analysis; pavement design; airport capacity analysis; and terminal and landside facilities design.

- **Airport Master Planning**—A comprehensive overview of the elements of airport master planning. Elements include community participation, goal setting, facilities assessment, forecasting, and capital planning.
- **Airport Land Use Compatibility Planning**—An introduction to the physical elements of consideration with respect to airports and surrounding land uses. Elements include obstruction evaluation/airport-airspace analysis; noise impact modeling; water and air quality assessments; and traffic impacts on areas surrounding the airport.

Technical electives pertinent for this track may include but are not limited to the following:

- **Engineering Design of Airports**—Advanced study of geometric design of the airfield. This covers enhanced topics, to include design of unusual airfield elements such as bridged runways, end-around taxiways, heliports, and other interesting examples.
- **Electrical Systems Design**—Study of the design of airport electrical systems, including the design of airfield electrical infrastructure and large building facilities.
- **Pavement Design**—Study of the design of airfield pavements, including soil and drainage analysis; design of rigid and flexible pavements; pavement maintenance and repair; and pavement friction and functional condition analysis. Use of FAARField FAA Pavement Design model.
- **Airport System Planning**—Study of the planning of a system of airports on the national, state, regional, and local levels.
- **Airport Construction and Project Planning**—Study of the process of planning and managing an airport construction project.
- **Airport Noise Modeling**—Advanced study of airport noise impact modeling using industry-accepted models (AEDT).
- **Land Use Planning**—Advanced study of airports and their economic, environmental, and transportation impact of surrounding land use.
- **Capacity and Delay Analysis**—Advanced study of airport capacity and delay analysis using industry-accepted models (SIMMOD, TAAM, etc.).
- **Terminal Design**—An advanced course in the historical, current, and future elements of designing a commercial airport terminal. Focus on terminal geometries; aircraft parking and gates; passenger facilities; and ground access infrastructure.
- **Design of Parking and Ground Access Systems**—A specialized course in parking facilities design and design of ground access systems, such as terminal curb fronts; intra-airport transit systems; rental car facilities; and facilities to accommodate new and future modes of transport, ranging from shared vehicle services to autonomous ground vehicles.
- **Computer Automated Design (CAD)**—Application of CAD to the design of airport airspace, airfield, terminal, and ground access facilities.
- **GIS**—Application of GIS to the airport planning environment, to include geographic layering, mapping, and data integration.
- **Designing for the User Experience**—Consideration of design thinking; passenger-experience preferences and the use of artificial intelligence, virtual reality, and other technologies in the design process.

Supporting the Curriculum with Culminating Experiences



A complete curriculum will be supported with activities that allow students to apply what they have learned in the classroom to practical real-world situations. Ways to achieve these culminating experiences are through capstone courses, internships, and extracurricular activities.

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Supporting curriculum with
enhanced delivery methods, internships, and extracurricular offerings

- Internships and co-ops
- Extracurricular offerings
- Online and blended curriculum delivery

Capstone Courses

Some of the most productive components of enhanced curricula are courses that allow students to apply what they have learned in their studies to a real-world issue over an entire term. These courses are typically known as capstone or culminating experience courses. They generally occur in the senior year of a program. Within these courses, students are grouped into small teams (usually 3 to 5 students per team). The team is then assigned a real-world issue to work on through the term.

These issues should come from the real world. Most often, an issue is brought to the class from industry. For example, an airport involved in a master planning process may consider a capstone course that provides students the opportunity to assist in the planning process in ways such as developing activity forecasts through data collection and modeling. Or perhaps, if an airport is interested in enhancing its social media strategy, students can perform an analysis of other airport social media platforms. The instructor should consider how best to support students, including making professional connections for student teams with airport industry personnel.

In these courses, students are responsible for both addressing the issue and presenting their findings at the end of the class. It may be possible to present the findings at the airport to the airport professionals with whom the students interacted. These presentations allow the students to refine their formal communications skills.

An example capstone course syllabus is provided in the resources section of the guidebook. As the syllabus illustrates, students are guided in the elements of completing term-long projects, including problem statement development, literature reviews, data collection, analysis, determining findings, and presenting results.

Experiential Learning Opportunities

Educational activities that take place outside of the classroom, known commonly as experiential learning, may be considered more enriching experiences than traditional classroom learning. Each of the previously described courses may include some component of experiential learning. Examples include real-world applications of the topics covered through course term projects, site visits to facilities of relevance to the topic covered, or integrated partnerships with industry professionals within courses. Experiential learning can also occur outside of courses themselves, including internships and co-ops, and extracurricular opportunities, such as participation in student organizations that focus on the airport industry; national research competitions; and program-sponsored guest lectures, symposia, and conferences.



Experiential learning is the practice of facilitating the education process by incorporating hands-on opportunities for students to apply what they have been taught in the classroom. There are many forms and formats for experiential learning. Some may be short term, lasting only a couple of hours or a day, while others may take weeks or longer.

The academic application of experiential learning can be incorporated into a course syllabus by offering a specific individual or team assignment. It can also be integrated into a lab associated with

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a course or as a standalone course. In most cases, experiential learning generally occurs outside the classroom setting but requires deliberate control of the learning outcome that will result.

Some programs offer an experiential learning opportunity by participating in the ACRP's University Design Competition as a one- or two-semester course. Other extended forms of experiential learning involve internships and co-ops.

Although creating an experiential learning opportunity for students requires unique considerations specific to the course, the following are general guidelines:

- Develop goals.
- Consider various experiential learning opportunities that will allow students to achieve these goals. Ideas include the following:
  - Volunteer as an airport ambassador
  - Assist with an airport master plan update
  - Assist with a full-scale emergency exercise
  - Ride along on airfield self-inspections
  - Assist with planning for an airshow
  - Support public hearings for a runway extension
  - Conduct passenger satisfaction surveys
  - Conduct research related to wildlife incidents, foreign object damage (FOD) incidents, or medical calls
  - Develop a training program that achieves regulatory compliance for all personnel
  - Assist with aircraft operations traffic count
  - Perform research of ACRP projects related to a specific topic
- Contact industry professionals to secure experiential learning opportunities for students.
- Assist with matching students (considering individual student interests) to specific experiential learning opportunities.

Innovative Curriculum Delivery Methods

As stated in Chapter 3, the research conducted for this guidebook revealed that the majority of aviation and airport-specific academic programs are delivered in the traditional manner of academia; classroom, lecture, and textbook-based over a defined academic calendar. However, this traditional template of teaching, both within and outside of aviation programming, is changing rapidly. There is no doubt that the delivery of enhanced airport academic programming will be very different in the near future.

Particularly, the new age of advanced communications and learning technologies, as well as the COVID-19 pandemic of 2020, are contributing to revolutionary changes in the delivery of higher education. It would be remiss to not consider innovative delivery methods when enhancing airport academic programming.

Traditional and New Technological Distance Learning Methods

Beyond the traditional college campus, some institutions with aviation programs have expanded their brand by providing degrees and courses through extended or satellite campuses. Many of these are associated with military bases and other remote locations. For example, while Embry-Riddle Aeronautical University has campuses located in Daytona Beach, Florida, and Prescott, Arizona, it also has an additional 130 campuses located globally and administered by a separate Embry-Riddle Worldwide program. Many courses through Embry-Riddle Worldwide are delivered in a traditional face-to-face setting and taught primarily by available adjunct faculty.

Historically, before the advent of modern communications technology, the only alternative to face-to-face instruction was the correspondence course. This form of course delivery



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involved a structured but self-paced sequence of readings from a course text and other materials (including VHS lectures) followed by written homework assignments that were sent by mail to the instructor. The instructor then graded the assignments, including corrections and other constructive comments, and returned them by mail to the student. The final exam may have been proctored through a prearranged monitor. Alternately, the course may have used other means to assess the degree of learning.

Although there are still institutions that offer correspondence courses, today's technology provides a vastly superior alternative medium for education.

With the rapid adoption of Internet technology beginning in the early 1990s, educational applications that took advantage of the capabilities exploded. Educational technology (also known as EdTech or EduTech) adapted the use of terminals tethered to central mainframe computers, and later, to personal computers to provide computer-based training (CBT) and learning programs.

Public access to Internet technology through personal computers was facilitated by services such as America Online™ (AOL), Compuserve™, and other companies that introduced email, bulletin boards, chat rooms, and newsfeeds. One of the first academic institutions to offer online courses was the University of Phoenix in the late 1980s. In 2002, the Massachusetts Institute of Technology was one of the first academic institutions to offer course material, including lecture videos and notes, online as open-source material through its OpenCourseWare project.

Since that time, the quality, content, and number of academic courses delivered online have evolved significantly.

Another phase in the evolution of online education was the emergence of massive open online courses (MOOC). In 2001, Stanford University pioneered MOOCs by offering an advanced computer applications course (artificial intelligence) online for free. Enrollment totaled over 160,000, with 20,000 completing the course. This experience inspired the creation of Udacity™, a commercial online education service. Coursera™ and EdX™ are similar platforms that offer free or fee-based collegiate-level courses.

Students choosing to enhance their level of knowledge, or pursue academic degrees, via online learning must possess additional skills to be successful. For example, students must be self-motivated. This includes checking email, engaging in discussion boards, and meeting assignment deadlines. Students must also exercise time management skills. Especially if enrolled in multiple courses simultaneously, students must remain abreast of assignment deadlines and stay on task to avoid getting behind. Students should also understand that the quality of online learning may vary from course to course. At the same time, students have the opportunity to use Internet tools to conduct research, for example, to enhance their learning experience.

Note: A brief search using the term "aviation" on each of these platforms resulted in only one course—Cyber Security Policy for Aviation and Internet Infrastructure—offered in collaboration with the University of Colorado as part of its Homeland Security and Cyber Security Specialization program.

Online Learning Through New Technologies



Historical distance learning platforms have evolved to a new level of maturity, particularly with the proliferation of high-speed Internet, powerful personal computers and mobile devices, and cloud computing platforms.

It is now relatively easy to establish a class; post syllabi and materials; present videos and other media; hold discussions; distribute and grade assignments and exams; perform extensive course

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projects; and otherwise completely manage an academic class through online technology. To facilitate this, most universities have subscribed to formal online Learning Management Systems (LMS), such as Blackboard, Google Classroom, or Canvas. Leveraging these LMS platforms to deliver airport curricula would certainly enhance course depth and breadth, enable the ability to at least provide "virtual" real-world applications, and increase accessibility, because students may be able to access the course and interact with the instructor and fellow students from anywhere in the world if they have sufficient Internet access.

Synchronous Versus Asynchronous Delivery

Online learning can be either synchronous or asynchronous. Synchronous learning takes place in real time, with students and the instructor logging in at the same time to attend an instructor lecture or have a class discussion through a video conferencing tool. Although students often perceive synchronous learning as more engaged, this form of online learning removes some of the flexibility inherent with online learning, that is, all students must plan to complete coursework (i.e., log on) at the same time.

On the other hand, asynchronous learning allows students to log on at any time to access course material. Typically, the instructor has recorded lectures or screencasts and uploaded these to the LMS to be viewed by students at their convenience. There are generally course deadlines to which students must adhere, but learning can take place at any time based on the student's preference.

The Internet technology has facilitated the ability for instructors and students to collaborate remotely. The expanded bandwidth and the field of available applications have profoundly enhanced this ability. Skype™, GoToMeeting™, Zoom™, Google Meet™, and Microsoft Teams™ are among the more popular applications that provide audio, video, and screen sharing capabilities for conducting synchronous teaching in class interactions.

Other technologies that have been used to enhance courses include the following:

MediaSite. MediaSite is a controlled video hosting service (much like YouTube but with additional protections for who can view or post videos). It also integrates directly with most LMS platforms.

TopHat. TopHat is a platform that allows for digital interaction among class instructors and participants through real-time surveys and discussion boards.

Voice Thread. This platform allows students the opportunity to upload a slide presentation and make voice comments on each slide. This allows a group of students to give a presentation remotely for viewing by an instructor and other students.

Note: The response by academia to the COVID-19 pandemic has served to accelerate the use of these applications. Many faculty who have had little experience with online teaching have been forced to adapt their material to prepare for and deliver their courses online, which has certain challenges. Fortunately, this unexpected transition occurred at a time in which technology was highly evolved, allowing faculty to transform a home office or bedroom into a classroom with highly engaging content.

Hybrid (Blended) Courses

Hybrid courses are designed to blend face-to-face and online learning, making use of the primary advantages of both delivery methods to enhance the student's learning experience. The pedagogy and methods are generally based on face-to-face course content delivered on campus

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but rely on opportunities to use online resources to make more efficient use of time and to improve the effectiveness of instruction, using additional opportunities for matching students' preferred learning styles beyond PowerPoint lectures.

As with online course development and preparation, hybrid courses require significant effort upfront by the instructor. During the course, the students are expected to invest in the same or even more work to contribute to their learning.

Extracurricular Opportunities

The education and experience students receive from participating in academically oriented extracurricular activities is as important as formal classroom study. These activities include interactions with national and regional professional organizations; program-sponsored field trips and site visits; attendance at program-hosted seminars, job fairs, symposia, and guest lectures; and attendance and participation in industry conferences.

In most curricula, extracurricular activities do not typically grant course credit toward graduation; however, they are important components of an academic education, and students are often strongly encouraged to participate.

Internship and Co-Op Opportunities

Perhaps one of the oldest and most common forms of experiential learning is through an internship or co-op opportunity. The concept of an internship or co-op is to provide a connection between what is taught in the classroom and real-world practices. During the internship experience, students learn through observation and application of their knowledge as it is practiced in the field, bringing that expanded perspective back to the classroom.

Although the terms co-op and internship are sometimes used interchangeably, there are some characteristic differences. A co-op usually involves a formal paid full-time (40 hours per week) employment position with assigned duties. An internship may be more informal, often less than full-time, and may or may not involve compensation. Some internships are not structured with assigned responsibilities, but rather the intern is rotated through departments acting more as an observer as opposed to a member of the team. Other internships may involve work on a specific project. In poorly structured experiences, interns may be given menial or make-work assignments, such as making copies or filing papers.

Most employers value candidates who have some relevant work experience; however, in the survey of aviation management programs, most do not require a co-op or internship. Most programs do offer the opportunity for students to earn academic credit for internships as technical elective credit.

Airport-related internships are designed to provide opportunities for students or recent graduates to gain valuable experience in airport management, before entering into full-time entry-level employment. Airport management benefits because interns are short-term additions to the workforce, and they are educated, focused, and enthusiastic people. In some cases, an internship acts as an intentional or de facto extended interview for potential candidates for a full-time position.



The internship position may be dedicated to a given department (such as administration, operations, or facilities) or it can be rotational and act more like a shadowing program for observing the various facets of airport activities. Rotational internships allow the intern to gain insights into all airport departments, providing the intern a more holistic view of the airport. On the other hand, spending time in one department provides more in-depth knowledge of that department.

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Airports located in cities where an aviation program exists will likely have at least one internship that draws from upper-division students. These internship positions will likely be highly coveted and competitive and will be exclusive to a select pool of outstanding students. Other airports may offer internships on a regional or national basis with no set preferences over any particular aviation program. These internships will also be highly competitive.

Programs that do not have established relationships with airports with internships will often recommend that students seek out their own internships. This can be problematic, however, especially if the academic program requires students to complete an internship before graduation. For this reason, it is most helpful to students if the academic program develops connections with local airports and connects these students to internship opportunities.

Developing and managing internship programs are not trivial activities. Guidance in doing so is provided in the form of an internship checklist and templates in Section IV.

Student Organizations with Affiliations with National and Regional Organizations

Formally recognized student organizations and clubs are some of the most common extracurricular offerings in academia. These clubs are wide-ranging in their activities and may include professional groups, social groups, and sports. Some student organizations aim to provide more exposure to an industry or career path. There are several aviation student organizations recognized throughout the academic community, including the student chapters of AAAE.

Affiliations with national and regional professional organizations provide many opportunities for interaction between academia and industry at many levels and offer support to the academic education process. Students and faculty who participate in these organizations are given a much broader and balanced perspective of contemporary policies and issues, thereby enhancing their airport educational pursuits.

Academic-oriented programming at AAAE national meetings includes formal sessions for students in topics ranging from the latest needs of the airport industry to how to create a professional resume. Meetings also offer career fair information sessions with industry and research presentation opportunities.

Aviation programs will often provide students with opportunities to participate in student chapters of national and regional aviation associations and other events. Aviation-oriented national organizations that support student chapters include the following:

- Alpha Eta Rho National Aviation Fraternity (AHP)
- American Association of Airport Executives (AAAE)
- The Flight Safety Foundation (FSF)
- The National Intercollegiate Flying Association (NIFA)
- Women in Aviation International (WAI)
- International Society of Air Safety Investigators (ISASI)

Of these organizations, the AAAE is widely recognized as the professional organization most directly partnered with academic institutions through their student chapter program. Becoming a recognized student chapter of the AAAE provides the student chapter (and its members) access to AAAE content, reduced registrations to AAAE meetings and conferences, assistance with professional development, and access to career and internship opportunities.

Members of student chapters may receive some of the same benefits as regular members, including the organization's publications and access to their website, which may contain member directories, resource documents, and other valuable information. Students attending national or regional conferences may also be offered complimentary or reduced registration fees. The

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students attending these events are often exposed to a rich array of educational and professional activities, including renowned speakers; expert panel discussions; and exhibits and displays of the industry's applied technologies, products, and services.

In most cases, student chapters are required to be officially recognized by the institution to receive financial support and other resources. The sponsoring organization may also require registration and the meeting of minimum criteria (i.e., members, annual report, etc.) for the student chapter to be recognized. Chapters will be organized with a charter, bylaws, and a slate of officers responsible for the governance of the chapter. A faculty or staff advisor is usually required as well to assist and guide students. Faculty advisors are encouraged to participate in AAAE activities through the organization's academic relations committee.

Regional chapters of AAAE are quasi-independent affiliates of the national organization, with separate leadership and financial resources. In addition to their own annual conference, they each host a specialty conference (e.g., Southeast Chapter—Finance and Administration; Northwest Chapter—Facilities; and Southwest Chapter—Airport Management Short Course) open to the national membership and others. Students are welcome to take advantage of these events, again with complimentary or reduced registration, sometimes in exchange for assisting staff with the preparation and support of the event activities.

The NBAA hosts their annual Business Aviation Convention & Exhibition (NBAA-BACE) every year, alternating between Las Vegas, Nevada, and Orlando, Florida. (Note: The annual event has grown so large that these are the only two venues in the nation large enough to accommodate the exhibits and have sufficient lodging for attendees.) As part of the annual conference agenda, NBAA offers a "Careers in Business Aviation Day" for students on the last day of the event. Besides a steeply discounted registration, students and educators are given full member access to the exhibition hall; the static displays of aircraft; and a special career development seminar with panel discussions and lunch provided. Many exhibitors at these and other similar events are FAA, state aviation organizations, and airports among other related product and service providers.

Some state organizations may foster a close association with in-state programs and provide similar benefits to students, particularly since many of the regular members may be alumni or supporters. The Ohio Aviation Association and the Texas Department of Transportation, for example, regularly support students through discounted access to conferences and routinely allocate one session of their annual meetings to student presentations of airport-related research.

Field Trips and Site Visits

Participation in organized field trips and attendance at aviation events are among other popular activities for students and faculty to develop closer working relationships with industry. Field trips to major airports, as well as airline and aerospace headquarters and facilities, provide an opportunity to observe and learn in the real-world aviation environment. Knowledgeable tour guides can offer answers to many questions that arise from seeing things firsthand.

Arranging field trips and site visits begins with creating or leveraging relationships with airport industry professionals. Visits to local facilities are more easily scheduled than longer distance trips that may require air travel or overnight stays.



Guest Lectures and Symposia

Hosting a guest lecturer or a panel of invited speakers in a symposium format to discuss current aviation issues is another common means to provide students and faculty with a direct connection to industry professionals. Alumni and local industry leaders are a readily available and

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accessible resource for guest speakers. Some organizations will have a speaker's bureau or similar list of people who have volunteered to be a speaker. A symposium, lecture series, or other special event may also be a forum to provide the program with access to experts in the field. These events may also be open to the public and provide opportunities to gain publicity about the program as well as a potential revenue source.

Career Fairs

Institutions and academic programs will organize and invite industry to attend one or more career fairs during the academic year. Many aviation programs will hold a career event that specifically targets aviation industries and organizations that may be interested in recruiting their students. Some events include the opportunity for panel discussions on career topics and personal interviews. Students greatly benefit from such aviation-specific events. For example, each year, The Ohio State University Center for Aviation Studies hosts an "Aviation Industry Night" event. During this event, a panel of aviation industry experts present their perspectives on the state of the aviation industry and participate in a discussion with attendees. Following the panel discussion, a networking reception is held, which includes interactive networking with industry organizations. This event is widely acclaimed with more than 300 attendees, most of whom are students, who find tremendous value in learning from industry in this forum.

Research Activities

Research is another way for industry and academia to develop partnering relationships. Research grants, sponsored research, and consultancies are other opportunities for industry and academia to develop partnering relationships in addition to supplementing revenue for the academic program.

Research grants usually involve competitive or solicited projects from governmental agencies or other organizations that involve subject matter for the benefit of the public. In many cases, academia and businesses will form a team to conduct the research. Faculty and students from academic programs have the opportunity to be part of this process. For example, the FAA has an office of research that manages both internal and external research efforts. One initiative where a certain amount of the FAA's research budget is allocated is to the ACRP to administer applied research projects and activities to address contemporary issues related to airport operations, administration, and development. ACRP regularly solicits research proposals for specific topics from qualified interests, and often academic/industry teams are developed to respond. There are many other research grant programs that provide opportunities to engage faculty and students in research. This guidebook, for example, was developed by a team of researchers from both academia and industry.

ACRP also administers the graduate research award (GRA) program as another opportunity for interaction between academic and industry interests. The GRA program solicits proposals from graduate students (master's, PhD) for scholarly research topics, usually a component of their thesis or dissertation, that are reviewed by industry professionals. Successful students are awarded a generous stipend to support their research and to ultimately prepare a scholarly article for publication in the *Transportation Research Record* or other journals.

Finding the Faculty Champion

Regardless of the activity, finding the right faculty member to lead a particular extracurricular effort is key to success. Faculty participate and support extracurricular activities in many ways, such as acting as the faculty advisor to a student organization, making connections with

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Championing the program with faculty leadership and industry partnerships

- Faculty champion
- Adjunct faculty appointments
- Industry bridge/pipeline programs
- Promoting the program to industry

industry on behalf of the students, teaching and assisting with administrative responsibilities (such as management of a student organization's finances), or simply generating excitement for these extracurricular activities.

Developing Faculty

There are not many established resources for developing faculty for supporting airport-related courses. One opportunity may be to actively recruit faculty from the ranks of airports and airport-related businesses to consider a teaching appointment. There are 20+ master's-level aviation programs in the country that may provide another resource for faculty candidates. Graduating students with airport experience may be recruited to teach.

Access to airport-focused faculty may also be leveraged through intercollegiate agreements that allow students to enroll in online airport courses for credit that are provided by one program and open to others. It is a common practice among NCAA athletic conferences (e.g., Big 10, Big 12, SEC) to share research, academic, and in some cases, financial resources, including faculty.

Following this model, there may be opportunities to offer several advanced aviation-oriented academic courses to aviation management students to help narrow this competitive gap and to develop better-qualified professionals for the aviation workforce.

One example may be to organize a consortium or collaborative agreement among aviation management programs to offer an extensive series of advanced academic courses on various aviation-related topics.

Such a consortium would provide access to aviation-focused educators and other available adjunct faculty currently teaching at various collegiate aviation programs. The course syllabi would be adapted to follow agreed-upon guidelines for upper-division academic coursework and hybrid or online delivery. The courses would be designed to provide an enhanced depth of knowledge so that graduates can compete successfully for entry-level career positions.

Qualified faculty resources for collegiate aviation programs will always be a struggle, especially for specialty areas such as airports. This will require greater attention on looking for opportunities to provide a pipeline for faculty to teach a broad array of airport-specific courses.

Because aviation is a small, diverse, and dynamic industry, it is often difficult for aviation programs to recruit and retain qualified faculty with the appropriate academic credentials and experience qualified to teach. The composition of faculty at a typical aviation management program may include 4 to 7 full-time faculty and several adjunct faculty. Like most institutions, full-time faculty come from a broad array of academic backgrounds and professional experience. In some cases, aviation faculty may have retired from a military or professional career. Many aviation program faculty will have experience as commercial or military pilots, while others may come from a business background. No two programs are like.

Other faculty may have risen through the academic ranks. They often begin as undergraduate aviation students, serve as flight instructors after graduation, and upon completing an advanced degree, become qualified to teach academic courses.

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Partnering with Industry

One of the great challenges in enhancing academic programming for airports is finding faculty with the education and experience to teach courses, advise extracurricular activities, and otherwise contribute to the program. One solution to this challenge is to actively engage with current industry professionals. Professionals working in the industry can provide a tremendous wealth of practical knowledge to the students, and with some preparation, guidance, and training, may be able to lead such efforts both in and out of the classroom. Such activities include conducting guest lectures or seminars; supporting capstone projects and research activities; advising student organizations or clubs as an industry liaison; or teaching courses as adjunct faculty.

Industry Professionals as Adjunct Faculty

Many industry professionals have little experience in academia, and hence should be guided on how to manage a class, deliver lectures, and create assignments and exams. It is good practice to bring these professionals to a place of comfort in the classroom in a strategic manner, first by offering one guest lecture, then a series of lectures, mentoring a project team, creating and supervising an assignment, developing a syllabus, and becoming comfortable with LMS before ultimately being given the complete responsibility for a course. Most often, adjunct faculty are actively mentored by a full-time member of the faculty.

Industry Bridge/Pipeline Programs

Some programs may have an established relationship with specific aviation employers to place their graduates. The relationship is usually based on the employers' experience with the quality of the program's graduates to fit their workforce needs, and they seek to hire the program's "best and brightest." Some employers may have a succession plan, under which successful candidates are promoted and replacements hired to continue the cycle of bringing in new personnel. In some cases, there may be competition among employers for these highly valued candidates.

Promoting the Expertise of an Academic Program

Academic programs can both enhance their curricula and promote their current expertise by a series of activities that benefits the program's students, faculty, and industry partners.

Not-for-Credit Short Courses and Seminars

As educational institutions, aviation programs can give back to the industry by hosting seminars, symposia, and short courses that address specific and relevant topics. They provide an opportunity to showcase the program's faculty as well as serving as continuing education opportunities for industry professionals. Specialty seminars and short courses can establish the program's expertise, and by bringing industry to the campus, provide networking opportunities that can extend to students and other faculty. Successful events can also serve as a revenue center for supporting the program and the institution financially. These short courses and seminars can also be offered to students. Student participation as assistants in managing these events is a great learning and networking experience as well.

Sponsored Research

Sponsored research projects typically involve specific business interests to support their mission and goals. Conducting academic research offers the sponsoring entity an opportunity

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to demonstrate independent investigations and observations conducted with scientific and ethical integrity. Sponsored research provides opportunities to financially support students, faculty, and additional resources the program may need. It also serves to bolster the program's academic "resume," showing their involvement in industry interests and expertise.

Consultancies

Many programs have faculty with expertise in specific airport-industry topics. These faculty may be called on by industry to collaborate in a consulting capacity.

Most institutions allow faculty to engage in extracurricular work activities as consultants as long as it does not appear to conflict or otherwise interfere with the institution's interests and faculty member's responsibilities. Many institutions encourage consultant work not only to enhance faculty compensation but also to enhance the prestige of the faculty member and the associated program. There may be restrictions on the use of the institution's facilities, equipment, or other resources; however, in many cases, this may be allowable as incidental use or through reimbursements. Some institutions have established policies to reduce overhead charges for faculty performing consulting to industry.

Depending on the project, students may be able to assist faculty on consulting work. This provides these students an excellent opportunity to gain industry experience, while also working on a real-world project with the faculty member.

The academic program benefits by (1) contributing to industry, (2) receiving financial and other support, and (3) providing a direct relationship to industry that in turn, provides opportunities to enhance the curriculum and provide faculty and students with practical experience in industry matters. Industry gains by tapping into the academic disciplines for the expertise of faculty, institutional resources, and energy provided by faculty and students. It also accesses timely, cost-effective independent research.

Job Placement Assistance Programs

The overarching goal of aviation management programs is to successfully place graduates into positions that are aligned with their career interests. Beyond preparing them academically to enter the workforce, institutions will provide additional resources and activities for matching qualified graduates with industry. It is common for the institution to have a centralized office to serve as a resource for providing a variety of services to students and employers. Services to students may include career counseling, job search assistance, resume preparation, and interview skill tips. Employers are offered direct access to students for internships and entry-level jobs.

Current employment opportunities will be posted by the institution's central career office to ensure that all students have access to the announcements. Some programs will have their own platform for job postings and require employers to register before posting positions. In these cases, registered employers are offered the use of facilities for recruiting visits, on-campus interviews, and so forth. In some cases, they may be allowed to review student resumes.



Another job posting/job search resource that institutions subscribe to is Handshake. Handshake is a commercially comprehensive job search database specifically designed to assist students and alumni in finding and applying for internships and degree-related jobs. Employers may also subscribe to Handshake to post positions that target specific programs or academic specialties.

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Summary

This chapter identifies methods of enhancing academic curricula to provide students more education about the airport industry. Opportunities include greater depth and breadth in course curricula that are of importance to the industry, particularly to the MCOs, and additional opportunities to infuse experiential learning that provides real-world learning experiences are described.

Enhanced course curriculum consists of a broad base of fundamental and cross-cutting topics in business; policy; engineering and design; and aviation, including introductory material to the world of airports, three tracks representing these three general topic areas, which consist of advanced core courses as well as a selection of technical electives, a capstone or culminating experience, and internship opportunities. Enhanced methods of delivery, focusing on online programming, using advanced LMS platforms, and multi-media technology were presented.

Opportunities to enhance airport education through extracurricular offerings include the following: student organizations, guest lectures, symposia, research opportunities as well as internships and co-op opportunities.

Section IV provides the resources to make this happen, including sample curriculum structures, syllabi, internship templates, and external resources.



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SECTION III

Guidance to Industry

Consider the common saying in the aviation sector, "If you've seen one airport, you've seen one airport." Academic programs, especially relevant to airports, are also not the same. On the other hand, there are some characteristics of academic programs that are similar throughout academia. Understanding how academia functions will certainly help industry practitioners navigate the system as well as more effectively contribute to enhancing airport academic programs for the benefit of future aviation professionals.

This section provides an overview of academia to the airport industry, including those working at an airport and at professional firms that support airports, such as engineering, planning, and management consultants. For these sectors, recruiting to fill entry-level positions often focuses on identifying and recruiting quality graduates from academic programs. Also included is a series of strategies that industry can use to engage with academia to educate more professionals for airports and the firms that support them.

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CHAPTER 5

How Academia Works for Non-Academics

Most airport professionals are likely to have been enrolled in a program of higher education at some point, whether it was learning a trade, or earning an associate, 4-year (bachelor's), or graduate (master's or doctorate) degree at a college or university. It is from these experiences that professionals gained insights into academia. However, there was likely limited exposure to the bureaucracy that envelops the process of creating and delivering degree-granting curricula. The following provides insights into how academia works, from the administrative and operations perspectives, with the goal of better understanding how industry can participate in, as well as influence, the process of enhancing curricula to better serve the airport industry.

The Administration of Academia

Understanding the placement of airport-relevant curricula is important to the overall industry seeking to collaborate and strengthen a wide spectrum of courses. Sometimes, the hierarchy is easily navigable; sometimes it may be quite burdensome. Enhancing academic programs, whether a small change to one course or the creation of an entire degree program, warrants review. The airport industry can and should participate within the administrative hierarchy to strengthen any curricula.

As illustrated in Figure 5.1, the hierarchy follows a standard top-down model that begins with a university board of trustees overseeing the overall direction of the institution and ends with the faculty. In between, there are administrative offices that have a hand in how academic programs are operated.

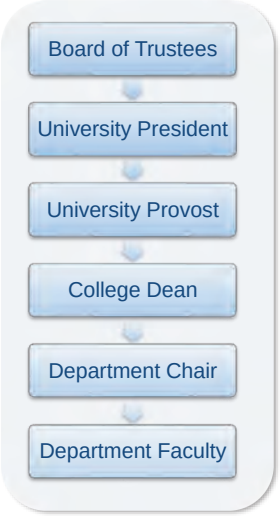


Figure 5.1. Hierarchy of academia.

The Faculty of an academic program develop and administer individual courses and are responsible for creating course syllabi, materials, and delivery methods. They are responsible for instruction and evaluation of student progress and meeting course objectives. For most aviation programs that offer airport courses, one or two members of the faculty are typically considered the subject matter experts. **These faculty members are often the first point of contact for industry members who are interested in contributing to the curricula.** For example, they may work with faculty to develop course topics, activities, and opportunities for industry participation, such as real-world case studies, guest lectures, and site visits.

The **Department Chair** is responsible for overseeing the direction of the department following policies created at the university or college level. These responsibilities include oversight of faculty course assignments, class scheduling, and higher-level metrics of department performance. This person should be kept abreast of any activities that a faculty member may be considering to enhance classes currently offered or planned. The Department Chair is often seen as the external point of contact for the programs in the department, and as such, may also be responsible for a department advisory board, which sets the direction for internship and

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capstone programs. The Chair may also lead strategies for development and engagement as well as seek philanthropic gifts for student scholarships and endowed faculty positions**. In this role, the Department Chair is a primary point of contact should industry professionals want to participate in department-wide activities.**

The **College Dean** is responsible for overseeing the college within which the aviation program resides. Primary responsibilities for this person include setting the mission and strategy of the college as a whole, understanding that there are often many programs that exist within the college. The Dean oversees all internal functions and is the public face of the college, interacting with alumni, industry, and the wider community. The Dean engages in large-scale changes to an academic program, while the Department Chair focuses on individual courses and subject matter. The Dean reviews and endorses major program changes, and they are proposed for approval by the Provost, President, and Board of Trustees.

The responsibility for overseeing the academic mission of an institution typically lies within the Office of the **Provost** and institution-wide leadership is held within the office of the **President**. The President reports to the **Board of Trustees** on all matters concerning the mission of the institution, including academics, student life, alumni relations, athletics, budget, fundraising, community affairs, and facilities planning. Most modifications to an academic program will eventually move to the offices of the Provost and the President for their review. Ultimately, the Board of Trustees formally approves program changes.

Academia Is Management by Committee

Along with their hierarchical structures, academic organizations are often managed by committee. Committees often made up of faculty members and sometimes members of industry exist to provide group thoughts and consensus-based recommendations to the hierarchy to aid in decisionmaking for all matters in academia.

For example, a **curriculum committee** often has the charge of evaluating and providing any recommendations to proposed curriculum enhancements, ranging from adding or removing topics in any given courses to creating new courses, specializations, minors, or major degree programs. This committee typically consists of faculty members only, but occasionally, the committee will invite outside members, including industry partners, to present their ideas at meetings.

To engage with industry, some committees invite business leaders to sit as committee members. One example is a **strategic planning committee**, which would have the charge of providing guidance to the department chair on strategic issues, such as faculty hiring, budgeting, and the overall mission of the department. An associated **external advisory committee** may also provide opportunities for input on the overall mission of the department. Airport professionals may participate in helping steer the direction of the program as well as to provide meaningful, long-lasting relationships between industry and the academic program.

Where Aviation/Airport Programs Fit Within the Academic Environment



Aviation/airport curricula may be found in numerous sections or programs at various institutions. Traditional academic topics, such as mathematics, economics, and English, are typically housed in similarly named departments (i.e., the Department of Mathematics). Wider academic disciplines, such as engineering, are often housed in colleges bearing that name. A College

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of Engineering, for example, may house Departments of Mechanical Engineering, Civil Engineering, and the like. A College of Natural Sciences may house Departments of Mathematics, Physics, and Chemistry. A College of Arts and Humanities may house Departments of English, Spanish, Religion, and so forth.

Airport curricula and aviation programs, in general, are often anomalies with respect to their fit in an academic department or college. These programs have been known to be housed either in colleges dedicated to "aviation" or in Colleges of Engineering, Business, Technology, Education, and Social Sciences. The location of the program is a sign of how the curriculum may be focused. For example, an airport curriculum housed within a college of business may focus its curricula on the business side of airports, with classes focusing on management; accounting and finance; and marketing, while a program housed within a College of Engineering may focus its curricula on topics such as airport planning, pavement design, and airspace analysis.

Academic Calendar

Most people are familiar with the start of the school year. Typically, new students enter their university programs during the fall and proceed by taking courses each term. A term at a university may range from 10 weeks (usually known as a quarter) to 15 weeks (usually known as a semester). There are usually breaks between terms, the longest of which are a winter break in December through the New Year, and in summer, where there may be around 12 weeks between the spring and fall terms.

Faculty structure their courses based on the institution's academic calendar. Academic calendars vary widely among institutions, including term start and end dates, break periods, and even calendar formats. Academic calendars operating on a quarter calendar will typically have three, 10-week terms during the academic year (fall, winter, and spring quarters). Those on a semester calendar have two, 15-week terms (fall and spring).

Typically, in the summer, at least one summer term occurs with fewer classes. Students may spend their summers on internships, at summer jobs outside their course of study, or doing other activities. Faculty often spend the summer working on research projects, preparing their materials for the upcoming term, or taking vacation. During this term, however, there may also be opportunities to create courses that are more flexible with scheduling and include experiential activities, such as site visits or larger project work.

Academia also runs on a fiscal cycle from July 1 through June 30. (Some colleges follow their state's fiscal year from October 1 through September 30. For example, the fiscal year for Texas colleges runs from September 1 through August 31.) The budget planning process for an upcoming fiscal year begins 6 months before the start of a new fiscal year. During the planning process, those seeking to enhance academic curricula should work with faculty to submit budgets for programs such as funding for site visits, guest lectures, and the like. The evaluation will include the program's need and budget for adjunct faculty (those hired on a part-time, per-course basis). Industry professionals who wish to be an adjunct faculty member for a particular course should be aware of this timing.

Structure of Academic Curricula

The structure of academic curricula is designed so that students receive education in topics at an increasing level of depth within their major as well as other courses for a well-rounded education.

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Course curricula consist of various categories:

- General education courses
- College core courses
- Major core courses
- Technical elective courses
- General elective courses
- Extracurricular activities

Each course is given a certain number of credit hours, commensurate with the level of activity necessary to meet the requirements of the course (typically calculated as a function of the number of hours per week the class meets in the classroom). For example, a course that meets 3 days per week for 1 hour per class would qualify as a 3-credit-hour class. Other factors that may contribute to the credit-hour calculation are the level of laboratory participation requirements and the number of actual hours required for an extracurricular activity. Summer internships may calculate credit hours based on the number of actual hours a student may work in the internship, a typical calculation of 1 credit hour for every 100 hours worked.

In general, academic curricula consist of approximately 120 credit hours. Students must complete a certain number of credit hours in the above categories to be awarded their degrees. The breakdown of credit hours by category varies among colleges. Table 5.1 is a breakdown of a 4-year bachelor's degree program.

Courses in each category are further classified by their level of technical advancement. A course numbering system may start at a "100 level" for introductory courses, and increase according to the depth of a course. Courses at the "400 level" may be more appropriate for 4th-year students and require taking lower-level courses as prerequisites. For example, Engineering 101 would be a course introducing the fundamentals of engineering, appropriate for 1st-year students, while Airport Engineering 453, Advanced Topics in Pavement Design, would be a higher-level course on this topic, often taken by 4th-year students. Each of the course categories may have courses at each level of advancement, with higher-level courses found in the major core, technical elective, and general elective categories.

Lower-level courses, including several college core and major core courses, are fairly rigid in design, with standard topics, assignments, and exams. As such, there is less flexibility in adding specialized airport content to these courses. Although these courses may allow for a guest lecture, curricula leadership may not be amenable to adding entirely new topics, assignments, or projects. Higher-level courses, on the other hand, particularly technical and general elective courses, have more flexibility to accommodate significant enhancements. Furthermore, new courses are often created as technical electives at higher-level course numbers (such as 300- and 400-level courses).

Table 5.1. Breakdown of a 4-year bachelor's degree program.

Course Category	Required Credit Hours	Courses
General Education	24	6
College Core	32	7
Major Core	32	8
Technical Elective	21	6
General Elective	9	4
Extracurricular Activity	2	1
Total	120	32



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Understanding a program's curricula in terms of courses in each category at each level, and the potential availability of adding a small amount of airport-oriented content through additional material in some courses and larger enhancements in others will be most helpful for industry professionals wishing to contribute to the enhancement of academic curricula. Detailed information about a program's curriculum may be found on the program website or in the institution's course catalog.

The details and structure of a particular course are provided in the course's syllabus. Although the course title and description are constants, the course delivery can vary from term to term and from instructor to instructor. Delivery methods often include the choice of materials and resources used (such as textbooks and other reading materials), specific assignments, exams, and term projects. More recently, delivery methods include in-class meetings and online learning. As long as the course delivery is intended to meet the learning objectives of the course, faculty teaching the course have considerable latitude in creating the course syllabus.

Academia and the Online Learning Environment

Increasingly, academic programs are offering online courses, which may or may not align with traditional academic calendars. Because online courses do not require physical space, nor do they require lectures to be delivered and attended by students at the same time (known as synchronous class delivery), the ability to be flexible is tremendous.

The inclusion of online courses in academic curricula, including aviation programs, is maturing, with widely prolific Internet access, well-established **LMS software platforms**, and student acceptance of this type of learning. In early 2020, online courses went mainstream with the onset of the COVID-19 pandemic, when nearly all institutions of higher education moved to online delivery for all curricula. Online courses will be a significant component of curricula for the foreseeable future.

Industry members have many opportunities to contribute to online courses, from the creation of videos that may be shared with the class to participation in online discussion forums with the students.

How Curricula Are Created

Academia admittedly moves slowly. Unlike the private sector in which a project may move from concept to implementation in days or weeks, significant enhancements to an academic curriculum are likely to take a minimum of months.

In some cases, minor enhancements to the curriculum, such as adding an assignment, guest lecture, or site visit to a given class, can be planned and implemented relatively quickly. Nevertheless, planning for weekly activities for a given class is typically planned out in 1 to 3 months before the class starts.

Creating new classes requires approximately 12 to 18 months from concept to implementation.

Here are the steps to this process:

1. **Concept creation.** This may be done by a faculty member who is interested in the topic, or through a department-motivated effort, based on the strategic plan for the curriculum. The concept document will include a class title and a short description of the course topics, objectives, prerequisites, and possible materials.



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2. **Draft syllabus development**. A draft syllabus consists of a course title, a short paragraph describing the learning objectives of the course, a suggested number of credit hours, prerequisite courses, and materials and resources to be used. Day-to-day instruction elements of the syllabus are usually reserved for the instructor to create each term. The proposed instructor or a small committee of faculty with subject matter expertise in the topic creates the initial draft syllabus.
3. **Review of the concept and syllabus by an internal department-level program curriculum committee.** The concept may be modified and sent back to the initial creators. This may be an iterative process.
4. **Next steps.** The fully vetted course is sent up for **review by a college-level academic programming committee**. This committee vets the course against larger strategic and operational constraints, such as content conflicts with other courses in the college. Once approved by the committee, **the course will typically be approved as a conditional class and allowed to be offered the following term**. Once the course has been offered multiple times over multiple years, the course is sent up to the university-level academic committee for recommendation for inclusion in the official course catalog, ultimately approved by the university's board of trustees. This approval often occurs 2 or more years from the course concept.

Role of Faculty and Staff

Faculty, staff, administrators, and students are crucial to the success of any academic program. The faculty are considered the subject matter experts and are responsible for developing course content and structure as well as delivering courses and working with students to give them the best opportunity to succeed in their educational pursuit and future careers. Each category of faculty has levels of authority and responsibility for contributing to academic programming.

Full-Time Faculty

The core educational foundation of any academic unit consists of the quantity and quality of its faculty. There will be at least one to three faculty members with doctoral degrees, including the department chair. Most full-time faculty have several years of professional and technical experience in their fields in addition to their academic credentials. Often the program is supported by part-time adjunct faculty, who teach specific courses. Depending on the institution, full-time faculty may be granted tenure or are engaged in limited-term, tenure-track appointed assignments. Some institutions may hire faculty on a long-or short-term contract basis.

Tenure

Tenure is a system of employing faculty for an indefinite term that can only be terminated for cause (e.g., criminal behavior, ethics violations, etc.), financial exigency, discontinuance of the program (for educational reasons), change in economic conditions, or other extraordinary circumstances. Tenure is designed to protect the academic freedom of educators, to allow them the opportunity to conduct research, and to publicly express their particular views (including open dissent) without fear of censorship or restriction. Tenure intends to preserve academia's core responsibilities to promote research to advance the body of knowledge, and to communicate that knowledge for the public good.



Tenure-Track Faculty

Tenure-track faculty are given a set period in which to demonstrate a record of scholarship through published research, teaching, and service to the institution, department, and industry.

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Faculty who fail to achieve a minimum acceptable standard of scholarship, usually determined by an established committee, are typically terminated at the end of their term.

Clinical (or Practice) Faculty

Clinical faculty at an institution that offers tenure may be assigned much of the same teaching and service responsibilities as other faculty. However, they are generally relieved of the independent research requirements the institution expects for tenure. Clinical faculty are usually given the responsibility to teach a full load of courses, often consisting of a minimum of 12 credit hours per semester (equivalent to four 3-credit-hour courses).

Adjunct Faculty

Adjunct faculty are commonly employed as part-time or contingency teachers, usually for general education courses or highly specialized courses beyond the expertise of full-time faculty.

Employment commitments to teach as adjunct faculty generally extend for only one semester. Over the past several years, institutions are relying on increasing numbers of adjunct faculty as a means to mitigate the cost of full-time faculty while ensuring that the curriculum is adequately delivered.

Academic Rank

Aside from employment status, faculty are usually given an academic title or rank commensurate with their experience, time in service, research production, and other measures of academic success. In addition to the recognition of the faculty member's scholarship, academic rank also serves as a basis for compensation and other benefits offered by the institution. The titles and general characteristics of traditional academic rank include the following.

Professor

As a formal title, a full professor is the terminal rank for faculty promotion and normally requires holding a doctoral degree. Prefixed titles such as "Distinguished" or "Endowed" describe special circumstances. The title "Emeritus" is conferred to honor retired faculty. Note: Faculty are often informally addressed by students as "professor" to recognize their status as faculty. However, it is generally understood that the use of the title does not represent their official academic rank recognized by the institution.

Associate Professor

The title of associate professor represents a mid-level rank for full-time faculty. They often hold a doctoral degree and tenure status if applicable.

Assistant Professor

Assistant professors are normally the entry-level academic rank for full-time tenure-track faculty. Although not always a requirement, they may hold a doctorate or are engaged in a doctoral program and eligible for promotion to associate professor upon completion of the degree.

Clinical Professor (Professor of Practice)

At institutions that offer tenure, faculty are given the title of clinical professor to recognize that they are not following a tenure-track appointment but are engaged in teaching specialized courses. A clinical professor will usually hold a doctorate or other terminal degree (e.g., J.D., M.D., etc.).



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Visiting Professor

Faculty with a limited-term contract (usually a 1-year appointment) and normally holding a doctorate are known as visiting professors. They may be from another institution and assigned to teaching and research responsibilities.

Instructor and Lecturer

Faculty holding a non-terminal degree with a full load of teaching assignments but no research responsibilities or expectations are instructors or lecturers.

Adjunct Professor

Limited part-time, contingency, or on-call faculty often with assignments to teach specific courses for a semester are adjunct professors.

Aviation Faculty

Because aviation is a small, diverse, and dynamic industry, aviation programs often face challenges recruiting and retaining qualified faculty with the appropriate academic credentials and experience to teach. The composition of faculty at a typical aviation management program may include four to seven full-time and several adjunct faculty. Like most institutions, full-time faculty come from a broad array of academic backgrounds and professional experience. In some cases, aviation faculty may have retired from a military or professional career. Many aviation program faculty have experience as commercial or military pilots, while others may have a business background. No two programs are alike.

Other faculty may have risen through the academic ranks—beginning as undergraduate aviation students, serving as flight instructors after graduation, and upon completing an advanced degree, becoming qualified to teach academic courses.

Although many faculty in an aviation program hold an advanced degree that may be aviation-specific (e.g., master's in aeronautical science, aviation business administration, etc.), few hold a doctorate in an aviation-related degree. As of this guidebook's publication, only Embry-Riddle Aeronautical University and the Florida Institute of Technology offer aviation-specific doctoral degree programs.

In cases where faculty come from the ranks of retired military or professionals, aviation programs can leverage their budget by relying on the faculty pensions or other retirement benefits to supplement compensation and benefit packages. It has been observed that the compensation for aviation faculty is lower compared with their peers at their institution's other departments. Also, the average age of faculty at aviation programs can exceed 65, with faculty relying on Medicare benefits for their primary medical insurance.

Some aviation programs encourage faculty to pursue research grants and consulting opportunities to bring in additional revenue to the program to support graduate research and teaching assistants as well as to provide supplemental income to the faculty member.

A common theme among aviation programs appears to be the passion shared by faculty for the industry and their enthusiasm for helping graduates achieve entry-level employment positions.

As subject matter experts, the faculty have the authority to add subject matter, assignments, and other activities to the class to expand learning opportunities. This includes the authority to

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invite members from industry to participate in course delivery. Faculty members often reach out to industry members to seek and disseminate the industry's perspective to their class. Faculty members often recognize that industry members have more recent and practical expertise in the subject matter. This is why industry participation in academia is so important.

How Industry Can Work with Academia

Professionals already in the airport industry can contribute significantly to enhancing academic curricula in many ways. Chapter 6 provides specific guidance on these types of opportunities.



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CHAPTER 6

Opportunities to Participate in the Academic Process

Academic Opportunities

A wide range of opportunities exists for members of the airport industry to contribute to the enhancement of academic programs for future airport professionals.

For any academic program, successfully providing a technical or professional education requires at least some connection with the industry that their graduates are seeking to enter. These relationships are vital to ensure that curricula remain current and relevant, providing a measure of practical and useful knowledge necessary for graduates to successfully compete for entry-level workforce positions. Industry connections also help students gain better perspectives of the real-world environment as well as provide opportunities for networking, internships, and a more direct path to employment.

Committing to Engage—Personal Steps

The first step in participating in the academic process is simply to commit to engagement. Becoming involved with an academic program is rewarding in many ways.

Perform a brief self-assessment. Ask yourself why you are interested.

- What skills or knowledge would you like to share?
- How could your airport or organization help you in your pursuit?
- Are there programs or schools for which you have an affinity, such as your alma mater?

Knowing what you can and want to bring to the relationship will form the foundation for your engagement.

Investigating the Possibilities

Next, investigate the possibilities. Determine which academic institutions you may be able to engage with most easily. Institutions to look for are those that (1) may be close to your current location, (2) have a program (aviation or otherwise) that may cover material that is in your area of expertise, and (3) have a faculty or staff person with whom you already have a working relationship.



Example 1. Local Focus. Community colleges throughout the United States offer 2-year associate degrees in a variety of general knowledge (rather than aviation- or airport-specific) subjects relevant to skill sets required at airports. Graduates of these programs are often prepared for entry-level technology-oriented and administrative positions. With guidance from industry partners, these graduates may be able to gain valuable knowledge of their learned craft as it applies to airports, making them better prepared for success in the airport industry.

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Example 2. Areas of Expertise. If you are an airport planner, programs in architecture, urban planning, policy, and government affairs are all relevant. Similarly, if you are in airport finance, marketing, or upper levels of administration, business schools are excellent programs to contact. Programs specific to aviation and airports are often open to enhancement by bringing to the classroom real-world knowledge and deeper discussion of the topics currently offered. Matching your skill sets with the needs of these programs will result in a best-fit for you to enhance their offerings.

Example 3. Your Alma Mater. Your college may have an AABI-accredited aviation program or a non-aviation technical program. As an alum, you will know the strengths of the program, and how you can contribute your industry expertise.

Making Contact

Once you have identified opportunities within one or more academic programs, the next step is to contact a person who will be most helpful in developing the relationship. Depending on your type of desired engagement, the initial contact will most likely be either a particular faculty member or the program's chair. Making initial contact with a faculty member would be most appropriate for some level of desired engagement on a classroom level. For example, offering to give a guest lecture on a course assignment topic or host a site visit would be welcomed by faculty members who are teaching courses relevant to your knowledge base. Alternatively, desiring to teach an entire course, offering internships, or participating on an advisory board may be best addressed with the chair. Faculty and program chairs are typically accessible and welcome industry participation.

Engagement Options

Engagement options have different levels of commitment, ranging from only a few hours per year to several hours per week. Some options offer direct interaction with students and faculty, while others provide more interaction with program administrators. These options do not require a significant capital investment. Typical costs associated with these opportunities are limited to time commitments, travel expenses (if the institution with which you are working is not local), and perhaps the production of some materials.

Teaching and Lecturing Opportunities

Students truly enjoy learning from those directly in industry. Your real-world experience provides insights into current issues with a perspective that enhances the topics learned in the traditional academic setting. Whether you engage once during a semester or every week, your presence as a professional within the industry would be a most welcome enhancement to any curriculum. These opportunities include adjunct faculty appointments, guest lecturer appointments, and site-visit hosts.

Adjunct Faculty Appointments

Many institutions rely on adjunct faculty. These part-time faculty do not earn tenure; they are paid per course. Generally, adjunct faculty are a cost-effective method to bring in additional expertise to benefit students. Airport professionals, especially those having earned a master's degree, can provide significant value to an aviation program by bringing real-world issues and applications into the classroom. This arrangement is often rewarding for the airport professional



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as well because they are giving back to the next generation, thereby doing their part to ensure a capable and knowledgeable future airport workforce.

Although it is the responsibility of full-time faculty and administrators to formally create and schedule courses within a curriculum, as an adjunct faculty, you will have much control over the management and delivery of the classes to which you are assigned. This includes the creation and delivery of lectures, assignments, examinations, projects, and other activities. Class sessions may occur several days a week, often during business hours. On some occasions, classes may be scheduled after traditional business hours, during lunch hours, or on a limited-days-per-week schedule. Courses meet for the number of hours per week according to the courses' number of credit hours. For example, a 3-credit-hour course will meet 3 hours per week.

As an adjunct faculty, be prepared to spend time, in addition to in-class lectures, preparing your materials, evaluating assignments and exams, and meeting with students during scheduled appointments. A common measure of time required is to add three times as many hours per week to the number of credit hours for a class. Thus, for a 3-credit-hour course, a total time commitment would be approximately 12 hours per week for your faculty responsibilities.

Guest Lecture Appointments

Another method for investing in the next generation of airport professionals is to serve as a guest lecturer. Aviation faculty are typically very accommodating, especially if an industry professional is interested in speaking to their students. This often requires initiative by the airport professional, as faculty usually are hesitant to bother industry professionals with such a request.

Site-Visit Hosts

Aviation students are eager to leave the classroom and visit industry sites of interest. Airports are well-received by students, especially if an Aircraft Rescue and Firefighting demonstration is included. If a local aviation program has not requested or participated in an airport tour, reach out to the program chair or aviation management/airport management faculty member. Offer to host the instructor and students at the airport for a morning or afternoon tour of the airfield, terminal building, and cargo facilities. Although such a site visit will likely require scheduling months in advance, it will be beneficial for the students and airport staff as well.

Capstone Project Sponsors

Aviation students are generally required to complete a capstone or senior project in their final year in the program. These projects benefit from a real-world focus, rather than a purely theoretical view. Students benefit when industry professionals invest in them and their projects. Be willing to assist a student, whether that requires participating in a phone interview, giving a tour of airport facilities, or simply acting as a sounding board. The result may provide far-reaching benefits for the industry.

Administrative and Philanthropic Opportunities



Advisory Board Membership

The mission of a program's advisory board is to provide advice or guidance to the strategic direction of the program. Industry representatives provide valuable insight into the current and future needs of airports and may suggest how best to address them in the academic program. The level of effort of advisory board members typically includes one or two in-person meetings

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per year, quarterly conference calls, and perhaps the writing of white papers. Advisory board members are usually chosen at the discretion of the program chair.

Personal and Financial Support

Any participation in enhancing academic curriculum is considered an act of giving. Philanthropic contributions also provide academic programs with opportunities to enhance curricula through the acquisition of new materials and infrastructure; funding to support scholarships; and paid internships as well as funds for students and faculty to travel for out-of-town site visits, attend industry conferences, and so forth. For example, some programs have received grants to provide all students with tablets that aid in on-site education. Others have used these resources to subscribe to educational journals and databases and to purchase field equipment.

Financial and in-kind contributions by industry and individuals provide much-needed support to help various elements of the aviation program. The institution will usually have a development office or similar unit that is responsible for accepting and disbursing contributions from industry, foundations, and individuals. Large donations may offer naming rights for a building, auditorium, or room, while smaller financial contributions are also recognized.

Many institutions host an annual Day of Giving as a focused event to solicit contributions from alumni and friends, often as a competition among other programs on campus. Contributions can be designated to specific purposes and may be matched by businesses that have a program to do so. In-kind contributions from industry can be in the form of equipment (including aircraft), hardware, tools, software, or other items that may be useful for training and education.

Scholarships provided directly by industry are another form of financial support that benefits aviation program students. Scholarships need not be specific to tuition assistance; they can be offered to help students and faculty attend industry events.



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SECTION IV

Reference Materials

This section provides reference materials and resources to use in enhancing curricula.

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Internship Checklists

ROTATIONAL INTERNSHIP CHECKLIST TEMPLATE (examples in italics)	
☐	**Identify Partnering Organizations** • Academic Organization: *XYZ College* • Industry Organization: *KABC Airport*
☐	**Determine Points of Contact** • University Faculty Member • University Academic Advisor • Industry Technical Lead(s) • Industry Administrative Lead
☐	**Create Overview of Internship Structure** *This is a 15-week, 20-hour-per-week (300 total hours) internship intended for students in the university's aviation program. This is a rotational internship with the student rotating among airside operations, terminal operations, finance & administration, and planning departments each month.* *University will grant 3.0 units of technical elective credit toward the student's aviation major upon successful completion of the internship.*
☐	**Determine Intern Eligibility** *This internship is open for application to students with at least junior status in the university's aviation program. Student must have at least a 3.0 GPA. Student must have a valid driver's license. Student must have own transportation to and from airport.*
☐	**Determine Application Process** *Interested applicants should complete an airport employee application form, provide a statement of purpose, an unofficial transcript, and a letter of recommendation from a faculty member. Airport will select intern(s) based on the received applications. Applications are due to airport 120 days before internship start date. Interns will be selected 90 days before start date.*
☐	**Determine Internship Schedule** • *Day 1: Welcome and Orientation, completion of paperwork, department visits* • *Week 1-4: Rotation 1: Airfield Operations* • *Week 5-8: Rotation 2: Terminal Operations* • *Week 9-11: Rotation 3: Finance & Admin* • *Week 12-14: Rotation 4: Planning* • *Week 15: Wrap up, presentations, assessment*
☐	**Determine Internship Deliverables and Assessment Methods** *Student will keep daily logs of activities. Student will deliver a presentation on their work in each department during the last week of the internship. Assessment of student performance will be conducted by department mentors. Student will provide their presentations and an internship summary write up to their university. University faculty member and advisor will determine level of accomplishment.*

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PROJECT-BASED INTERNSHIP CHECKLIST TEMPLATE (examples in italics)	
☐	**Identify Partnering Organizations** • Academic Organization: *XYZ College* • Industry Organization: *KGGG Airport*
☐	**Determine Points of Contact** • University Faculty Member • University Academic Advisor • Industry Technical Lead(s) • Industry Administrative Lead
☐	**Create Overview of Internship Structure** *This is a 15-week, 20-hour-per-week (300 total hours) internship intended for students in the university's aviation program. This is a project-based internship with the student focusing on assisting the airport with developing its master plan update.* *University will grant 3.0 units of technical elective credit toward the student's aviation major upon successful completion of the internship.*
☐	**Determine Intern Eligibility** *This internship is open for application to students with at least junior status in the university's aviation program. Student must have at least a 3.0 GPA. Student must have a valid driver's license. Student must have own transportation to and from airport.*
☐	**Determine Application Process** *Interested applicants should complete an airport employee application form, provide a statement of purpose, an unofficial transcript, and a letter of recommendation from a faculty member. Airport will select intern(s) based on the received applications. Applications are due to airport 120 days before internship start date. Interns will be selected 90 days before start date.*
☐	**Determine Internship Schedule** • *Day 1: Welcome and Orientation, completion of paperwork, department visits* • *Week 1-8: Assist in performing inventory of facilities* • *Week 9-12: Assist in performing activity forecasts* • *Week 13-14: Assist in conducting community engagement meetings* • *Week 15: Wrap up, presentations, assessment*
☐	**Determine Internship Deliverables and Assessment Methods** *Student will keep daily logs of activities. Student will deliver a presentation on their work in each department during the last week of the internship. Assessment of student performance will be conducted by department mentors. Student will provide their presentations and an internship summary write up to their university. University faculty member and advisor will determine level of accomplishment.*

Sample Syllabi

COURSE SYLLABUS

Intro to Aviation

Course Objective: Introduction to the aviation industry, including its elements, components, and structures.

Topics include introductions to the national airspace system; aircraft; airlines; cargo operators; corporate and general aviation; airspace and air traffic control; and government oversight of aviation.

Learning Outcomes: Upon completion of this course, students should be able to successfully understand and discuss the following:

- Significant milestones in the history of civil aviation
- Common aviation terminology, definitions, and acronyms

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- Aviation industry structure, including the role of air carriers, general aviation, and corporate aviation
- Organization and operation of the national airspace system and air traffic control
- Basic aircraft systems, instruments, and components of a conventional airplane
- Basic principles of flight
- Basic weather principles and their impact on aviation operations
- Role of airports in the aviation system
- Careers in aviation and the sources of information on those careers

COURSE SYLLABUS

Airports 101

Course Objective: Airports 101 is designed to provide the student with a basic understanding of the airport industry, including essential elements of airport administration, operations, finance, and development. For the student interested in airport management as a career field, this course will serve as a foundation to prepare for further study, an internship, and entry-level position at an airport or related business. Whether a student is specifically interested in airport management as a potential career choice, or in another segment of aviation, such as airlines, corporate aviation, or government, it is important to understand that airports are a vital component of the global aviation transportation system, and their interface with other facets of the aviation industry is an integral part of the collegiate aviation education experience.

Learning Outcomes: Upon completion of this course, students should be able to successfully understand, discuss, and demonstrate the following:

- Understand the role and requirements of airport management
- Discuss significant legislation and changes in technology affecting historical airport development
- Compare and contrast the various airport types and their roles in the airport system
- Identify airport facilities on the airfield, in the terminal, and on the landside
- Discuss the importance of airport security
- Explain current ground access issues at major airports
- Identify current federal advisories and regulations regarding airport development and operation
- Understand qualitative and quantitative methods of evaluating airfield and terminal operations
- Describe the master planning process and its importance to airport management
- Discuss the economic, social, and political impacts of airport development on the local vicinity
- Describe various financial and organizational strategies employed by airport management
- Evaluate research and other resources that focus on current airport management issues

COURSE SYLLABUS

Airport Capstone





Course Objective: The Airport Capstone course is designed to assess and build on skills acquired in earlier courses and emphasize situations and challenges that exist in industry. Capstone provides an opportunity for students to demonstrate and refine a range of professional competencies and communication skills. Using computer and business models, case studies, and research projects, students develop and apply critical thinking and problem-solving skills, goals that are key to a student's education and success in Industry.

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Working with a faculty capstone coordinator, students are organized into teams to define and address various research- and issue-related topics. Over the course of a semester, student teams plan their approach and assign responsibilities. Teams meet at least weekly to monitor progress. Deliverables are scheduled and interim reports are expected. Toward the end of the semester, each team prepares a poster presentation and a more detailed final report to their industry sponsor, faculty, and the other teams.

Learning Outcomes: Upon completion of this course, students should have obtained the following:

- Comprehensive knowledge gained from a major area of study in a manner aimed to assist students to positively address an industry need
- Experience working in a professional group environment, providing peer direction, and peer assessment
- Enhanced interpersonal and professional communication skills

COURSE SYLLABUS

Airport Co-Op/Internship

Course Objective: This course is designed to provide an opportunity for the student to participate in a direct hands-on experience with airports and other airport-related interests that offer co-op and intern positions. The student will have the opportunity to enhance their education, applying the knowledge and skills learned in the classroom through a total immersion experiential learning environment. The intern experience helps the student understand more clearly, the operation, administration, and development of airports while also providing the opportunity to develop professional working relationships. In many cases, the internship may serve as a screening tool and extended interview that could lead to a full-time entry-level position.

Learning Outcomes: Upon successful completion of the co-op or internship, students should be able to understand, discuss, and demonstrate the following:

- Demonstrate the ability to perform satisfactorily in a professional work environment
- Compare and contrast the knowledge and skills learned in the classroom with the actual work setting
- Describe in detail the responsibilities, functions, and activities of the various airport departments and professionals
- Demonstrate examples of professional oral and written communication skills
- Discuss available career path opportunities
- Provide feedback and discuss areas for improving the internship program
- Demonstrate a satisfactory job performance evaluation that reflects favorably on the student, the program, and the institution

Note: Some internships may be organized to provide exposure to several airport departments through a rotational assignment schedule. In contrast, co-ops and other internships may be focused on temporary or limited-term employment to participate as staff of a specific department. In either case, the focus of the course is on the experiential learning that takes place while immersed in the working environment.

COURSE SYLLABUS

Airport Policy and Administration

Course Objective: Airport Policy and Administration is an advanced study of the organizational, political, and financial administration of public and private civil use airports. Areas of

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emphasis include public relations management; safety and security issues; employee organizational structures; economic impacts of airport operations; airport performance measurement standards; and current trends and issues of direct concern to airport administrators.

This course is designed as an advanced survey of airport policies and administrative strategies for those students who have achieved a basic understanding of airport operations from relevant coursework or previous work experience. Emphasis in the course is placed on the financial strategies employed by airports under a variety of organizational structures. Airports are an integral part of aviation because they interact with all other facets of the industry. As such, airport managers must be familiar with the policies and regulations imposed by industry agencies. This course will examine relevant policies imposed by the FAA, commercial airlines, and other users of a civil use airport.

Learning Outcomes: Upon successful completion of this course, students should be able to understand, discuss, and demonstrate the following:

- Understand the relevant policies and regulations specified by the FAA as it relates to airport operations
- Explain the strategies employed by airport management when it comes to working with the commercial airline industry
- Explain the impact of various financial strategies employed by airport management
- Explain the concept of airport privatization and its effects on airport management policies and administration
- Measure the performance of an airport using a variety of performance measurement methodologies
- Discuss environmental issues specific to airports and strategies used to address such issues
- Discuss various marketing methods used by airport management
- Understand the relationship between airport management and the community the airport serves

COURSE SYLLABUS

Airport Master Planning

Course Objective: Airport Master Planning introduces the student to the requirements, issues, and processes involved in the planning of airport development. The course includes an in-depth study of the sources of aviation data, forecasting methods, the airport master planning process, and environmental issues and requirements.

The course is designed to provide the student with a basic understanding of the planning and development of airports, and it plays an important role in rounding out the aviation course offerings to students in any of the college's aviation degree programs. Whether a student is specifically interested in airport planning and development as a potential career choice, or in another segment of aviation, such as airport management, airlines, corporate aviation, or government, the role of airports as a fundamental component of the aviation system and how airports interface with other facets of the aviation industry are integral parts of the collegiate aviation education experience.





Learning Outcomes: Upon successful completion of this course, students should be able to understand and thoroughly discuss the following:

- Objectives and processes for airport planning from the federal, state, regional, and local perspective
- Forecasting concepts, processes, and various methodologies for forecasting aviation activity

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- Various factors affecting the airport's capacity to accommodate aircraft demand and manage delays when demand exceeds capacity
- Relationship between the requirements of the airside and landside facilities of the airport
- National Environmental Protection Act (NEPA) process and methodologies for evaluating the impact of proposed projects on the environment
- Function and characteristics of the airport's airspace (FAR Part 77 and TERPS) and the need to protect airspace from encroachment
- Process for evaluating development alternatives from operational, environmental, and financial perspectives

COURSE SYLLABUS

Airports and the Environment

Course Objective: Airports and the Environment is designed to provide the student with a broad perspective on aircraft noise and compatible land use; air and water quality; climate change; and energy efficiency issues directly associated with airports. The course will provide the student with the background of how environmental concerns evolved with the advent of commercial jets and increased levels of air travel. It will also provide a detailed overview of how airports respond to and manage contemporary environmental concerns regarding aircraft and other activity, including impact studies and the implementation of operational measures, land use compatibility controls, and capital projects. Sustainability and climate change policies and programs are also covered.

Learning Outcomes: Upon successful completion of this course, students should be able to understand and competently discuss the following:

- Historical and legislative perspectives regarding environmental concerns relevant to aviation activity on or near airports
- Purpose and intent of the NEPA and other current laws and regulations
- NEPA compliance policies and practices relevant to airports, including environmental impact assessments and Part 150 Noise Compatibility Programs
- Metrics, thresholds, and mitigation strategies for addressing aircraft noise
- Metrics, thresholds, and mitigation strategies for addressing compatible land use around airports
- Metrics, thresholds, and mitigation strategies for addressing ambient air quality on and around airports
- Metrics, thresholds, and mitigation strategies for addressing water quality on and around airports
- Balanced wildlife hazard management and mitigation strategies
- Climate change theory, supporting data, policies, and plans for adapting to future impacts
- Sustainability policies, plans, and initiatives, including energy conservation, renewable energy resources, carbon footprint reduction, and waste management

COURSE SYLLABUS

Airport Design Practicum

Course Objective: The Airport Design Practicum is designed to provide students with an opportunity to research applied technologies to develop an innovative and practical solution for contemporary issues facing airports. It culminates in the submission of a proposal that meets the guidelines of the ACRP University Design Competition (an annual event).

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The ACRP University Design Competition focuses on four broad areas that involve issues relevant to airports and include:

- Airport Operation and Maintenance
- Runway Safety/Runway Incursions/Runway Excursions
- Airport Environmental Interactions
- Airport Management and Planning

The practicum can also serve as a culminating experience that meets the accreditation requirements of the institution and the program.

Learning Outcomes: Upon successful completion of this course, students should be able to demonstrate the following:

- Conduct a thorough and relevant review of literature to identify gaps and opportunities for developing innovation and creative solutions
- Develop and conduct technical analyses using accepted methodologies
- Understand aviation regulatory policies and practices
- Understand the practical application of the FAA's SMS
- Effective interactions with industry professionals
- Develop a business case for proposals that include benefit-cost considerations
- Provide meaningful contributions as a member of a team in a professional manner
- Prepare and submit a professional quality proposal that is responsive to instructions

COURSE SYLLABUS

Professional Development and Communications

Course Objective: This course focuses on developing professional skills needed for success in the airport industry, to include the development of oral, written, and presentation communications skills, project planning, time management, and an understanding of professional culture.

The objective of this course is to provide the students with a fundamental overview of the essential elements to effectively communicate and work in a professional environment. Because this course is intended to be taken by students pursuing an education in the airport industry, it will provide applications of communications principles to the airport industry.

Learning Outcomes: Upon successful completion of this course, students should be able to demonstrate the following abilities:

- Deliver a simple professional presentation
- Write a simple yet professional piece of correspondence
- Develop a professional style resume
- Develop a professional style Internet profile, such as a LinkedIn page
- Create and stick to a simple calendar and project plan
- Understand and appreciate diversity, inclusion, and equity in the professional environment





COURSE SYLLABUS

Fundamentals of Data Analytics

Course Objective: This course provides an introduction to and application of data analytics tools, methods, and purposes. It will involve applying data analytics software tools (spreadsheets,

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visualization tools, programming packages) to airport data to seek business and operational insights.

The objective of this course is to provide the students with a fundamental overview of the essential elements of data analytics. Because this course is intended to be taken by students pursuing an education in the airport industry, it will provide applications of data analytics principles to the airport industry.

Learning Outcomes: Upon successful completion of this course, students should be able to demonstrate the following:

- A fundamental understanding of the concept and purpose of data analytics
- A fundamental understanding of data sources
- A fundamental understanding of analytical methods including creating tables, graphs, charts, and basic statistical models
- A fundamental understanding of data analytics software tools and platforms
- The ability to perform a basic data analytics exercise

COURSE SYLLABUS

Fundamentals of Engineering Technology

Course Objective: This course introduces engineering principles with a focus on properties of mechanics, aerodynamics, electrical systems, IT, and an overview of engineering specializations to include mechanical, civil, and electrical engineering, and IT.

The objective of this course is to provide the students with a fundamental overview of the essential elements of engineering technology. Because this course is intended to be taken by students pursuing an education in the airport industry, it will provide applications of engineering technology principles to the airport industry.

Learning Outcomes: Upon successful completion of this course, students should be able to demonstrate the following:

- A fundamental understanding of essential elements of mechanical engineering technology, including basic statics, dynamics of forces, and tensions
- A fundamental understanding of the essential elements of electrical engineering technology, including fundamentals of circuits, direct and alternating currents, and power transformers
- A fundamental understanding of the essential elements of civil engineering, including structures, foundations, pavements, construction, and transportation systems
- A fundamental understanding of IT, including hardware and software platforms; digital networks; server-client architecture; and cloud infrastructure

COURSE SYLLABUS

Fundamentals of Public Policy

Course Objective: This course introduces how public policy operates, to include high-level education on federal, state, and local governments, laws and regulations, and governmental procedures.

The objective of this course is to provide students with a fundamental overview of the essential elements of public policy. Because this course is intended to be taken by students pursuing an education in the airport industry, it will provide applications of policy, government relations, and community engagement principles to the airport industry.

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Learning Outcomes: Upon successful completion of this course, students should be able to demonstrate the following:

- A fundamental understanding of how policy is created
- An understanding of regulatory bodies on local, regional, and federal levels
- An understanding of the basics of creating rules, regulations, operating procedures, and culture
- An understanding of the politics of policymaking

COURSE SYLLABUS

Fundamentals of Business

Course Objective: This course introduces the elements of a business environment, to include intro to economics, finance, accounting, marketing, communications, organizational structures, operations, customer service, human resources management, and how to be a professional in industry.

The objective of this course is to provide students with a fundamental overview of the essential elements of operating within a business enterprise. Because this course is intended to be taken by students pursuing an education in the airport industry, it will provide applications of business principles to the airport industry.

Learning Outcomes: Upon successful completion of this course, students should be able to demonstrate the following:

- A fundamental understanding of the operation of a business enterprise
- An understanding of the elements of business operations
- An understanding of the basics of business accounting and finance
- An understanding of business marketing, advertising, and communications
- A familiarization with organizational structures and human resources operations
- How to read business plans, strategic plans, and annual reports

Resources for Course Content

Airport Management

1. ACRP Research Report 16: Guidebook for Managing Small Airports—Second Edition (2019)
2. ACRP Web-Only Document 24: Challenges to Implementing Successful Land Use Strategies at Airports (2019)
3. ACRP Research Report 213: Estimating Market Value and Establishing Market Rent at Small Airports (2020)
4. ACRP Research Report 192: Airport Management Guide for Providing Aircraft Fueling Services (2019)
5. ACRP Research Report 184: Executive Summary for the Guidebook on Understanding FAA Grant Assurance Obligations (2018)
6. ACRP Synthesis 86: Airport Operator Options for Delivery of FBO Services (2018)
7. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 1: Guidebook (2018)
8. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 2: Technical Appendices (2018)
9. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 3: Research Report (2018)
10. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 4: Summary of AIP Grant Assurance Requirements (2018)
11. ACRP Synthesis 30: Airport Insurance Coverage and Risk Management Practices (2011)
12. ACRP Legal Research Digest 37: Legal Issues Relating to Airports Promoting Competition (2019)
13. ACRP Report 58: Airport Industry Familiarization and Training for Part-Time Airport Policy Makers (2011)

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14. ACRP Report 59: Information Technology Systems at Airports—A Primer (2012)
15. ACRP Synthesis 19: Airport Revenue Diversification (2010)
16. ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook (2012)
17. ACRP Synthesis 56: Understanding the Value of Social Media at Airports for Customer Engagement (2014)
18. ACRP Legal Research Digest 30: Contract Risk Management for Airport Agreements (2016)
19. ACRP Legal Research Digest 23: A Guide for Compliance with Grant Agreement Obligations to Provide Reasonable Access to an AIP-Funded Public Use General Aviation Airport (2015)
20. ACRP Synthesis 39: Airport Wildlife Population Management (2013)
21. ACRP Synthesis 27: Airport Self-Inspection Practices (2011)
22. ACRP Synthesis 41: Conducting Aeronautical Special Events at Airports (2013)
23. ACRP Report 77: Guidebook for Developing General Aviation Airport Business Plans (2012)
24. ACRP Report 47: Guidebook for Developing and Leasing Airport Property (2011)
25. ACRP Legal Research Digest 2: Theory and Law of Airport Revenue Diversion (2008)
26. ACRP Legal Research Digest 7: Airport Governance and Ownership (2009)
27. ACRP Report 150: NextGen for Airports, Volume 5: Airport Planning and Development (2017)
28. ACRP Report 159: Pavement Maintenance Guidelines for General Aviation Airport Management (2016)
29. ACRP Report 138: Preventive Maintenance at General Aviation Airports, Volume 1: Primer (2015)
30. ACRP Report 113: Guidebook on General Aviation Facility Planning (2014)
31. AC 150/5070-7 - The Airport System Planning Process https://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5070-7
32. AC 150/5070-6B - Airport Master Plans https://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5070-6
33. AC 150/5050-4A - Community Involvement in Airport Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036772
34. AC 150/5360-13A - Airport Terminal Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1033908
35. AC 150/5000-17 - Critical Aircraft and Regular Use Determination https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1031186
36. AC 150/5300-13A - Airport Design https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1020359
37. AC 150/5100-13C - Development of State Aviation Standards for Airport Pavement Construction https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036953
38. AC 150/5340-18G - Standards for Airport Sign Systems https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035909
39. AC 150/5340-1M - Standards for Airport Markings https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035908
40. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/
41. FAA Order 5050.4B, National Environmental Policy Act (NEPA) Implementing Instructions for Airport Projects https://www.faa.gov/airports/resources/publications/orders/environmental_5050_4/
42. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
43. FAA Order 5200.11 - FAA Airports (ARP) Safety Management System (SMS) https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/323070
44. FAA Order 5500.1 - Passenger Facility Charge https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/12947

Airport Environmental Planning

1. ACRP Legal Research Digest 22: The Role of the Airport Sponsor in Airport Planning and Environmental Reviews of Proposed Development Projects Under the National Environmental Policy Act (NEPA) and State Mini-NEPA Laws (2014)
2. ACRP Synthesis 17: Approaches to Integrating Airport Development and Federal Environmental Review Processes (2009)
3. ACRP Synthesis 66: Lessons Learned from Airport Sustainability Plans (2015)
4. ACRP Research Report 209: Integrating Sustainability Planning and the Environmental Review Process (2019)
5. ACRP Synthesis 44: Environmental Management System Development Process (2013)
6. ACRP Report 15: Aircraft Noise: A Toolkit for Managing Community Expectations (2009)

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7. ACRP Report 84: Guidebook for Preparing Airport Emissions Inventories for State Implementation Plans (2013)
8. ACRP Research Report 168: Runway Protection Zones (RPZs) Risk Assessment Tool Users' Guide (2016)
9. ACRP Report 43: Guidebook of Practices for Improving Environmental Performance at Small Airports (2010)
10. ACRP Web-Only Document 1: Analysis and Recommendations for Developing Integrated Airport Information Systems (2008)
11. ACRP Synthesis 16: Compilation of Noise Programs in Areas Outside DNL 65 (2009)
12. ACRP Report 149: Improving Ground Support Equipment Operational Data for Airport Emissions Modeling (2015)
13. ACRP Web-Only Document 36: Enhanced AEDT Modeling of Aircraft Arrival and Departure Profiles, Volume 2: Research Report (2018)
14. ACRP Report 7: Aircraft and Airport-Related Hazardous Air Pollutants: Research Needs and Analysis (2008)
15. ACRP Web-Only Document 26: Methodology to Improve AEDT Quantification of Aircraft Taxi/Idle Emissions (2016)
16. ACRP Web-Only Document 11: A Comprehensive Development Plan for a Multimodal Noise and Emissions Model (2010)
17. ACRP Synthesis 24: Strategies and Financing Opportunities for Airport Environmental Programs (2011)
18. ACRP Research Report 155: Guidebook for Advanced Computerized Maintenance Management System Integration at Airports (2018)
19. ACRP Research Report 195: Best Practices for Airport Obstruction Management Guidebook (2019)
20. ACRP Research Report 183: User Guides for Noise Modeling of Commercial Space Operations—RUMBLE and PCBoom (2018)
21. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/
22. FAA Order 5050.4B, National Environmental Policy Act (NEPA) Implementing Instructions for Airport Projects https://www.faa.gov/airports/resources/publications/orders/environmental_5050_4/
23. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
24. AC 150/5300-13A - Airport Design https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1020359
25. AC 150/5100-13C - Development of State Aviation Standards for Airport Pavement Construction https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036953
26. AC 150/5340-18G - Standards for Airport Sign Systems https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035909
27. AC 150/5340-1M - Standards for Airport Markings https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035908

Airport Finance

1. ACRP Legal Research Digest 2: Theory and Law of Airport Revenue Diversion (2008)
2. ACRP Web-Only Document 5: Development of a Guidebook for Managing Small Airports (2009)
3. ACRP Research Report 16: Guidebook for Managing Small Airports—Second Edition (2019)
4. ACRP Synthesis 1: Innovative Finance and Alternative Sources of Revenue for Airports (2007)
5. ACRP Synthesis 19: Airport Revenue Diversification (2010)
6. ACRP Report 90: Impact of Regulatory Compliance Costs on Small Airports (2013)
7. ACRP Report 121: Innovative Revenue Strategies—An Airport Guide (2015)
8. ACRP Research Report 213: Estimating Market Value and Establishing Market Rent at Small Airports (2020)
9. ACRP Report 66: Considering and Evaluating Airport Privatization (2012)
10. ACRP Research Report 215: Transportation Network Companies (TNCs): Impacts to Airport Revenues and Operations—Reference Guide (2020)
11. ACRP Research Report 176: Generating Revenue from Commercial Development On or Adjacent to Airports (2017)
12. ACRP Report 49: Collaborative Airport Capital Planning Handbook (2011)
13. ACRP Report 120: Airport Capital Improvements: A Business Planning and Decision-Making Approach (2014)
14. ACRP Report 47: Guidebook for Developing and Leasing Airport Property (2011)
15. ACRP Synthesis 30: Airport Insurance Coverage and Risk Management Practices (2011)
16. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/

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17. FAA Order 5050.4B, National Environmental Policy Act (NEPA) Implementing Instructions for Airport Projects https://www.faa.gov/airports/resources/publications/orders/environmental_5050_4/
18. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
19. FAA Order 5500.1 - Passenger Facility Charge https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/12947

Airport Global Networks

1. ACRP Research Report 161: Guidelines for Improving Airport Services for International Customers (2016)
2. ACRP Web-Only Document 46: Recovering International Recyclables from In-Flight Service (2020)
3. ACRP Report 61: Elimination or Reduction of Baggage Recheck for Arriving International Passengers (2012)
4. ACRP Insight Event Conference Proceedings 55: Airport Roles in Reducing Transmission of Communicable Diseases (2019)
5. ACRP Legal Research Digest 24: Sovereign Immunity for Public Airport Operations (2015)
6. ACRP Report 143: Guidebook for Air Cargo Facility Planning and Development (2015)
7. ACRP Report 111: A Guidebook for Airport-Airline Consortiums (2014)
8. ACRP Research Report 190: Common Performance Metrics for Airport Infrastructure and Operational Planning (2018)
9. ACRP Report 68: Strategies for Maintaining Air Service (2015)
10. ACRP Report 57: The Carbon Market: A Primer for Airports (2011)
11. ACRP Research Report 194: Using Disaggregated Socioeconomic Data in Air Passenger Demand Studies (2019)
12. ACRP Report 118: Integrating Aviation and Passenger Rail Planning (2015)
13. ACRP Research Report 191: A Primer to Prepare for the Connected Airport and the Internet of Things (2018)
14. ACRP Report 10: Innovations for Airport Terminal Facilities (2008)
15. ACRP Synthesis 31: Airline and Airline–Airport Consortiums to Manage Terminals and Equipment (2011)
16. ACRP Report 91: Infectious Disease Mitigation in Airports and on Aircraft (2013)
17. ACRP Legal Research Digest 37: Legal Issues Relating to Airports Promoting Competition (2019)
18. ACRP Report 36: Airport/Airline Agreements—Practices and Characteristics (2010)
19. ACRP Report 65: Guidebook for Airport Irregular Operations (IROPS) Contingency Planning (2012)
20. ACRP Report 48: Impact of Jet Fuel Price Uncertainty on Airport Planning and Development (2011)
21. ACRP Report 5: Quarantine Facilities for Arriving Air Travelers: Identification of Planning Needs and Costs (2008)
22. AC 129-3 - Foreign Air Carrier Security https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/22525
23. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/
24. FAA Order 5050.4B, National Environmental Policy Act (NEPA) Implementing Instructions for Airport Projects https://www.faa.gov/airports/resources/publications/orders/environmental_5050_4/
25. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
26. FAA Order 5200.11 - FAA Airports (ARP) Safety Management System (SMS) https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/323070
27. FAA Order 5500.1 - Passenger Facility Charge https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/12947

Airport Design

1. ACRP Report 39: Recommended Guidelines for the Collection and Use of Geospatially Referenced Data for Airfield Pavement Management (2010)
2. ACRP Report 59: Integrating Airport Geographic Information System (GIS) Data with Public Agency GIS (2014)
3. ACRP Research Report 203: Guidelines for Collecting, Applying, and Maintaining Pavement Condition Data at Airports (2019)
4. ACRP Report 96: Apron Planning and Design Guidebook (2013)
5. ACRP Research Report 214: BIM Beyond Design Guidebook (2020)
6. AC 150/5360-13A - Airport Terminal Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1033908

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7. AC 150/5000-17 - Critical Aircraft and Regular Use Determination https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1031186
8. AC 150/5300-13A - Airport Design https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1020359
9. AC 150/5100-13C - Development of State Aviation Standards for Airport Pavement Construction https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036953
10. AC 150/5340-18G - Standards for Airport Sign Systems https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035909
11. AC 150/5340-1M - Standards for Airport Markings https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035908

Airport Advanced Computer Applications

1. ACRP Research Report 155: Guidebook for Advanced Computerized Maintenance Management System Integration at Airports (2018)
2. ACRP Report 128: Alternative IT Delivery Methods and Best Practices for Small Airports (2015)
3. ACRP Research Report 195: Best Practices for Airport Obstruction Management Guidebook (2019)
4. ACRP Research Report 183: User Guides for Noise Modeling of Commercial Space Operations—RUMBLE and PCBoom (2018)
5. ACRP Web-Only Document 11: A Comprehensive Development Plan for a Multimodal Noise and Emissions Model (2010)
6. ACRP Web-Only Document 33: Commercial Space Operations Noise and Sonic Boom Modeling and Analysis (2018)
7. ACRP Web-Only Document 36: Enhanced AEDT Modeling of Aircraft Arrival and Departure Profiles, Volume 1: Guidance (2018)
8. ACRP Web-Only Document 36: Enhanced AEDT Modeling of Aircraft Arrival and Departure Profiles, Volume 2: Research Report (2018)
9. ACRP Synthesis 70: Building Information Modeling for Airports (2016)
10. ACRP Web-Only Document 32: Improving AEDT Noise Modeling of Mixed Ground Surfaces (2017)
11. ACRP Report 50: Improved Models for Risk Assessment of Runway Safety Areas (2011)
12. ACRP Research Report 180: Guidebook for Quantifying Airport Ground Access Vehicle Activity for Emissions Modeling (2017)
13. ACRP Synthesis 98: Simulation Options for Airport Planning (2019)
14. ACRP Web-Only Document 30: Development of a NOx Chemistry Module for EDMS/AEDT to Predict NO2 Concentrations (2017)
15. ACRP Web-Only Document 41: Alternative Jet Fuels Emissions: Quantification Methods Creation and Validation Report (2019)
16. AC 150/5360-13A - Airport Terminal Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1033908
17. AC 150/5000-17 - Critical Aircraft and Regular Use Determination https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1031186
18. AC 150/5300-13A - Airport Design https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1020359
19. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/
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21. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
22. FAA Order 5200.11 - FAA Airports (ARP) Safety Management System (SMS) https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/323070

Airport Administration

1. ACRP Research Report 16: Guidebook for Managing Small Airports—Second Edition (2019)
2. ACRP Conference Proceedings on the Web 24: Challenges to Implementing Successful Land Use Strategies at Airports (2019)
3. ACRP Research Report 213: Estimating Market Value and Establishing Market Rent at Small Airports (2020)
4. ACRP Research Report 192: Airport Management Guide for Providing Aircraft Fueling Services (2019)

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5. ACRP Research Report 184: Executive Summary for the Guidebook on Understanding FAA Grant Assurance Obligations (2018)
6. ACRP Synthesis 86: Airport Operator Options for Delivery of FBO Services (2018)
7. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 1: Guidebook (2018)
8. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 2: Technical Appendices (2018)
9. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 3: Research Report (2018)
10. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 4: Summary of AIP Grant Assurance Requirements (2018)
11. ACRP Synthesis 30: Airport Insurance Coverage and Risk Management Practices (2011)
12. ACRP Legal Research Digest 37: Legal Issues Relating to Airports Promoting Competition (2019)
13. ACRP Report 58: Airport Industry Familiarization and Training for Part-Time Airport Policy Makers (2011)
14. ACRP Report 59: Information Technology Systems at Airports—A Primer (2012)
15. ACRP Synthesis 19: Airport Revenue Diversification (2010)
16. ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook (2012)
17. ACRP Synthesis 56: Understanding the Value of Social Media at Airports for Customer Engagement (2014)
18. ACRP Legal Research Digest 30: Contract Risk Management for Airport Agreements (2016)
19. ACRP Legal Research Digest 23: A Guide for Compliance with Grant Agreement Obligations to Provide Reasonable Access to an AIP-Funded Public Use General Aviation Airport (2015)
20. ACRP Synthesis 39: Airport Wildlife Population Management (2013)
21. ACRP Synthesis 27: Airport Self-Inspection Practices (2011)
22. ACRP Synthesis 41: Conducting Aeronautical Special Events at Airports (2013)
23. ACRP Report 77: Guidebook for Developing General Aviation Airport Business Plans (2012)
24. ACRP Report 47: Guidebook for Developing and Leasing Airport Property (2011)
25. ACRP Legal Research Digest 2: Theory and Law of Airport Revenue Diversion (2008)
26. ACRP Legal Research Digest 7: Airport Governance and Ownership (2009)
27. ACRP Report 150: NextGen for Airports, Volume 5: Airport Planning and Development (2017)
28. ACRP Report 159: Pavement Maintenance Guidelines for General Aviation Airport Management (2016)
29. ACRP Report 138: Preventive Maintenance at General Aviation Airports, Volume 1: Primer (2015)
30. ACRP Report 113: Guidebook on General Aviation Facility Planning (2014)
31. AC 150/5070-7 - The Airport System Planning Process https://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5070-7
32. AC 150/5070-6B - Airport Master Plans https://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5070-6
33. AC 150/5050-4A - Community Involvement in Airport Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036772
34. AC 150/5360-13A - Airport Terminal Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1033908
35. AC 150/5000-17 - Critical Aircraft and Regular Use Determination https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1031186
36. AC 150/5300-13A - Airport Design https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1020359
37. AC 150/5100-13C - Development of State Aviation Standards for Airport Pavement Construction https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036953
38. AC 150/5340-18G - Standards for Airport Sign Systems https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035909
39. AC 150/5340-1M - Standards for Airport Markings https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035908
40. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/
41. FAA Order 5050.4B, National Environmental Policy Act (NEPA) Implementing Instructions for Airport Projects https://www.faa.gov/airports/resources/publications/orders/environmental_5050_4/
42. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
43. FAA Order 5200.11 - FAA Airports (ARP) Safety Management System (SMS) https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/323070
44. FAA Order 5500.1 - Passenger Facility Charge https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/12947

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Airport Planning

1. ACRP Research Report 216: Guidebook for Assessing Collaborative Planning Efforts Among Airport and Public Planning Agencies (2020)
2. ACRP Research Report 190: Common Performance Metrics for Airport Infrastructure and Operational Planning (2018)
3. ACRP Report 49: Collaborative Airport Capital Planning Handbook (2011)
4. ACRP Report 130: Guidebook for Airport Terminal Restroom Planning and Design (2015)
5. ACRP Report 20: Strategic Planning in the Airport Industry (2009)
6. ACRP Report 25: Airport Passenger Terminal Planning and Design, Volume 1: Guidebook (2010)
7. ACRP Report 37: Guidebook for Planning and Implementing Automated People Mover Systems at Airports (2010)
8. ACRP Report 118: Integrating Aviation and Passenger Rail Planning (2015)
9. ACRP Research Report 206: Guidebook on Effective Land Use Compatibility Planning Strategies for General Aviation Airports (2019)
10. ACRP Research Report 16: Guidebook for Managing Small Airports—Second Edition (2019)
11. ACRP Conference Proceedings on the Web 24: Challenges to Implementing Successful Land Use Strategies at Airports (2019)
12. ACRP Research Report 213: Estimating Market Value and Establishing Market Rent at Small Airports (2020)
13. ACRP Research Report 192: Airport Management Guide for Providing Aircraft Fueling Services (2019)
14. ACRP Research Report 184: Executive Summary for the Guidebook on Understanding FAA Grant Assurance Obligations (2018)
15. ACRP Synthesis 86: Airport Operator Options for Delivery of FBO Services (2018)
16. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 1: Guidebook (2018)
17. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 2: Technical Appendices (2018)
18. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 3: Research Report (2018)
19. ACRP Web-Only Document 44: Understanding FAA Grant Assurance Obligations, Volume 4: Summary of AIP Grant Assurance Requirements (2018)
20. ACRP Synthesis 30: Airport Insurance Coverage and Risk Management Practices (2011)
21. ACRP Legal Research Digest 37: Legal Issues Relating to Airports Promoting Competition (2019)
22. ACRP Report 58: Airport Industry Familiarization and Training for Part-Time Airport Policy Makers (2011)
23. ACRP Report 59: Information Technology Systems at Airports—A Primer (2012)
24. ACRP Synthesis 19: Airport Revenue Diversification (2010)
25. ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook (2012)
26. ACRP Synthesis 56: Understanding the Value of Social Media at Airports for Customer Engagement (2014)
27. ACRP Legal Research Digest 30: Contract Risk Management for Airport Agreements (2016)
28. ACRP Legal Research Digest 23: A Guide for Compliance with Grant Agreement Obligations to Provide Reasonable Access to an AIP-Funded Public Use General Aviation Airport (2015)
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31. ACRP Synthesis 41: Conducting Aeronautical Special Events at Airports (2013)
32. ACRP Report 77: Guidebook for Developing General Aviation Airport Business Plans (2012)
33. ACRP Report 47: Guidebook for Developing and Leasing Airport Property (2011)
34. ACRP Legal Research Digest 2: Theory and Law of Airport Revenue Diversion (2008)
35. ACRP Legal Research Digest 7: Airport Governance and Ownership (2009)
36. ACRP Report 150: NextGen for Airports, Volume 5: Airport Planning and Development (2017)
37. ACRP Report 159: Pavement Maintenance Guidelines for General Aviation Airport Management (2016)
38. ACRP Report 138: Preventive Maintenance at General Aviation Airports, Volume 1: Primer (2015)
39. ACRP Report 113: Guidebook on General Aviation Facility Planning (2014)
40. AC 150/5070-7 - The Airport System Planning Process https://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5070-7
41. AC 150/5070-6B - Airport Master Plans https://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.current/documentNumber/150_5070-6
42. AC 150/5050-4A - Community Involvement in Airport Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036772
43. AC 150/5360-13A - Airport Terminal Planning https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1033908
44. AC 150/5000-17 - Critical Aircraft and Regular Use Determination https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1031186

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45. AC 150/5300-13A - Airport Design https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1020359
46. AC 150/5100-13C - Development of State Aviation Standards for Airport Pavement Construction https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1036953
47. AC 150/5340-18G - Standards for Airport Sign Systems https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035909
48. AC 150/5340-1M - Standards for Airport Markings https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1035908
49. FAA Order 5100.38D Airport Improvement Program (AIP) https://www.faa.gov/airports/aip/aip_handbook/
50. FAA Order 5050.4B, National Environmental Policy Act (NEPA) Implementing Instructions for Airport Projects https://www.faa.gov/airports/resources/publications/orders/environmental_5050_4/
51. FAA Order 5190.6B FAA Airport Compliance Manual https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/99721
52. FAA Order 5200.11 - FAA Airports (ARP) Safety Management System (SMS) https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/323070
53. FAA Order 5500.1 - Passenger Facility Charge https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/12947

Other External Resources (Annotated Bibliography)

TRB Publications

ACRP Publications: Reports and Syntheses

http://www.trb.org/Publications/PubsACRPPublications.aspx

- Findings from individual research projects on a range of topics are published in ACRP publications including reports, syntheses of practice, research results digests, web-only documents, legal research digests, and impacts on practice. Reports, often in the form of guidebooks or manuals, are the main product of ACRP research projects. Syntheses are focused on documenting the state of practice in critical areas based on literature reviews and surveys of recent activities. Syntheses also share innovations that are being used by others in the airport industry to solve current problems.

TRB TRID (Transport Research International Documentation)

https://trid.trb.org/

- TRID is a comprehensive resource for transportation research with over 1.25 million references to books, technical reports, conference proceedings, and journal articles. Publications cover all modes and disciplines of transportation. Users can sort by keyword(s) or view references for recent hot topics.

Academic Journals

Journal of Airport Management

https://www.henrystewartpublications.com/jam

- The *Journal of Airport Management* is a quarterly journal that publishes in-depth, peer-reviewed articles and case studies focused on airport management; finance; development; ground handling; and airport-airline relations and regulations. Each journal issue focuses on the latest research in the field, including detailed analysis, new thinking, lessons learned, and how-to guides applicable to airport management professionals; airlines and regulators; service providers; advisors; and researchers and academics.



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Journal of Air Transport Management

https://www.journals.elsevier.com/journal-of-air-transport-management

- The *Journal of Air Transport Management* is an international journal offering research articles and commentary focused on the major economic, management, and policy issues in the air transport industry. The journal is intended for both practitioners and academics and covers all major sectors of the air transport industry, including airlines, airports, and air traffic management. The journal provides analyses of topic areas, including policy; regulation and law; strategy; operations; marketing; economics and finance; and sustainability as well as industry developments and trends.

Collegiate Aviation Review International (CARi)

https://ojs.library.okstate.edu/osu/index.php/CARI/index

- CARi is a peer-reviewed journal of the University Aviation Association (UAA), focused on publishing scholarly articles from academia in the areas of aviation, aerospace, and aeronautics. The journal includes original research, position papers, literature reviews, book reviews, and editorials as well as proceedings from the UAA Annual Conference.

Journal of Aviation/Aerospace Education and Research (JAAER)

https://commons.erau.edu/jaaer/

- JAAER is a peer-reviewed journal for educators, researchers, and professionals in the aviation and aerospace industry. The journal covers topics relevant to colleges and universities, with an emphasis on how the aviation and aerospace community influence the educational process as well as the effect of education on the industry. Articles are focused on aviation/aerospace education research, curriculum development, innovative methods of instruction, industry research, and partnerships between education and industry.

Trade Publications

Airport Business

https://www.aviationpros.com/magazine/5409cf3dc827f1be4bb4840f

- *Airport Business* is a magazine targeted to professionals who manage airports, airport-based businesses, and corporate flight facilities. Each issue includes case studies of what others are doing successfully in the North American air transportation system, along with industry news and expert analysis.

Airport Improvement

https://airportimprovement.com/

- *Airport Improvement* is focused on airport improvements, processes, and the people involved in these projects. Each issue of Airport Improvement features case studies of recently completed project work at airports (e.g., general aviation projects, airside projects, security projects, etc.). Each case study includes information on project details, challenges, lessons learned, and potential applications to future projects.



Airport Magazine

https://airportmagazine.net/

- *Airport Magazine* is published by the AAAE, the largest professional organization for airport executives and airport management personnel. Airport Magazine articles focus on a wide variety of critical issues facing today's airports.

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Airports International

https://airportsinternational.keypublishing.com/

- *Airports International* is targeted toward airport industry professionals across the globe. The magazine aims to provide information on the services, finance, and management of airports and their development projects. Each issue includes coverage of the latest industry headlines; technological developments and innovations; and new products and services.

Passenger Terminal Today

https://www.passengerterminaltoday.com/online-magazines

- *Passenger Terminal Today* is focused on providing the latest news for airport operators and airport management professionals working in the airport terminal industry. The magazine is focused on airport terminal design and technology and provides analysis of the latest developments. The magazine's website (www.passengerterminaltoday.com) offers daily news and updates.

Airport Technology

https://www.airport-technology.com/

- *Airport Technology* includes articles focused on projects and trends in airports, facilities, security, technology, and more. Content is geared toward airport executives, procurement professionals, and managers at airports, airlines, and other relevant organizations.

Airport Business (ACI Europe)

http://www.airport-business.com/

- *Airport Business* provides air transport news published by Airports Council International (ACI) Europe. The magazine includes news and interviews that cover topics including airport development, operations, commercial, security, and government aviation policy.

Airport Experience News

https://airportxnews.com/

- *Airport Experience News* provides coverage on issues relevant to companies that do business in airports. Airport News provides news and information geared toward individuals working in food and beverage, retail, development, duty-free, and passenger services at airports.

Security magazine (Airports)

https://www.securitymagazine.com/keywords/airport%20security

- *Security* magazine provides cybersecurity professionals with information on important enterprise security issues, including risk management, cybersecurity, and leadership and management. The magazine addresses a variety of business sectors, including transportation, construction, hospitality, and more.

Business Airport International

https://www.businessairportinternational.com/

- *Business Airport International* provides news, interviews, and other information geared toward business airports, FBOs, and ground handling partners. Each issue includes articles that have a unique location focus, case study, region or city guide, and more.



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Professional Organization Training Materials

AAAE: Study materials for AAE, CM, AMF, AFO, and ACE programs

https://www.aaae.org/studymaterials

- AAAE offers several different accreditations and certifications to members. Students and recent graduates may be interested in becoming Certified Members (CMs). The CM consists of a multiple-choice exam that examines a candidate's comprehensive knowledge of airport management. Individuals working in airport management who are looking to further their career may be interested in completing the Accredited Airport Executive Program (AAE), which includes a multiple-choice exam, research paper or case study, and an interview with a panel of AAEs. The Airport Certified Employee (ACE) program offers students and individuals in the aviation industry the opportunity to earn a certification in specific disciplines including Law Enforcement; Airport Finance; Airfield Communications; Airfield Lighting Maintenance; Airport Operations; Airport Safety and Management Systems; and Airport Security. Other accreditations and certifications offered by AAAE include the Airport Master Firefighter (AMF) program and Airport Fire Officer (AFO) program. AAAE offers study materials for each of the certification programs.

ACI Global Training Programs

https://aci.aero/global-training/

- ACI offers a variety of in-house and web-based training programs for airport management and operations professionals. Courses and programs are offered in topic areas including the following:
  - **Customer experience**—courses on how to provide a pleasant and seamless airport experience to customers
  - **Economics**—courses on financial management; business development and planning; revenue generation; and more to help airports make long-term planning decisions
  - **Environment**—courses focused on ways to reduce the environmental impact of airports, such as energy management and carbon management
  - **Leadership and management**—courses to develop and enhance airport professionals' skills in areas including project management; communications and public relations; leadership; and enterprise risk management
  - **Operational and technical**—courses focused on different aspects of the airport including terminal and airside, and airport business operations
  - **Safety**—courses to help airport operators and executives ensure the safety of their customers and employees, such as learning about safety management systems; airside safety and operations; and emergency planning and crisis management
  - **Security**—courses to provide airports with up to date knowledge to ensure compliance with current and new security regulations
  - **Cybersecurity**—courses for airport managers and specialists focused on the security of airport systems, networks, and data

Academic Non-Degree Programs





University of California (UC) Berkeley Institute of Transportation Studies Airport Planning Short Course

https://its.berkeley.edu/nextor/airportcourse

- The UC Berkeley Institute of Transportation Studies offers a 4-week course led by practitioners in the industry. The course provides an overview of airport planning and design. It is

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intended for airport engineers, planners, and airport and airline management personnel as well as individuals involved in teaching transportation engineering courses. A range of topics are covered in the course, including airport and airline economics and demand forecasting; airport capacity analysis; environmental planning; airfield design; and passenger terminal planning.

Textbooks and Additional Materials

Prather, D. (2015). *Airport Management*. Aviation Supplies & Academics, Inc.

https://www.asa2fly.com/Airport-Management-P2295.aspx

- This textbook provides a comprehensive overview and insight into all aspects of airports, with an emphasis on how to practically apply this knowledge for successful management and operation of airports. *Airport Management* provides students with an introduction to a career in airports. The textbook is structured with learning objectives, key terms, questions for review, and additional suggested resources. Case studies and scenarios that highlight contemporary issues faced by airport professionals on the job are also included.

Young, S. B., and Wells, A. T. (2019). *Airport Planning & Management* (7th ed.). McGraw-Hill Education.

https://www.amazon.com/Airport-Planning-Management-Seventh-Young/dp/1260143325/

- This textbook covers important aspects of airport infrastructure, ranging from airfield and runway planning and management to air traffic control and terminal and security systems. The 7th edition of *Airport Planning & Management* explains the FAA's National Plan of Integrated Airport Systems (NPIAS) and relevant legislation and regulations, including FAR Part 139, and covers cutting-edge concepts in airport planning and management, such as automation, enhanced security, and analytics.

Price, J. C. & Forrest, J. S. (2016). *Practical Aviation Security: Predicting and Preventing Future Threats* (3rd ed.). Butterworth-Heinemann.

https://www.elsevier.com/books/practical-aviation-security/price/978-0-12-804293-9

- This textbook is a comprehensive guide to the aviation security system, with a focus on both current, cutting-edge technologies that are shaping the future in aviation and the critical historical events; policies; policymakers; and terrorists and criminal acts leading up to the present day. The textbook is intended for students looking to enter into the industry and professionals currently working in airport security or other aviation management roles. It provides readers with knowledge of how to implement effective security programs, meet international guidelines, and protect facilities of any size.

Horonjeff, R., McKelvey, F., Sproule, W., and Young, S. B. (2010). *Planning and Design of Airports* (5th ed.). McGraw-Hill Education.

https://www.amazon.com/Planning-Design-Airports-Robert-Horonjeff/dp/0071446419

- This textbook provides guidance on all aspects of planning, design, engineering, and renovating airports and terminals. *Planning and Design of Airports* includes current practices, techniques, and innovations in airport planning and design, including coverage of the latest technologies in aircraft and air traffic management, and passenger processing.

Ashford, N. J., Mumayiz, S., and Wright, P. H. (2011). *Airport Engineering: Planning, Design, and Development of 21st Century Airports* (4th ed.). John Wiley & Sons, Inc.

https://onlinelibrary.wiley.com/doi/book/10.1002/9780470950074

- This textbook provides fundamental guidance and best practices for individuals interested in airport engineering and transportation planning. The current edition of *Airport Engineering*

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reflects the shift in the growth of airports globally, along with the wane of construction of new airports in the United States, with a focus on the role of the International Civil Aviation Organization (ICAO).

Ashford, N. J., Coutu, P., and Beasley, J. R. (2012). *Airport Operations* (3rd ed.). McGraw-Hill Education.

https://www.mhprofessional.com/9780071775847-usa-airport-operations-third-edition-group

- This textbook provides strategies and best practices for efficiently managing airport functions. *Airport Operations* is geared toward students and educators, along with aviation managers, airport planners and designers, and other airport personnel. The textbook includes coverage of topics such as operational readiness, passenger terminal operations, airport security, airport technical services, airport access, operational administration, airport safety management systems, and sustainable development.

Stephen Quilty Relevant Research

Quilty, S. M. (2005). Skill and Trait Identification for Entry-Level Airport Operations and Management Personnel. *International Journal of Applied Aviation Studies, 5*(1), 183–194.

- The author surveyed 106 airport managers and airfield operations personnel to identify the important skills, traits, and capabilities required for individuals working in airport operations and management positions. The findings of this paper are intended to inform the development of curricula for aviation management and operations programs at colleges and universities as well as the development of tailored training materials.

Quilty, S. M. (2004). Airport Management Program and Curriculum Issues at 2- and 4-Year Aviation Colleges and Universities. *Collegiate Aviation Review International, 22*(1), 57–70.

- This paper outlines issues regarding the development of an airport/aviation management curriculum for 2- and 4-year colleges and university programs. Issues discussed include the courses that should make up the core of a program, the content and learning objectives of these courses, and how to meet marketplace demands for graduates.

David Byers Dissertation

Byers, D. A. (2004). *The Making of the Modern Airport Executive: Causal Connections among Key Attributes in Career Development, Compromise, and Satisfaction in Airport Management*. ProQuest Dissertations Publishing, 3127101.

- The author identified specific career development attributes of contemporary senior-level airport executives and evaluated the relationship of these attributes to the level of satisfaction airport executives have in their career choice. Attribute sets that were examined included early aviation interests, health factors, psychological factors, demographic factors, formal education, and other aviation-related experiences. A hypothesized causal career path model was developed that expressed the relationship these attributes had on the satisfaction with airport management as a career choice. The results of the path analysis indicated that four attribute sets (early aviation interests, health factors, formal education, and other aviation-related experiences) had both direct and indirect significant effects on career satisfaction.



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ACRP. (2017). Impacts on Practice. ACRP and Higher Education: Building the Next Generation of Airport Professionals. February. http://www.trb.org/ACRP/Blurbs/175653.aspx.

Byers, D. A. (2004). *The Making of the Modern Airport Executive: Causal Connections among Key Attributes in Career Development, Compromise, and Satisfaction in Airport Management*. ProQuest Dissertations Publishing, 3127101.

Byers, D. A. (2016). The Aviation Workforce of Tomorrow: Where Are They Needed—and Where Will They Come from? *TR News,* No. 304, July–August 2016. http://onlinepubs.trb.org/Onlinepubs/trnews/trnews304feature.pdf.

Cronin, C. B., A. Alexander, E. Majumdar, C. Riches, J. Jenkins, S. Van Beek, A. Bisker, B. Heinen, and C. Lewis. (2016). *ACRP Web-Only Document 28: Identifying and Evaluating Airport Workforce Requirements*. Transportation Research Board, Washington, DC.

ICF. (2018). *ACRP Research Report 186: Guidebook on Building Airport Workforce Capacity*. Transportation Research Board, Washington, DC.

Phillips, E. D., J. Ruiz, and H. Mehta. (2006). Industry Members Evaluate the Strengths and Weaknesses of Aviation Management Graduates. *Collegiate Aviation Review International, 24*.

Prather, C. D., and California Baptist University. (2019). *ACRP Synthesis 103: Promoting Aviation Career Education in High Schools and Community Colleges.* Transportation Research Board, Washington, DC.

Quilty, S. M. (2005). Study Results on Knowledge Requirements for Entry-Level Airport Operations and Management Personnel. *Journal of Air Transportation, 10*(3), 100-122.

Young, S. B. (2010). *ACRP Synthesis 18: Aviation Workforce Development Practices.* Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/14368.

Young, S. B., M. Price, Columbus Regional Airport Authority, and HNTB, Inc. (2013). *ACRP Report 75: Airport Leadership Development Program.* Transportation Research Board of the National Academies, Washington, DC. https://doi.org/10.17226/22574.

Wensveen, J. (2017). How to solve aviation's talent gap problem. *International Airport Review*. https://www.internationalairportreview.com/article/26445/aviation-talent-gap-john-wensveen/.

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APPENDIX A

Database of Programs

A fundamental component of the scope of this project is identifying and evaluating the characteristics of various institutions offering a baccalaureate aviation management degree. General information, such as the institution's location, form of ownership (i.e., whether public or private), organizational structure, and tuition was collected. For most programs, detailed curriculum data was investigated to determine the number of courses and credit hours committed to aviation-related and airport-specific topics. These courses were identified by course number, credit hours, course description, and categorized along generalized aviation-related topical areas including the following:

- Management and/or Administration
- Operations
- Planning
- Development
- Economics and Finance
- Marketing and Communications
- FBO and Corporate Management
- ATC and Airspace
- Safety and/or Security
- Environment
- Regulations and Law
- Capstone or Culminating Experience
- Career Development
- Internship (Required)
- Other Technical Courses

These aviation-related courses were also identified if they were specific to airports (if the term airport appeared in the course title or course description). The cumulative credit hours for aviation-related and airport-specific courses in the curriculum were compared with the total required credit hours for graduation to assess the depth and breadth of the students' understanding of airports as a fundamental piece of the National Air Transportation System.

Aviation Management Programs

Two primary resources were used to identify institutions offering a degree in aviation management. These programs were found among the listings of 4-year colleges from the UAA's 2019 Aviation Directory and the Aircraft Owners and Pilots Association's (AOPA) 2020 Collegiate Aviation Program Directory. From these two directories, a total of 96 separate colleges and universities were identified.

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These programs were assessed to determine if their curriculum included at least one airport-specific course. A total of 65 programs (68%) met the initial screening criteria for a more detailed investigation of the program's curriculum. Program curriculum maps that describe the sequence that courses are taken were used to compare the representative programs.

Other identifying characteristics included 22 aviation management programs that have been accredited by the AABI. See Appendix B for more information on accreditation standards. Also, 26 programs were identified as having a current affiliation with the AAAE through an established student chapter.

Table A-1 presents a summary of the 65 aviation management programs that were included in the curriculum evaluation.

Curriculum Summary

The curriculum of each program was reviewed to determine the total number of credit hours required for graduation as well as aviation-related and airport-specific coursework. Open (unrestricted) elective hours were also included in the evaluation to assess the opportunity for enhancing airport-specific coursework. Table A-2 presents a summary of curriculum of the 65 programs.

Tuition Summary

Tuition is the cost of attending the program each semester and is usually represented as a flat fee based on the student taking a minimum of 12 credit hours. Tuition beyond the minimum course hours is based on a per-credit-hour rate (a standard 3-credit-hour course will be three times the per-credit-hour rate). Some programs may not have a flat semester tuition fee but each course is added a la carte to the cost of attendance. Most programs will have additional fees for each course broken out to pay for course expenses and other university budget items.

Most public institutions are subsidized by federal and state aid programs and other resources and thus, have lower tuition rates than private institutions that do not have access to those benefits. Also, public institutions subsidized by state tax revenues will generally have much higher tuition rates for non-resident students since state funds are used for the exclusive benefit of in-state residents. Many institutions participate in regional tuition reciprocity or discount programs to offset out-of-state tuition rates. As an example, the Midwest Student Exchange Program has nine member states, including Illinois, Indiana, Kansas, Minnesota, Missouri, Nebraska, North Dakota, Ohio, and Wisconsin. Non-residents attending public institutions in these states pay 150% of the in-state tuition compared with the full out-of-state rate.

Tuition rates at most private institutions follow the same conventions (flat semester rate) as public institutions; however, the rates are significantly higher, and these institutions do not differentiate students by state of residency. Table A-3 presents a summary of tuition rates sorted from low to high and distinguished by public versus private institutions.

Airport-Specific Course Content

Airport-specific courses offered by the various aviation management programs were examined by topic and content. In many cases, programs offered only one or two airport courses, usually entitled airport management, airport operations, or airport administration. Other programs included more airport-related courses, some with airport management concentrations within the degree major.



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Table A-1. Aviation management program summary.

ID	Institution	City	State	Public or Private	Degree	Major (Concentration)	AABI Accreditation	AAAE Student Chapter
ASU	Arizona State University (Polytechnic) *	Tempe	AZ	Pub	BS	Aeronautical Management Technology (Air Transportation)	X	
AUB	Auburn University *	Auburn	AL	Pub	BS	Aviation Management	X	
BGSU	Bowling Green State University *	Bowling Green	OH	Pub	BS	Aviation: Aviation Management and Operations		
CBU	California Baptist University *	Riverside	CA	Pvt	BS	Aviation Management		X
CSULA	California State University - Los Angeles *	Los Angeles	CA	Pub	BS	Aviation Administration		X
CWU	Central Washington University *	Ellensburg	WA	Pub	BS	Aviation Management		X
DSU	Delta State University *	Cleveland	MS	Pub	BCA	Aviation Management		
EKU	Eastern Kentucky University *	Richmond	KY	Pub	BS	Aviation (Aerospace Management Concentration)		X
ERAUDB	Embry-Riddle Aeronautical University	Daytona Beach	FL	Pvt	BS	Aviation Business Administration	X	X
ERAUPC	Embry-Riddle Aeronautical University	Prescott	AZ	Pvt	BS	Aviation Business Administration	X	X
FSC	Farmingdale State College *	Farmingdale	NY	Pub	BS	Aviation Administration	X	X
FIT	Florida Institute of Technology *	Melbourne	FL	Pvt	BS	Aviation Management	X	X
FMU	Florida Memorial University *	Miami Gardens	FL	Pvt	BS	Aviation Management	X	
HU	Hampton University *	Hampton	VA	Pvt	BS	Aviation Management	X	
ISU	Indiana State University *	Terre Haute	IN	Pub	BS	Aviation Management		X
JAX	Jacksonville University *	Jacksonville	FL	Pvt	BS	Aviation Management	X	
KSU	Kansas State University (Polytechnic) *	Salina	KS	Pub	BS	Aeronautical Technology (Airport Management Option)		X
KSOH	Kent State University *	Kent	OH	Pub	BS	Aeronautics/Aviation Management	X	X
LU	Lewis University *	Romeoville	IL	Pvt	BS	Aviation Administration		X
LTU	Louisiana Tech University *	Ruston	LA	Pub	BS	Aviation Management	X	
MSUD	Metropolitan State University of Denver *	Denver	CO	Pub	BS	Aviation & Aerospace Management		X
MGSU	Middle Georgia State University (MCN & ESM)	Macon	GA	Pub	BS	Aviation Science and Management		
MTSU	Middle Tennessee State University *	Murfreesboro	TN	Pub	BS	Aviation Management	X	
MKT	Minnesota State University - Mankato *	Mankato	MN	Pub	BS	Aviation Management		
OKSU	Oklahoma State University *	Stillwater	OK	Pub	BS	Aerospace Administration and Operations: Aviation	X	X
PU	Purdue University *	Lafayette	IN	Pub	BS	Aviation Management	X	X
RMC	Rocky Mountain College *	Billings	MT	Pvt	BS	Aviation Management	X	
SLU	Saint Louis University *	St. Louis	MO	Pvt	BS	Aviation Management	X	
SIU	Southern Illinois University *	Carbondale	IL	Pub	BS	Aviation Management		
SNHU	Southern New Hampshire University *	Manchester	NH	Pvt	BS	Aviation Management		X

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TSU	Texas Southern University *	Houston	TX	Pub	BS	Aviation Science Management		X
OSU	The Ohio State University *	Columbus	OH	Pub	BS	Business Administration: Aviation Management	X	X
UAK	University of Alaska - Anchorage *	Anchorage	AK	Pub	BS	Aviation Technology (Aviation Management Emphasis)		
UCM	University of Central Missouri *	Warrensburg	MO	Pub	BS	Aviation Management – Airport Management Option	X	X
DBQ	University of Dubuque *	Dubuque	IA	Pvt	BS	Aviation Management	X	X
UNO	University of Nebraska - Omaha *	Omaha	NE	Pub	BS	Aviation	X	
UND	University of North Dakota *	Grand Forks	ND	Pub	BBA	Airport Management		X
UOK	University of Oklahoma *	Norman	OK	Pub	BS	Aviation Management	X	X
UVU	Utah Valley University *	Provo	UT	Pub	BS	Aviation Management	X	X
VC	Vaughn College *	Flushing	NY	Pvt	BS	Airport Management		X
WMU	Western Michigan University *	Kalamazoo	MI	Pub	BS	Aviation Management and Operations	X	X
AVU	Averett University	Danville	VA	Pvt	BS	Aerospace Management		
BU	Baylor University	Waco	TX	Pvt	BS	Aviation Sciences – Aviation Administration		
CAU	California Aeronautical University	Bakersfield	CA	Pvt	BS	Aviation Business Administration		
CUNY	CUNY Aviation Institute at York College	New York	NY	Pub	BS	Aviation Management		
DESU	Delaware State University	Dover	DE	Pub	BS	Aviation Management		
EMU	Eastern Michigan University	Detroit	MI	Pub	BS	Aviation Management Technology		
ENM	Eastern New Mexico University	Portales	NM	Pub	BS	Aviation Science		
ECSU	Elizabeth City State University	Elizabeth City	NC	Pub	BS	Aviation Science		
ERAU DB	Embry-Riddle Aeronautical University	Daytona Beach	FL	Pvt	BS	Aviation Business Administration		
EU	Everglades University	Boca Raton	FL	Pvt	BS	Aviation/Aerospace		
FSUWV	Fairmont State University	Clarksburg	WV	Pub	BS	Aviation Technology		
GRC	Green River College	Auburn	WA	Pub	BAS	Aeronautical Science		
LeTU	LeTourneau University	Longview	TX	Pvt	BS	Aviation Management		
MU	Marywood University	Wilkes-Barre	PA	Pvt	BS	Aviation Management		
PLK	Polk State College	Lakeland	FL	Pub	BS	Aerospace Sciences		
QU	Quincy University	Quincy	IL	Pvt	BS	Aviation Management		
SDCC	San Diego Christian College	San Diego	CA	Pvt	BS	Aviation		
TNSU	Tennessee State University	Nashville	TN	Pub	BS	Aeronautical and Industrial Technology (Aviation)		
TXAM	Texas A&M University - Central Texas	Killeen	TX	Pub	BS	Aviation Science		
TESC	Thomas Edison State College	Trenton	NJ	Pub	BS	Aviation Management		
UMES	University of Maryland Eastern Shore	Salisbury	MD	Pub	BS	Aviation Science		
UNK	University of Nebraska Kearney	Kearney	NE	Pub	BS	Aviation Systems Management		
UNT	University of North Texas	Denton	TX	Pub	BS	Aviation Logistics		X
WLM	Wilmington University	Wilmington	DE	Pvt	BS	Business Administration (Professional Aeronautics)		

* Representative programs.

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Table A-2. Aviation management program curriculum summary.

ID	Institution	Public/ Private	Total Credit Hours	Aviation-Related Credit Hours	Airport-Specific Credit Hours	Open Electives	Internship Required
KSU	Kansas State University (Polytechnic) *	Pub	120	43	**27**	-	No
CWU	Central Washington University *	Pub	120	58	**18**	-	No
FIT	Florida Institute of Technology *	Pub	123	53	**18**	2	No
FSC	Farmingdale State College *	Pvt	122	45	**15**	-	No
HU	Hampton University *	Pub	120	68	**15**	-	No
UCM	University of Central Missouri *	Pub	120	39	**13**	6	Yes
BGSU	Bowling Green State University *	Pub	121	39	**12**	12	Yes
CSULA	California State University - Los Angeles *	Pub	120	36	**12**	9	No
ERAUWW	Embry-Riddle Aeronautical University Worldwide	Pvt	120	36	**12**	-	No
MGSU	Middle Georgia State University (MCN & ESM) *	Pvt	120	54	**12**	24	No
MTSU	Middle Tennessee State University *	Pub	120	60	**12**	-	Yes
SNHU	Southern New Hampshire University *	Pvt	121	43	**10**	18	Yes
CBU	California Baptist University *	Pvt	126	42	**9**	22	No
ERAUPC	Embry-Riddle Aeronautical University (PRC) *	Pvt	120	30	**9**	18	No
EU	Everglades University	Pub	123	72	**9**	-	No
PU	Purdue University *	Pvt	120	55	**9**	9	No
TXAM	Texas A&M University - Central Texas	Pub	120	42	**9**	-	No
UNO	University of Nebraska - Omaha *	Pub	120	60	**9**	-	Yes
WMU	Western Michigan University *	Pvt	123	53	**7**	6	No
AUB	Auburn University *	Pub	126	31	**6**	20	No
AVU	Averett University	Pub	120	30	**6**	-	Yes
CUNY	CUNY Aviation Institute at York College	Pub	120	60	**6**	-	No
EKU	Eastern Kentucky University *	Pub	120	45	**6**	14	No
FSUWV	Fairmont State University	Pub	120	45	**6**	-	No
ISU	Indiana State University *	Pub	120	48	**6**	27	No
JAX	Jacksonville University *	Pub	122	40	**6**	-	No
KSOH	Kent State University *	Pvt	120	51	**6**	3	No
LU	Lewis University *	Pvt	128	56	**6**	-	No
MU	Marywood University	Pub	120	24	**6**	-	No
MSUD	Metropolitan State University of Denver *	Pvt	121	63	**6**	6	No
MKT	Minnesota State University - Mankato *	Pub	120	31	**6**	-	No
PLK	Polk State College	Pub	120	45	**6**	-	No
SIU	Southern Illinois University *	Pub	120	48	**6**	-	No
OSU	The Ohio State University *	Pub	120	32	**6**	6	No
UMES	University of Maryland Eastern Shore	Pvt	120	64	**6**	-	No
UND	University of North Dakota *	Pub	121	31	**6**	5	No
VC	Vaughn College *	Pub	127	44	**6**	-	Yes
GRC	Green River College	Pub	120	60	**5**	-	No
CAU	California Aeronautical University	Pub	121	34	**4**	-	Yes
ASU	Arizona State University (Polytechnic) *	Pvt	123	54	**3**	15	Yes
BU	Baylor University	Pub	126	44	**3**	-	No
DESU	Delaware State University	Pvt	120	37	**3**	-	No

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Table A-2. (Continued).

ID	Institution	Public/ Private	Total Credit Hours	Aviation-Related Credit Hours	Airport-Specific Credit Hours	Open Electives	Internship Required
DSU	Delta State University *	Pvt	124	39	**3**	10	No
EMU	Eastern Michigan University	Pvt	124	-	**3**	-	No
ENM	Eastern New Mexico University	Pub	120	-	**3**	-	No
ECSU	Elizabeth City State University	Pub	120	-	**3**	-	No
ERAUDB	Embry-Riddle Aeronautical University (DAB) *	Pub	122	28	**3**	15	No
FMU	Florida Memorial University *	Pub	120	38	**3**	21	No
LeTU	LeTourneau University	Pub	122	-	**3**	-	No
LTU	Louisiana Tech University *	Pvt	121	30	**3**	5	Yes
OKSU	Oklahoma State University *	Pvt	120	57	**3**	13	No
QU	Quincy University	Pub	125	-	**3**	-	No
RMC	Rocky Mountain College *	Pub	121	39	**3**	27	Yes
SLU	Saint Louis University *	Pvt	121	64	**3**	9	Yes
SDCC	San Diego Christian College	Pvt	124	47	**3**	15	No
TNSU	Tennessee State University	Pub	120	46	**3**	9	No
TSU	Texas Southern University *	Pvt	120	46	**3**	12	Yes
TESC	Thomas Edison State College	Pvt	120	-	**3**	-	No
UAK	University of Alaska - Anchorage *	Pub	120	48	**3**	9	No
DBQ	University of Dubuque *	Pub	121	35	**3**	15	No
UNK	University of Nebraska Kearney	Pub	120	-	**3**	-	No
UNT	University of North Texas	Pub	120	39	**3**	4	Yes
UOK	University of Oklahoma *	Pub	125	34	**3**	14	No
UVU	Utah Valley University *	Pub	120	67	**3**	15	No
WLM	Wilmington University	Pvt	120	15	**3**	-	No

* Representative programs.

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Table A-3. Aviation management program tuition summary.

ID	Institution	Public/ Private	Tuition Resident (Semester)	Tuition Resident (Credit Hour)	Tuition Non-Resident (Semester)	Tuition Non-Resident (Credit Hour)
PLK	Polk State College	Pub	1,488	124	5,640	470
ECSU	Elizabeth City State University	Pub	1,500	125	7,500	625
UNT	University of North Texas	Pub	1,500	100	14,160	944
ENM	Eastern New Mexico University	Pub	1,998	269	2,997	352
CWU	Central Washington University	Pub	2,105	421	7,333	1,467
UOK	University of Oklahoma *	Pub	2,394	200	10,085	840
UNO	University of Nebraska - Omaha *	Pub	2,436	229	8,616	718
UAK	University of Alaska - Anchorage *	Pub	2,676	223	9,468	789
UMES	University of Maryland Eastern Shore	Pub	2,709	224	7,914	583
UCM	University of Central Missouri *	Pub	2,820	235	5,700	475
CSULA	California State University - Los Angeles *	Pub	3,384	282	9,324	777
GRC	Green River College	Pub	3,444	287	8,232	686
CUNY	CUNY Aviation Institute at York College	Pub	3,465	305	9,300	620
MTSU	Middle Tennessee State University *	Pub	3,468	300	12,936	1,078
FSC	Farmingdale State College *	Pub	3,535	295	8,490	708
MKT	Minnesota State University - Mankato *	Pub	3,697	289	7,843	622
DSU	Delta State University *	Pub	3,750	313	-	-
FSUWV	Fairmont State University	Pub	3,756	313	8,304	692
TXAM	Texas A&M University - Central Texas	Pub	3,783	192	7,372	614
UND	University of North Dakota *	Pub	4,106	342	12,318	1,027
DESU	Delaware State University	Pub	4,129	344	8,647	721
TNSU	Tennessee State University	Pub	4,212	351	5,892	491
OKSU	Oklahoma State University *	Pub	4,509	376	12,269	1,022
ISU	Indiana State University *	Pub	4,518	327	9,980	706
MSUD	Metropolitan State University of Denver *	Pub	4,575	268	12,212	834
PU	Purdue University *	Pub	4,604	329	9,401	600
EKU	Eastern Kentucky University *	Pub	4,633	386	9,669	806
KSOH	Kent State University *	Pub	4,633	422	8,703	792
BGSU	Bowling Green State University *	Pub	4,639	387	8,633	719
UVU	Utah Valley University *	Pub	4,728	394	14,052	1,171
TESC	Thomas Edison State College	Pub	4,788	399	6,108	509
AUB	Auburn University *	Pub	4,812	401	14,436	1,203
SIU	Southern Illinois University *	Pub	4,819	321	4,819	321
TSU	Texas Southern University *	Pub	4,835	348	11,587	670
KSU	Kansas State University (Polytechnic) *	Pub	5,064	293	9,468	789
EMU	Eastern Michigan University	Pub	5,250	438	12,840	1,070
WMU	Western Michigan University *	Pub	6,047	504	7,559	630
UNK	University of Nebraska Kearney	Pub	6,090	203	13,290	443
MGSU	Middle Georgia State University (MCN & ESM)	Pub	6,336	528	10,008	834
OSU	The Ohio State University *	Pub	6,377	570	16,866	1,444
LTU	Louisiana Tech University *	Pub	7,944	662	11,627	662
ASU	Arizona State University (Polytechnic) *	Pub	10,710	893	28,800	2,400

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Table A-3. (Continued).

ID	Institution	Public/ Private	Tuition Resident (Semester)	Tuition Resident (Credit Hour)	Tuition Non-Resident (Semester)	Tuition Non-Resident (Credit Hour)
MU	Marywood University	Pvt	3,216	1,423	-	-
WLM	Wilmington University	Pvt	4,572	381	-	-
FMU	Florida Memorial University *	Pvt	6,288	524	-	-
EU	Everglades University	Pvt	8,760	730	-	-
CAU	California Aeronautical University	Pvt	8,820	392	-	-
HU	Hampton University *	Pvt	12,475	1,040	-	-
VC	Vaughn College *	Pvt	12,650	840	-	-
RMC	Rocky Mountain College *	Pvt	14,988	1,249	-	-
QU	Quincy University	Pvt	15,580	1,298	-	-
SDCC	San Diego Christian College	Pvt	15,840	1,362	-	-
LeTU	LeTourneau University	Pvt	15,870	578	-	-
LU	Lewis University *	Pvt	16,635	996	-	-
CBU	California Baptist University *	Pvt	16,965	1,305	-	-
DBQ	University of Dubuque *	Pvt	17,035	1,020	-	-
ERAUDB	Embry-Riddle Aeronautical University	Pvt	17,712	1,476	-	-
AVU	Averett University	Pvt	17,725	1,477	-	-
ERAUPC	Embry-Riddle Aeronautical University	Pvt	18,228	1,519	-	-
ERAUWW	Embry-Riddle Aeronautical University	Pvt	18,228	1,519	-	-
FIT	Florida Institute of Technology *	Pvt	19,010	1,170	-	-
JAX	Jacksonville University *	Pvt	19,070	1,275	-	-
BU	Baylor University	Pvt	22,272	1,856	-	-
SLU	Saint Louis University *	Pvt	22,350	1,560	-	-
SNHU	Southern New Hampshire University *	Pvt	30,756	1,282	-	-

* Representative programs.
Sorted by public/private and low to high semester tuition, data current as of 2019.

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Table A-4. Airport-specific course content for aviation management programs.

AIRPORT TOPIC	1st Order	2nd Order	Total Courses	Programs w/Content	Percentage of Programs
Management	38	12	50	46	71%
Planning	35	2	37	34	52%
Operations	23	3	26	25	38%
Design	7	9	16	15	23%
Administration	10	4	14	12	18%
Security	8	2	10	9	14%
Finance	5	4	9	9	14%
Career	5	-	5	3	5%
Environment	4	1	5	5	8%
Marketing	2	1	3	3	5%
Safety	1	2	3	3	5%
Law	2	-	2	2	3%
Technical	2	-	2	1	2%
International	1	-	1	1	2%
TOTAL	**143**	**40**	**183**		

Note: Some courses have multiple topics, such as airport planning and management, and are counted as 1st Order or 2nd Order as listed.

Many courses include more than one topic, such as airport planning and management or airport planning and design, and they appear to be organized around a specific text. Table A-4 includes an analysis of airport-specific course content among the various programs. The 1st order terms reflect the primary topic of a course. The 2nd order terms reflect the secondary topic of a course. For example, 50 courses among 46 programs were found to have a course where the airport management is either a primary (38 courses) or secondary (12 courses) topic. A course in airport management would be considered a course with management as a 1st order topic, while it would be considered a 2nd order topic in a course in airport planning and management.

Relevant Airport/Aviation Curriculum Summary

A database containing the 65 aviation management programs was compiled for use in the analysis of descriptive and comparative factors as well as the detailed evaluation of the curriculum. The analysis identified several specific aviation management programs that offered a significant number of airport-related courses as well those with only one or two courses. Also, as given in Table A-4, there is a wide variety of airport-related course content among these programs. Although a basic information profile was developed for all 65 programs, the 41 representative programs have a detailed map of their curriculum, including the course number and title of each aviation-related and airport-specific course. Table A-5 contains a matrix of relevant airport-specific courses as well as aviation-related courses that have some significance to airport courses offered by each program.



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Table A-5. Relevant airport/aviation curriculum matrix for aviation management programs.

ID	Institution	Representative Program	Aviation-Related Credit Hours	Airport -Specific Credit Hours	Management and/or Administration	Operations	Planning	Development	Economics and Finance	Marketing and Communications	FBO and Corporate Management	ATC and Airspace	Safety and/or Security	Environment	Regulations and Law	Capstone and/or Culminating Experience	Career Development	Internship	Other Technical Courses
ASU	Arizona State University (Polytechnic) *	Y	54	3	C		C					X	X		X	X	X	X	
AUB	Auburn University *	Y	31	6	X		X	X				X	X		X		X		
BGSU	Bowling Green State University *	Y	39	12	X	X	C	C				X	X		X	X		X	
CBU	California Baptist University *	Y	42	9	X	X	C	C	X	X	X		X		X	X			
CSULA	California State University - Los Angeles *	Y	36	12	X	X	X		X				X		X				
CWU	Central Washington University *	Y	58	18	C	X	C	C	C	X			X		X		X	X	
DSU	Delta State University *	Y	35	3	X				X		X	X	X		X				
EKU	Eastern Kentucky University *	Y	39	3	X	C						X	X		X				
ERAUDB	Embry-Riddle Aeronautical University	Y	45	6	X	C			C			X	C		X				
ERAUPC	Embry-Riddle Aeronautical University	Y	28	3	X														
FSC	Farmingdale State College *	Y	30	9	X	X			X					X					
FIT	Florida Institute of Technology *	Y	53	18	X		X	X					X		X		X		X
FMU	Florida Memorial University *	Y	38	3	X							X	X		X	X			
HU	Hampton University *	Y	45	15	X	X	C		C	C		X	X		X	X			
ISU	Indiana State University *	Y	68	15	X	X	X	X	X			X	X		X	X			
JAX	Jacksonville University *	Y	48	6		X	X					X	X		X		X		
KSU	Kansas State University (Polytechnic) *	Y	40	6	X	C	C		X		C	X			X	X			
KSOH	Kent State University *	Y	51	6	X	X					X	X	X		X	X	X		
LU	Lewis University *	Y	43	27	X	X	X	X					X	X	X	X			X
LTU	Louisiana Tech University *	Y	30	3	C	X	C				X	X	X		X		X	X	
MSUD	Metropolitan State University of Denver *	Y	56	6	X				C			X	X		X		X		
MGSU	Middle Georgia State University	Y	54	12	X	C	C	C		C			C	C	C				
MTSU	Middle Tennessee State University *	Y	31	6	X	X							X		X				
MKT	Minnesota State University - Mankato *	Y	63	6	C	X	C						X		X	X			
OKSU	Oklahoma State University *	Y	60	12	X	X	X				X		X		X	X			X
PU	Purdue University *	Y	57	3	C		C		X	X	X		X		X	X			

(*continued on next page*)

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Table A-5. (Continued).

ID	Institution	Representative Program	Aviation-Related Credit Hours	Airport -Specific Credit Hours	Management and/or Administration	Operations	Planning	Development	Economics and Finance	Marketing and Communications	FBO and Corporate Management	ATC and Airspace	Safety and/or Security	Environment	Regulations and Law	Capstone and/or Culminating Experience	Career Development	Internship	Other Technical Courses
RMC	Rocky Mountain College *	Y	32	6	X		C	C		X					X	X			
SLU	Saint Louis University *	Y	55	9	X	X			X				X		X	X			X
SIU	Southern Illinois University *	Y	39	3	C		C				X		X		X		X	X	
SNHU	Southern New Hampshire University *	Y	48	6	X		X		X	X	X	X	X		X			X	
TSU	Texas Southern University *	Y	64	3		C			X		C		X		X	X		X	
OSU	The Ohio State University *	Y	43	10	C	X	C		X		X	X	X		X			X	
UAK	University of Alaska - Anchorage *	Y	46	3	X	X					X	X	X		X	X			
UCM	University of Central Missouri *	Y	48	3	X	X					X	X	X		X				
DBQ	University of Dubuque *	Y	39	13	X	X							X		X		X	X	
UNO	University of Nebraska - Omaha *	Y	31	6	C	C	C				X	X	X		X	X	X		
UND	University of North Dakota *	Y	33	9	X	X	X				X		X		X			X	
UOK	University of Oklahoma *	Y	34	3	C	C							X		X	X	X		
UVU	Utah Valley University *	Y	67	3	X	X			X	X	X	X	X		X	X	X		
VC	Vaughn College *	Y	44	6	X			X				X	X	X	X	X		X	
WMU	Western Michigan University *	Y	53	7	C	C	C				X		X		X	X			
AVU	Averett University	N	n/a	6	X		X												
BU	Baylor University	N	n/a	3	X														
CAU	California Aeronautical University	N	n/a	4	X														
CUNY	CUNY Aviation Institute at York College	N	n/a	6	C		C						X						
DESU	Delaware State University	N	n/a	3	C		C												

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EMU	Eastern Michigan University	N	n/a	3	X														
ENM	Eastern New Mexico University	N	n/a	3		X													
ECSU	Elizabeth City State University	N	n/a	3	C		C												
ERAUW	Embry-Riddle Aeronautical University	N	n/a	12	X		C	C	X	X									
EU	Everglades University	N	n/a	9	X		X						X						
FSUWV	Fairmont State University	N	n/a	6	X		C	C											
GRC	Green River College	N	n/a	5		C	C												
LeTU	LeTourneau University	N	n/a	3	C	C													
MU	Marywood University	N	n/a	6	X								X						
PLK	Polk State College	N	n/a	6	X	X	X												
QU	Quincy University	N	n/a	3	X														
SDCC	San Diego Christian College	N	n/a	3	X														
TNSU	Tennessee State University	N	n/a	3	X														
TXAM	Texas A&M University - Central Texas	N	n/a	9	X	X								X					
TESC	Thomas Edison State College	N	n/a	3	X														
UMES	University of Maryland Eastern Shore	N	n/a	6	X		X												
UNK	University of Nebraska Kearney	N	n/a	3	X														
UNT	University of North Texas	N	n/a	3			X												
WLM	Wilmington University	N	n/a	3	X														

* Representative programs.
Y = Representative “benchmarked” program.
N = A program researched in this study that was not part of the list of representative programs.
X = Course on this topic found in the curriculum.
C = Course that includes multiple topics in the column list (usually affiliated with another C in a given row)
Darker shaded boxes are considered airport-specific courses. (These are purple in the electronic version.)
Lighter shaded boxes are courses that are not airport specific but are aviation-related and contain airport content. (These are yellow in the electronic version.)

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APPENDIX B

Understanding Accreditation

Important Note: The following discussion on accreditation does not reflect an endorsement of any accrediting organization by ACRP or the authors but is provided to illustrate specific characteristics and criteria by which most programs can be assessed whether or not they are accredited.

Most institutions of higher learning subscribe to a form of accreditation as a means to ensure the quality of their programs of study. Criteria for accreditation are set by independent associations that may be aligned along regional, national, or specific professional interests. Public colleges and universities generally subscribe to regional accreditation bodies such as the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC), the Middle States Commission on Higher Education (MSCHE), and the Northwest Commission on Colleges and Universities (NWCCU).

Nationally accredited schools are generally characterized as for-profit and vocational/technical programs. The Accrediting Commission on Career Schools and Colleges and the American Academy for Liberal Education are two examples of national accreditation agencies.

Many accrediting organizations are aligned with specific industries. The accreditation standards are generally based on recognized educational foundations but also focus on specific learning outcomes embedded in the curriculum relevant to the particular industry's interests. The goal of the accreditation organization is to evaluate the institution and its programs using several different criteria and to verify that the recognized standards are being met, and in doing so, promote a standard of quality.

Many collegiate aviation programs are accredited not only by their university-level regional accrediting board but also by the AABI. AABI was founded in 1988 with 14 charter members. The mission of AABI is to promote excellence and self-improvement among collegiate aviation programs and, by doing so, increase the credibility, integrity, and acceptance of these programs within the academic community and the aviation industry.

AABI accredits all levels of collegiate aviation programs, from associate degrees to doctorates. The types of collegiate aviation programs that are eligible for accreditation include the following:

- Aviation Management
- Flight Education
- Aviation Maintenance
- Safety Science
- Aviation Electronics/Avionics
- Air Traffic Control
- Aviation Studies
- Unmanned Aircraft Systems

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To date, there are 94 accredited aviation programs hosted by 34 colleges and universities throughout the United States, including 4 associate degrees, 84 baccalaureate degrees, 3 master's degrees, and 1 doctoral degree. From these, there are 29 accredited aviation management degree programs.

AABI promotes specific aviation-related educational outcomes that must be demonstrated by the program graduate's ability to

1. Describe the professional attributes; requirements or certifications; and planning applicable to aviation careers.
2. Describe the principles of aircraft design; performance and operating characteristics; and the regulations related to the maintenance of aircraft and associated systems.
3. Evaluate aviation safety and the impact of human factors on safety.
4. Discuss the impact on aviation operations of international aviation law, including applicable ICAO or other international standards and practices; and applicable national aviation law, regulations, and labor issues.
5. Explain the integration of airports, airspace, and air traffic control in managing the National Airspace System.
6. Discuss the impact of meteorology and environmental issues on aviation operations.

The ten major components evaluated by AABI for accrediting a degree program include the following:

1. *Students*—student goals consistent with the mission and educational goals of the program and institution.
2. *Program Mission and Educational Goals*—mission statement and goals that reflect the program's educational philosophy, purpose, and general intent.
3. *Student Learning Outcomes* (General/Aviation Core)—process to ensure student's ability to demonstrate competency in meeting general and aviation-specific outcomes.
4. *Curriculum*—courses available to support each learning outcome and prepare graduates for successful aviation careers.
5. *Faculty and Staff*—adequate number of qualified faculty and staff are available to achieve the learning outcomes.
6. *Facilities, Equipment, and Services*—adequate resources are available to accomplish the program goals in an appropriate learning environment.
7. *Institutional Structure and Support*—financial and administrative support to assure the continuity of the program.
8. *Aviation Safety Culture and Program*—formal policies, procedures, and practices in place that emulate SMS principles to foster a culture of safety.
9. *Relations with Industry*—established framework for ensuring that the program remains relevant to the aviation industry. Industry Advisory Boards with members representing a diverse mix of various segments of aviation are common.
10. *Continuous Assessment and Improvement*—formal mechanism to assess the effectiveness of each program component and measures taken for improvement.

To retain their status, AABI-accredited programs are reviewed every 5 years and must be able to provide evidence that the outcomes are being successfully achieved, and there is a process in place to continually assess and improve the program.

AABI does not have an accreditation program specifically for airport management but airport-focused curricula are found within other accredited programs, most often in the aviation management or aviation studies degrees. Opportunities may exist to engage in further development of a standard for airport-oriented education, adding a new accreditation category among AABI's other aviation academic programs.



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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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ISBN 978-0-309-67385-3

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